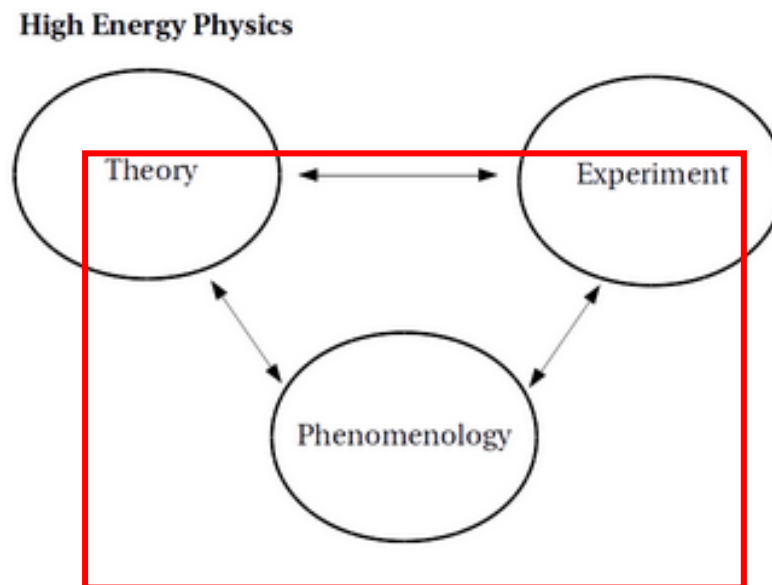


# Particles and Nuclei 2

- First Day 1/15/26
- Announcements:
- Canvas - available
- HW 1 Not finalized but could start thinking of those Griffiths problems.
- INITIAL READING from Tuesday Griffiths Ch 1 by ~Thursday. Mann 1.2 and 1.3.
- Continue w/ Ch 2 over the weekend. (Finish by middle of next week)
- Preview of some HW 1 problems available will post by tomorrow.

# Particles and Nuclei 2 - Theory

- OU Nuclear grad courses (traditionally):
  - Particles and Nuclei 1 -- Nuclear Physics
  - Particles and Nuclei 2 -- Particle Physics
- The course will be about  $\frac{3}{4}$  basics of various aspects of “Particle” Physics “theory”, much more so “phenomenology”
- The other  $\sim \frac{1}{4}$  will be experiment

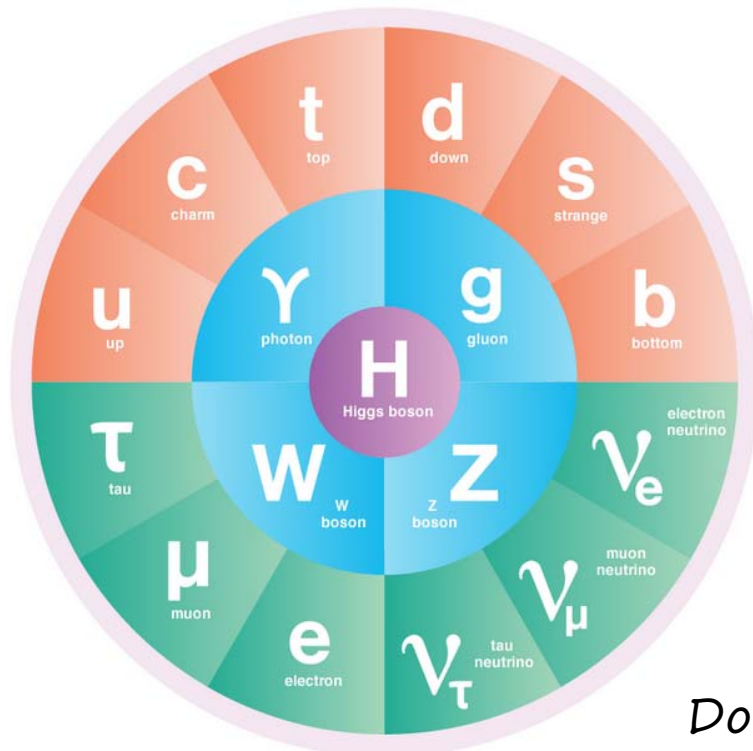


Some theorists  
due to ego use  
“phenomonology”  
pejoratively –  
we won’t!

- I am an experimentalist not a theorist, we will be avoiding field theory as much as possible – but there will still be (quite?) a bit

# Standard Model – Particles

- “Most elementary”
- So far no sign of size (“point particles”) or “sub”structure
- Some more exotic theories propose quark structure (“techniquarks”)
- But the 2 leading BSM (Beyond Standard Model) theories
  - Super symmetry / String Theory -- ~both consider all elementary

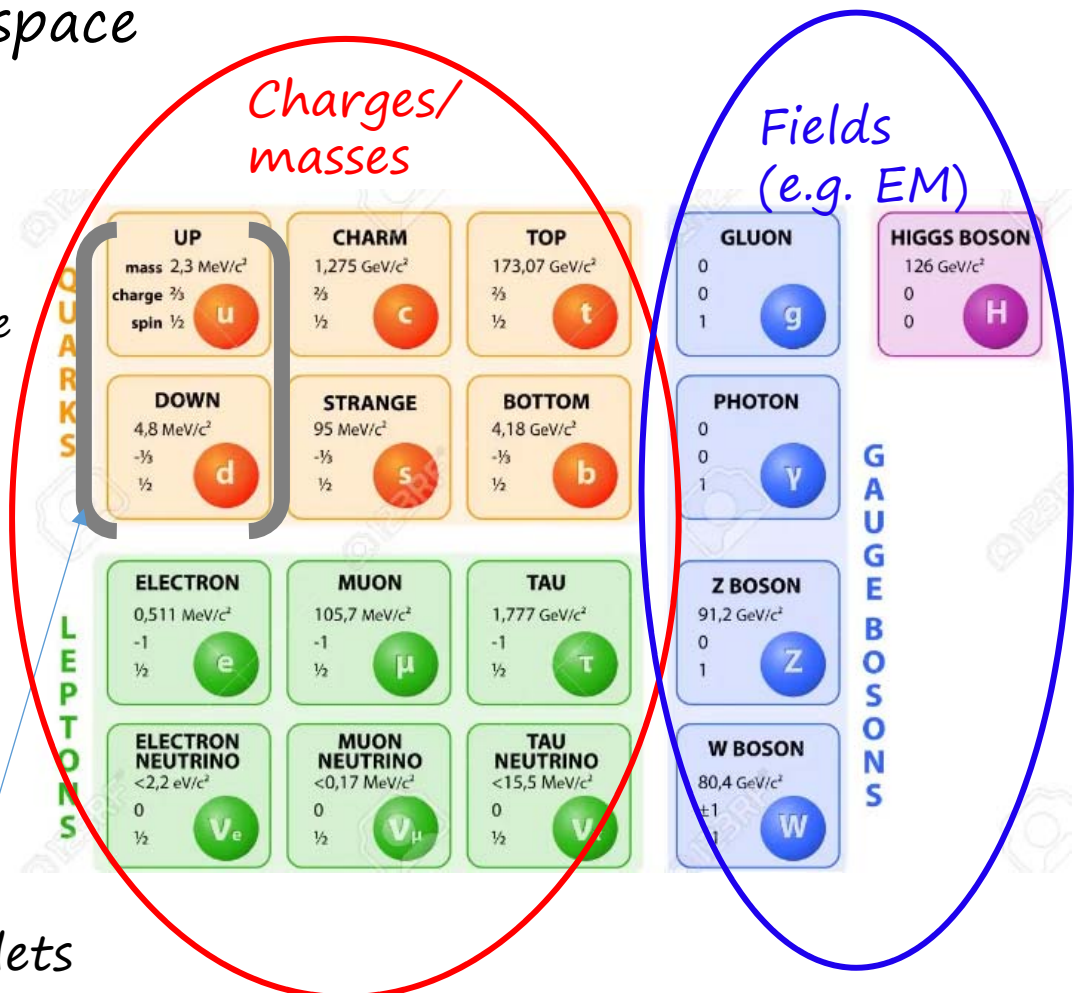
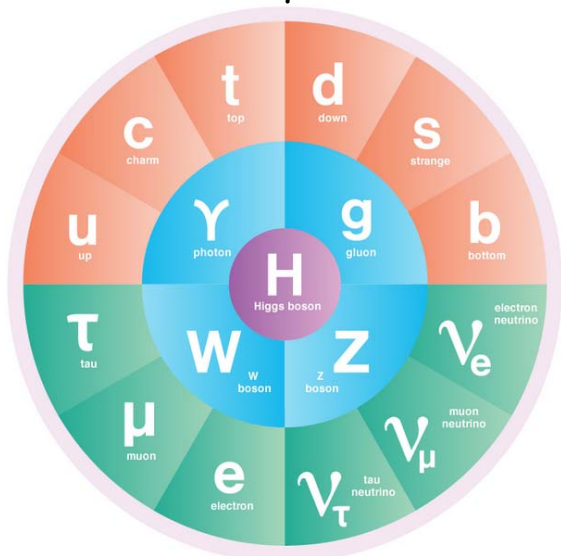


|        |                                                                                                                                                                         |                                                                                                                                                                          |                                                                                                                                                                        |              |                                                                                                                                                                       |                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUARKS | <div><b>UP</b><br/>mass 2,3 MeV/c<sup>2</sup><br/>charge 2/3<br/>spin 1/2<br/></div> | <div><b>CHARM</b><br/>1,275 GeV/c<sup>2</sup><br/>2/3<br/>1/2<br/></div>              | <div><b>TOP</b><br/>173,07 GeV/c<sup>2</sup><br/>2/3<br/>1/2<br/></div>             | GAUGE BOSONS | <div><b>GLUON</b><br/>0<br/>0<br/>1<br/></div>                                     | <div><b>HIGGS BOSON</b><br/>126 GeV/c<sup>2</sup><br/>0<br/>0<br/></div> |
|        | <div><b>DOWN</b><br/>4,8 MeV/c<sup>2</sup><br/>-1/3<br/>1/2<br/></div>              | <div><b>STRANGE</b><br/>95 MeV/c<sup>2</sup><br/>-1/3<br/>1/2<br/></div>             | <div><b>BOTTOM</b><br/>4,18 GeV/c<sup>2</sup><br/>-1/3<br/>1/2<br/></div>          |              | <div><b>PHOTON</b><br/>0<br/>0<br/>1<br/></div>                                   |                                                                                                                                                             |
|        | LEPTONS                                                                                                                                                                 | <div><b>ELECTRON</b><br/>0,511 MeV/c<sup>2</sup><br/>-1<br/>1/2<br/></div>          | <div><b>MUON</b><br/>105,7 MeV/c<sup>2</sup><br/>-1<br/>1/2<br/></div>            |              | <div><b>TAU</b><br/>1,777 GeV/c<sup>2</sup><br/>-1<br/>1/2<br/></div>            | <div><b>Z BOSON</b><br/>91,2 GeV/c<sup>2</sup><br/>0<br/>1<br/></div>  |
|        |                                                                                                                                                                         | <div><b>ELECTRON NEUTRINO</b><br/>&lt;2,2 eV/c<sup>2</sup><br/>0<br/>1/2<br/></div> | <div><b>MUON NEUTRINO</b><br/>&lt;0,17 MeV/c<sup>2</sup><br/>0<br/>1/2<br/></div> |              | <div><b>TAU NEUTRINO</b><br/>&lt;15,5 MeV/c<sup>2</sup><br/>0<br/>1/2<br/></div> | <div><b>W BOSON</b><br/>80,4 GeV/c<sup>2</sup><br/>±1<br/>1<br/></div> |
|        |                                                                                                                                                                         |                                                                                                                                                                          |                                                                                                                                                                        |              |                                                                                                                                                                       |                                                                                                                                                             |
|        |                                                                                                                                                                         |                                                                                                                                                                          |                                                                                                                                                                        |              |                                                                                                                                                                       |                                                                                                                                                             |

Doublets

# Standard Model – Particles

- Relation to Classical Understandings
  - Fields and Charges/Masses – Occupying space & @ specific locations respectively  $\rightarrow$  Fields mediate forces btw charges
- But at quantum sizes/level both are **Probability**  
Fields existing over all space
- Q.F.T. ~mostly removes these distinctions
  - e.g. all fields everywhere
  - e.g. photons can interact with  $e$  other via  $q, e^{+/-}$  “mediators”



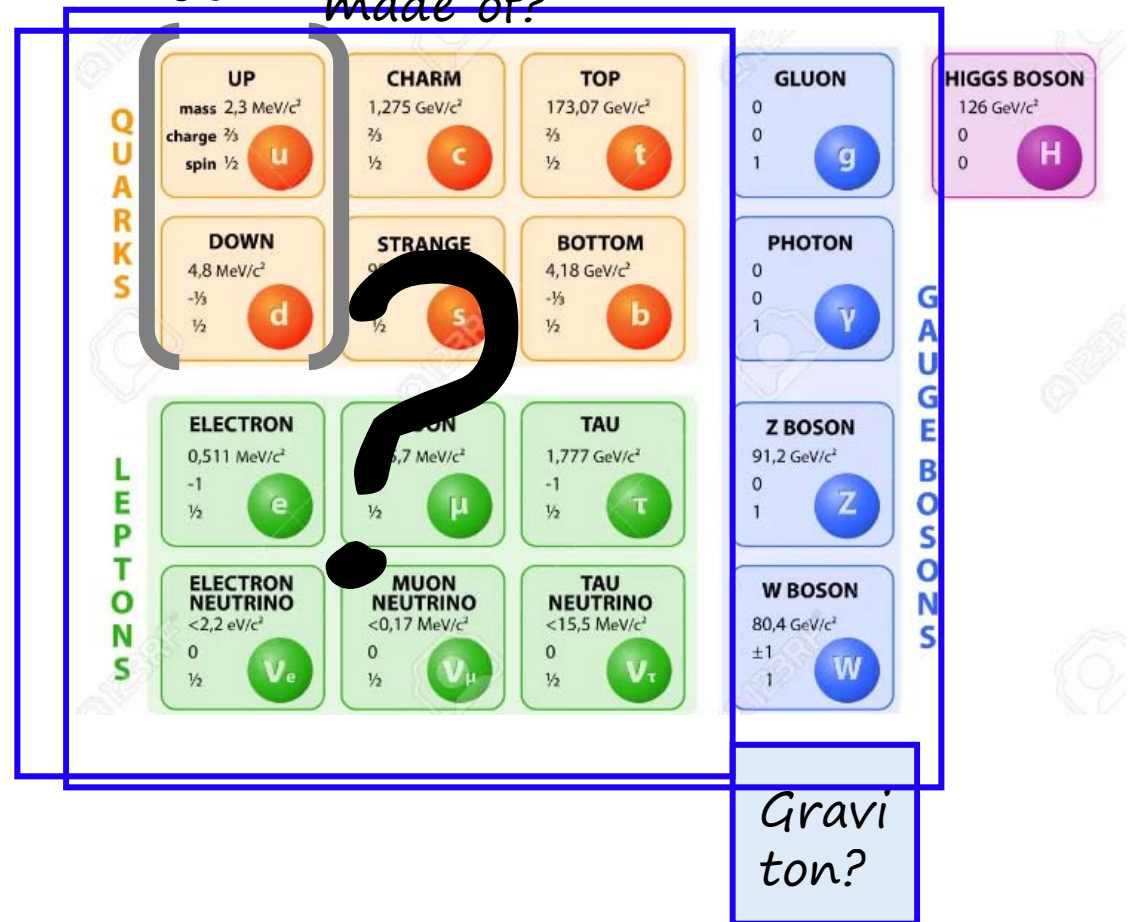


# Standard Model – Higgs/Gravity

- Graviton?
- Higgs  $\rightarrow$  Gravity ?
- Both related directly to mass?
- no, gravity depends on energy

Gravity – General relativity  
 “classical” field is Curvature of spacetime  
 Gravitational waves  
 $\rightarrow$  not gravitons themselves but made of?

*Need theory at least that resolves quantum uncertainty in position and curvature doesn't need to have graviton*



# Preview: Main Particles (Esp In Nuclear)

- These are in practice as good as supermultiplets

| Baryons     | Quark composition |
|-------------|-------------------|
| P           | uud               |
| N           | Udd               |
| $\Sigma^+$  | suu               |
| $\Sigma^0$  | sud               |
| $\Sigma^-$  | sdd               |
| $\Xi^0$     | ssu               |
| $\Xi^-$     | ssd               |
| $\Lambda^0$ | sud               |
| $\Omega^-$  | sss               |

| Mesons      | Quark composition                              |
|-------------|------------------------------------------------|
| $\pi^+$     | $u\bar{d}$                                     |
| $\pi^0$     | $(u\bar{u} - d\bar{d}) / \sqrt{2}$             |
| $\pi^-$     | $d\bar{u}$                                     |
| $K^+$       | $s\bar{u}$                                     |
| $K^0$       | $d\bar{s}$                                     |
| $K^-$       | $\bar{u}s$                                     |
| $\bar{K}^0$ | $\bar{d}s$                                     |
| $\eta^0$    | $(u\bar{u} + d\bar{d} - 2s\bar{s}) / \sqrt{6}$ |
|             |                                                |

chegg.com

- Wikipedia has very good lists:
  - [https://en.wikipedia.org/wiki/List\\_of\\_baryons](https://en.wikipedia.org/wiki/List_of_baryons)
  - [https://en.wikipedia.org/wiki/List\\_of\\_mesons](https://en.wikipedia.org/wiki/List_of_mesons)

# Standard Model (SM) – 3 QFT's

- The Standard Model is “three” Quantum Field Theories (QFT) combined

- QED Q Electrodynamics
- QFD Q Flavordynamics
  - (QGD, QAD) (Weak F)
  - (Greek flavor gueifsi- aroma)



Electroweak theory -Two parts of a single theory/set of particles connected/generated from combinations of shared set

- QCD Q. Chromodynamics (Strong F)

# Units : Caveats

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- “Reason” (?)
- $m_{\text{proton}} = \sim 1 \text{ GeV}$
- use GeV for every energy/momentum scale
  - Occasionally others
- Nominally [Length] = [Time] = 1/GeV
- However Some Important Caveats

$$\sigma_x \sigma_p \leq \frac{\hbar}{2} \leq \hbar$$

$$\sigma_t \sigma_E \leq \hbar$$

## Heavy-ions at the LHC

- Energy per nucleon in a  $^{208}_{82}\text{Pb-Pb}$  collision at the LHC (Run 1):
  - Total collision energy in Pb-Pb:  
 $\sqrt{s} = 574 \text{ TeV}$
  - Run 2:  $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$  and thus  
 $\sqrt{s} = 1.04 \text{ PeV}$

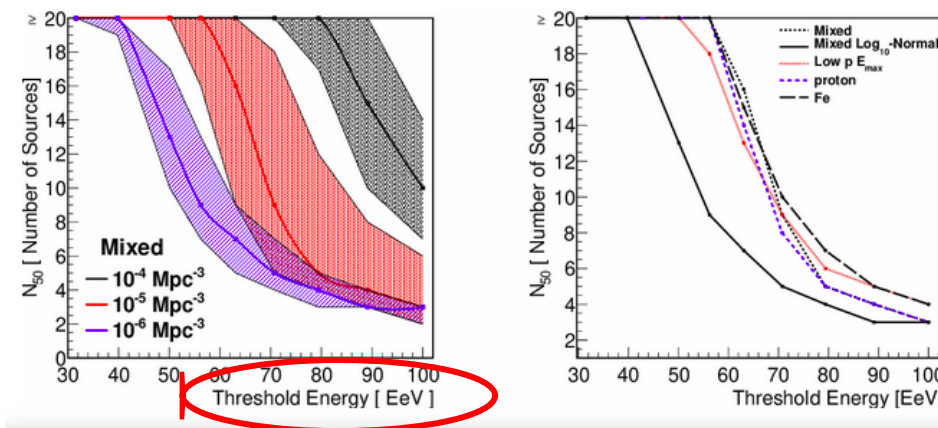
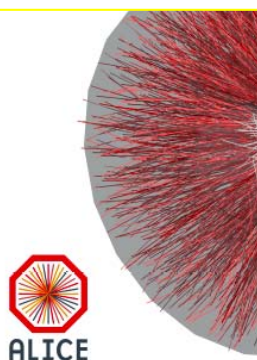
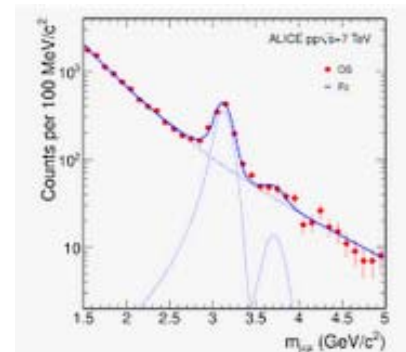
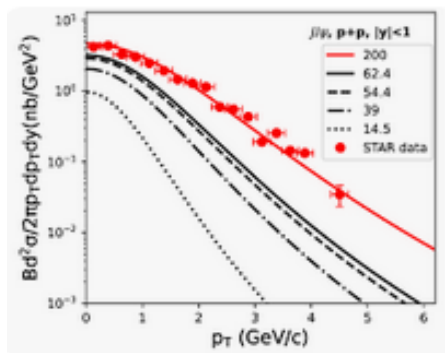


Figure 1: Number of sources contributing to the flux of UHECRs observed on Earth above a given energy  $E$ , as a function of that energy (in abscissa), for different UHECR source scenarios. Left: for a



# Units – Caveats I

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- use GeV for every energy/momentum scale  $\sigma_x \sigma_p \leq \frac{\hbar}{2} \leq \hbar$
- Nominally [Length] = [Time] = 1/GeV (Heisenberg)  $\sigma_t \sigma_E \leq \hbar$
- Caveats I
- Even though  $c = 1$ , units for momentum *have traditionally been* written explicitly as  $[p] = \text{GeV}/c$ 
  - To distinguish? *LHC: (recent) new standard -- NOT using the c's!*
- For mass, mostly quoted in GeV or MeV but for plots  $[m] = \text{GeV}/c^2$ 
  - LHC: (recent) new standard -- NOT using the c's!*
- (Just because two quantities have the “same” units in this system, doesn't mean they ARE the same )



# Units – Caveats fm $\hbar c$

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- use GeV for every energy/momentum scale
- Nominally [Length] = [Time] = 1/GeV (Heisenberg)

$$\sigma_a \sigma_p \leq \frac{\hbar}{2} \leq \hbar$$

$$\sigma_t \sigma_E \leq \hbar$$

- **Caveats**
- For HEP the above [L] / [T] may be fine in many circumstances
- For NP for length we use the **fm** (femtometer, but ~always called “Fermi”  $10^{-15}$  m) (= size of nucleon/p/n – again unit is “nucleon”)
- Time (especially important for RHIC  $\rightarrow$  time evolution of QGP) will be in:
  - **fm/c**
- Then there is this VERY IMPORTANT RELATION that must be employed carefully
  - **$\hbar c \approx 0.197 \text{ GeV} \cdot \text{fm}$**
  - “factors of two are for chumps!”

# Units – Caveats time

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- use GeV for every energy/momentum scale
- use GeV for every energy/momentum scale
- Nominally [Length] = [Time] = ~~1/GeV~~ (Heisenberg)
- $\hbar c \approx 0.197 \text{ GeV} \cdot \text{fm}$
  
- Nucleon size  $\sim r = 0.85 \approx 1 \text{ fm}$
- Au Nuclei size (e.g.)  $r = 6 \text{ fm}$
- “Fastest” Decaying Hadrons (via Strong force) Particles  
lifetime:  $1e-24 \text{ s} \rightarrow c\tau = 0.3 \text{ fm}$        $1e-23 \text{ s} \rightarrow 3 \text{ fm}$
- Apparently Z / W boson decay  $\tau$  currently  $1e-25$  (?)  $0.03 \text{ fm}$

# Particle Properties - $c\tau$

- $c\tau$  scale of distance particle travels ( $1/e\dots$ ) before decaying
- assuming  $\sim \text{GeV}$  scale for most particles. technically  $E$  such that speed  $\sim c$ )
- But  $\gamma$  greater than 1 also means time dilation so
- $t = \gamma\tau$

- So actual decay length is

$$L_{\text{decay}} = \gamma c\tau$$

We will see that  
the easiest way to  
calc  $\gamma$  is from  
 $\gamma = E / m$



# Zach Meisel's Slide with this

- Review

## Nuclear Forces, mechanisms for binding & transmutation

- Fundamental forces of nature:

| Force           | Range (m)  | Relative strength      | Force carrier, X |
|-----------------|------------|------------------------|------------------|
| Gravitational   | $\infty$   | $10^{-38}$             | Graviton         |
| Weak            | $10^{-18}$ | $10^{-5}$              | $W^{\pm}, Z^0$   |
| Electromagnetic | $\infty$   | $\alpha \approx 1/137$ | Photon           |
| Strong          | $10^{-15}$ | 1                      | Gluon, Pion      |

- The range comes from the uncertainty principle:

- When converting reactant(s) to product(s), energy is spent on the mass of the force carrier:  $\Delta E \geq M_X c^2$
- Violating energy conservation like this is only allowed due to the uncertainty principle:  $\Delta E \Delta t \geq \hbar/2$
- And therefore the force carrier must be reabsorbed before:  $t \leq \hbar/(2\Delta E)$
- Since a particle can't be moving faster than the speed of light,  $c$ , the furthest range a force can be mediated is:  
 $r \leq tc = \hbar c/(2\Delta E) = \hbar c/(2M_X c^2) = \hbar/(2M_X c)$
- Thus, since factors of 2 are for chumps, the range of an interaction is defined as:  **$R \equiv \hbar/(M_X c)$**
- Since the graviton and photon are massless,  $R = \infty$ .
- For the weak force,  $M_X \sim 100\text{GeV}/c^2$ , so  $R_{\text{weak}} \sim 10^{-3}\text{ fm}$ .
- Similarly, since  $M_{\pi} \sim 100\text{MeV}/c^2$ ,  $R_{\text{strong}} \sim 1\text{ fm}$ .

*The trick for these calculations is that  $\hbar c \approx 197\text{MeV}\cdot\text{fm}$ .*

*Note force range relative to nucleon size.*



# Units – Caveats time

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- use GeV for every energy/momentum scale
- Nominally [Length] = [Time] = ~~1/GeV~~ (Heisenberg)
- $\hbar c \approx 0.197 \text{ GeV} \cdot \text{fm}$

## Another example

From Mann

For example in natural units the Compton wavelength  $\lambda = 1/m$ . To convert this to physical units we set  $\lambda \rightarrow \lambda/\hbar c$  and  $m \rightarrow mc^2$ , giving  $\lambda \rightarrow \lambda/\hbar c = 1/mc^2$  or  $\lambda = \hbar/mc$ , which is the standard formula.

# Units – Caveats time – exercise

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- use GeV for every energy/momentum scale
- use GeV for every energy/momentum scale
- Nominally [Length] = [Time] = ~~1/GeV~~ (Heisenberg)
- $\hbar c \approx 0.197 \text{ GeV} \cdot \text{fm}$
- $\hbar \approx 0.197 \text{ GeV} \cdot \text{fm} / c$

$$\sigma_x \sigma_p \leq \frac{\hbar}{2} \leq \hbar$$
$$\sigma_t \sigma_E \leq \hbar$$

## EXERCISE

- At RHIC the many particles produced are produced with an average Energy of around 500 MeV. What is an estimate for the formation time of the particles on average?
  - Put answer in tophat

# Units – Caveats time – exercise

- Natural Units  $\hbar = c = \dots 1$
- GeV for energy (eV... MeV and TeV also common)
- use GeV for every energy/momentum scale
- use GeV for every energy/momentum scale
- Nominally [Length] = [Time] = ~~1/GeV~~ (Heisenberg)
- $\hbar c \approx 0.197 \text{ GeV} \cdot \text{fm}$
- $\hbar \approx 0.197 \text{ GeV} \cdot \text{fm} / c$

$$\sigma_x \sigma_p \leq \frac{\hbar}{2} \leq \hbar$$

$$\sigma_t \sigma_E \leq \hbar$$

## EXERCISE

- At RHIC the many particles produced are produced with an average Energy of around 500 MeV. What is an estimate for the formation time of the particles on average?

- Put answer in tophat

$$\Delta t = \frac{\hbar}{0.5 \text{ GeV}} = \frac{0.2 \text{ GeV} \cdot \text{fm}/c}{0.5} = \sim 0.3 \text{ fm}/c$$

# Particles and Nuclei 2

- Third Day 1/20/26
- Announcements:
- Canvas - available
- HW 1 FINALIZED: Due ~Monday/Tues Morning.
- READING Thru Ch 4.3 (+ Mann) by next time (can SKIP Ch3 Griff), then Lancaster & Blundell (L&B) Ch 3, 4, [handout pdf, slightly less than that]:
- Then Start on Chapter 3.
- Preview of some HW 1 problems available will post by tomorrow.

# Roadmap

- BASICS (Ch1,2,4)
  - Particles (Properties : mass, ctau, charges1, ..., spin,...)
  - Interactions
  - QCD Complications
- Relativistic Kinematics (Ch3 + Rapidity + )
- A little bit more on basics (Ch 4/5)
- Feynman diagram calculations (Ch 6 +....)

# Standard Model (SM) – QFT's

- QFT's:

- Fields for every particle, existing everywhere
- same treatment force carriers (e.g. photons / electric field) and interacting particles other than spin characteristics
- explicit rep. of all fields involve a<sup>†</sup> a creation/annihilation operators making quanta explicit...vector/group theory properties built in

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left( a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^{\dagger} e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}} \right)$$

Act on generalized  
number states  $|0\rangle, |1\rangle,$   
...



# QFT's – Quick Bkg Prep

- Field operators are used to construct the “usual” Lagrangian or Hamiltonian form ( $L = T - V, T + V$ )
  - relativistic/field forms but similar to  $x, \dot{x}, q, \dot{q}$  ones

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_j} \right) = \frac{\partial L}{\partial q_j}$$

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^\dagger e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}})$$



$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2$$

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} F_{\mu\nu}^a F_a^{\mu\nu} + m_f \bar{\Psi} \Psi + \bar{\Psi} (\partial_\mu - g_s t^a \bar{A}_\mu^a - g_e A_\mu) \gamma^\mu \Psi$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

- Apply e.g. minimization of action / solve eq. motion
  - Luckily for us Feynman path way of doing Quantum Mech + P. T. simplifies expressions a lot (but they are still very complicated)

# QFT's – Cool Stuff – Some focus on

- QFT's Explain:
  - different yet indistinguishable particles
  - boson vs fermion behavior:
  - **PAULI EXCLUSION PRINCIPLE --**
  - anti-particles , force carrier exchange
  - Noethers theorem charges  $\rightarrow$  symmetries



| HELIUM 1s ORBITAL ELECTRON SPIN                                                       |                                                                                       |                                                                                       |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |
| X                                                                                     | X                                                                                     | ✓                                                                                     |

Hund's Rule, the Pauli Exclusion Principle...

I will cover some of these highlights of "pure QFT" to the best of my ability ...Peskin other reading...

# QFT's – Quick Bkg Prep

- Field operators are used to construct the “usual” Lagrangian or Hamiltonian form ( $L = T - V, T + V$ )
  - relativistic/field forms but similar to  $x, \dot{x}, q, \dot{q}$  ones

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_j} \right) = \frac{\partial L}{\partial q_j}$$

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^\dagger e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}})$$



$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2$$

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} F_{\mu\nu}^a F_a^{\mu\nu} + m_f \bar{\Psi} \Psi + \bar{\Psi} (\partial_\mu - g_s t^a \bar{A}_\mu^a - g_e A_\mu) \gamma^\mu \Psi$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

- Apply e.g. minimization of action / solve eq. motion
  - Luckily for us Feynman path way of doing Quantum Mech + P. T. simplifies expressions a lot (but they are still very complicated)

# Perturbation Theory

- Free particle part:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}F_{\mu\nu}^a F_a^{\mu\nu} + m_f \bar{\Psi}\Psi + \bar{\Psi}(\partial_\mu - g_s t^a \bar{A}_\mu^a - g_e A_\mu)\gamma^\mu \Psi$$

- F.P. Solutions:

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left( a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^\dagger e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}} \right)$$

Act on generalized  
number states  $|0\rangle, |1\rangle,$   
...

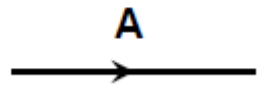
- P.T. Theory “Matrix Elements” Sandwiches of the  $|0\rangle$  :
  - L&B reading starts to elucidate this w/ more detail
  - Algebra/tricks of a  $a^\dagger$
  - What the Feynman Diagrams Represent in terms of these states

# QFT's – Quick Bkg Prep – The Approximation

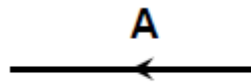
- These methods result in *Pert. Theory* expansion
  - “Tree level” leading order LO term + “next”/ 2nd order, NLO + NNLO + NNNLO + .....N<sup>i</sup>LO
- Successively more complicated Integrations for higher and higher order terms
  - Order parameter is  $\propto$  naturalized coupling  $g^2$  ( $\alpha = 1/137$  for EM)
- Each order is  $1/\alpha$  smaller
  - -e.g. for QED  $< 1\%$  of previous
  - - for QCD at low energy  $\alpha \rightarrow 1$
- Some schemes to “resum” parts to all orders – “resummation”
- Next to Leading Log (NLL) ?
  - e.g. N<sup>3</sup>LO NLL... (for QCD ~all calcs are at at least NLO)

# Feynman Diagrams

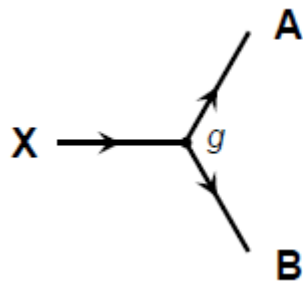
- Following Krieger (U Toronto) slides



Stable particle (A) in free space (moving or at rest)



Stable anti-particle ( $\bar{A}$ ) in free space (moving or at rest)

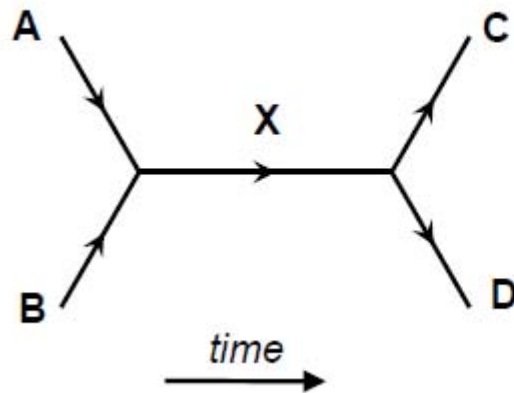


Particle decay  $X \rightarrow A B$

Strength of interaction given by coupling constant  $g$

# Basics

- (Kreiger)

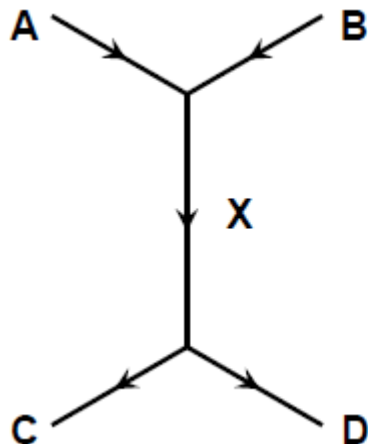


if for example, X also couples C to D  
(e.g. there is an XCD vertex)

Combine two vertex primitives to make  
a (lowest order) scattering diagram

$$A + B \rightarrow C + D$$

*s type*



Rotate diagram by 90°

$$A + \bar{C} \rightarrow \bar{B} + D$$

Amplitudes for these two processes  
are related (crossing symmetry).

*t type*

*Related but  
not same,  
as this is a  
different  
physical  
process*

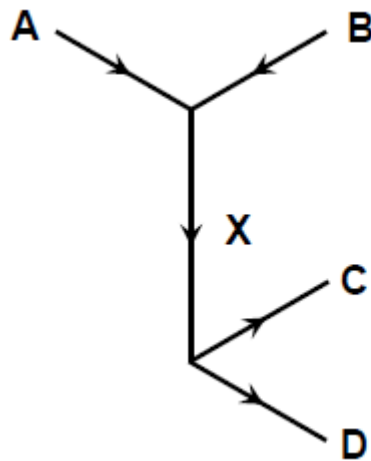


# Feynman Diagrams

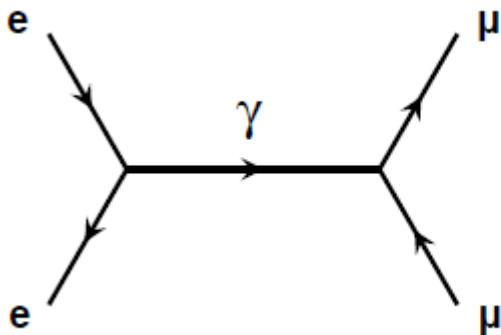
- Assuming this is already known: Each part of the diagram corresponds to a mathematical operation (integral) where each part contributes specific factors that will get integrated.
- Most importantly each vertex contributes a factor of coupling “ $g$ ” which for QED is  $e$  (fund.charge) and so the tree level diagrams with two vertices are  $\sim \text{propto } e^2$   
 $\sim \text{propto } \alpha_{EM} = 1/137$

## Others - EM

- These represent the LO terms in the QFT series proportional to  $\alpha = 1/137$

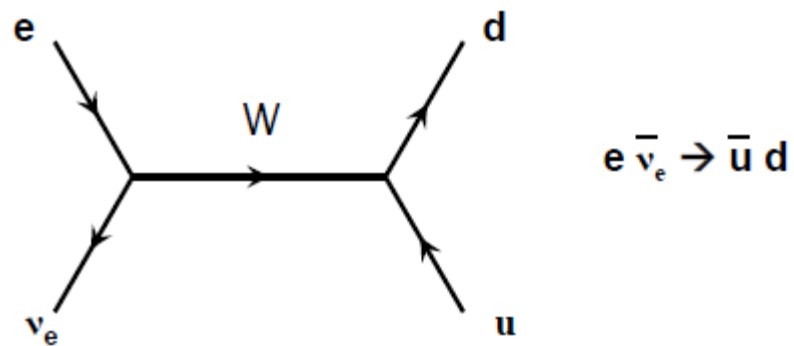
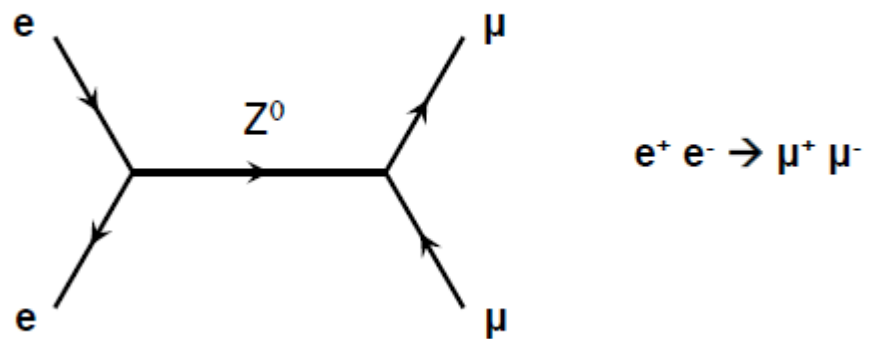


$A \rightarrow \bar{B} C D$   
Decay of particle A



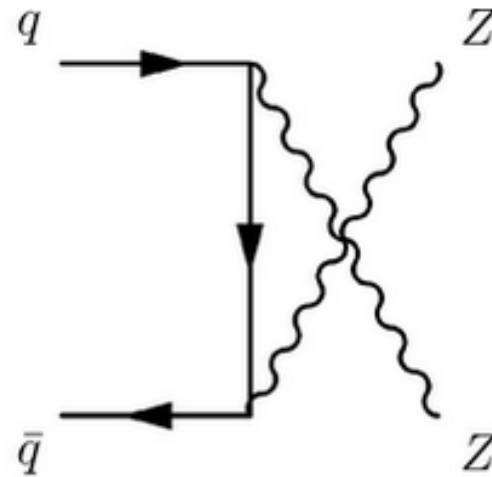
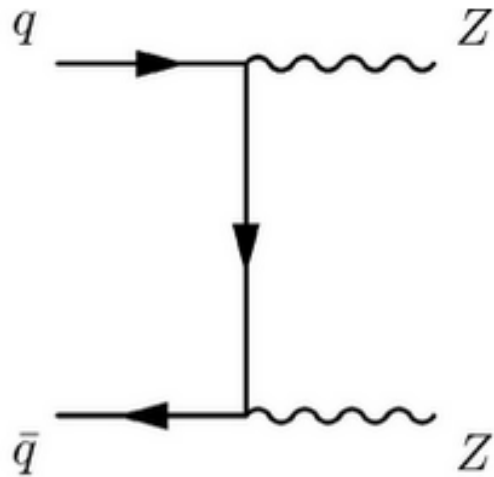
$e^+ e^- \rightarrow \mu^+ \mu^-$

# Weak



# Crossed Diagrams

- Different Process physically how to describe?



$$A + B \rightarrow \frac{C}{Z_1} + \frac{D}{Z_2}$$

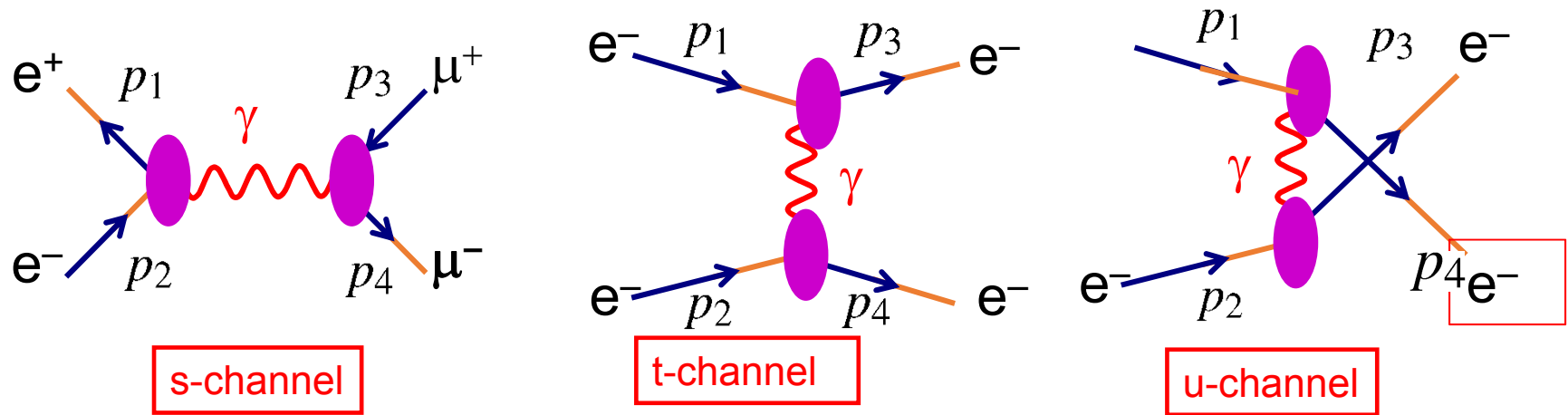
$$A + B \rightarrow \frac{D}{Z_1} + \frac{C}{Z_2}$$

It is two different physical situations and different cross section calcs will result, but why?

First some terminology to help..

# Some Terminology

- The kind of scattering goes by “s,t,u” which are kinematic variables that will be introduced later
- For now consider it just terminology

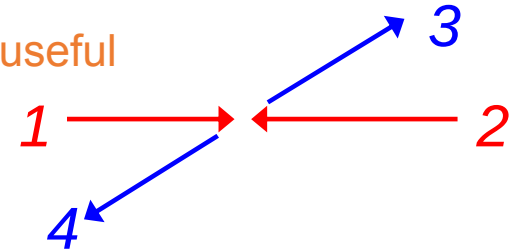


- $s$  describes annihilation to photon and then photon “decaying”
- $t$  and  $u$  describe classic “interaction”:  $e^-$  on top emits photon (exerts force) absorbed by  $e^-$  on bottom.

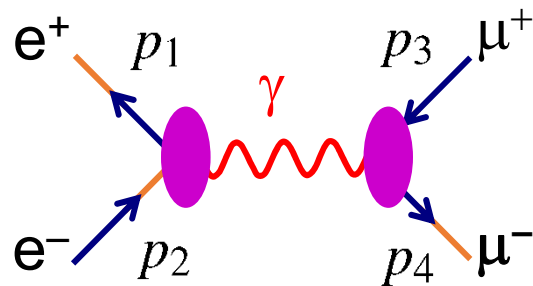
# Mandelstam $s$ , $t$ and $u$

- ★ In particle scattering/annihilation there are three particularly useful Lorentz Invariant quantities:  $s$ ,  $t$  and  $u$

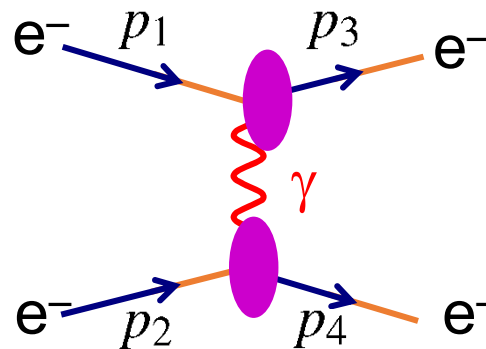
- ★ Consider the scattering process  $1 + 2 \rightarrow 3 + 4$



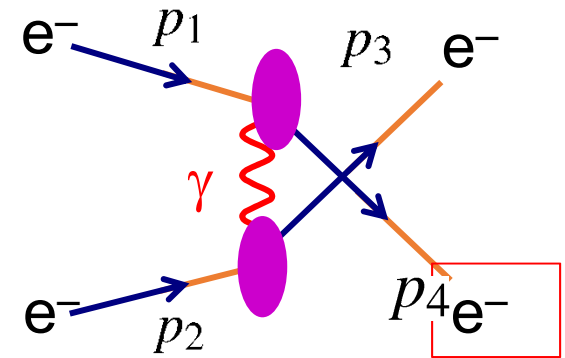
- ★ (Simple) Feynman diagrams can be categorised according to the four-momentum of the exchanged particle



s-channel



t-channel



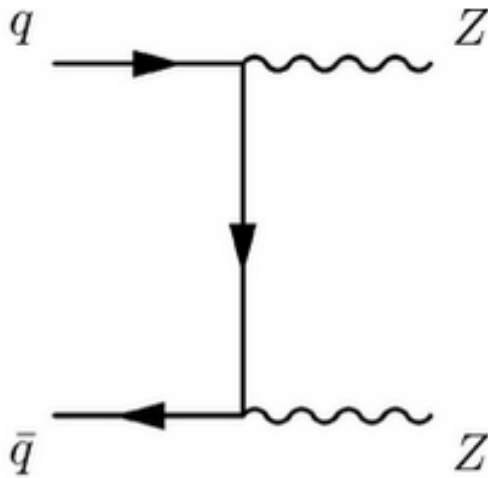
u-channel

- Can define **three** kinematic variables:  $s$ ,  $t$  and  $u$  from the following four vector scalar products (squared four-momentum of exchanged particle)

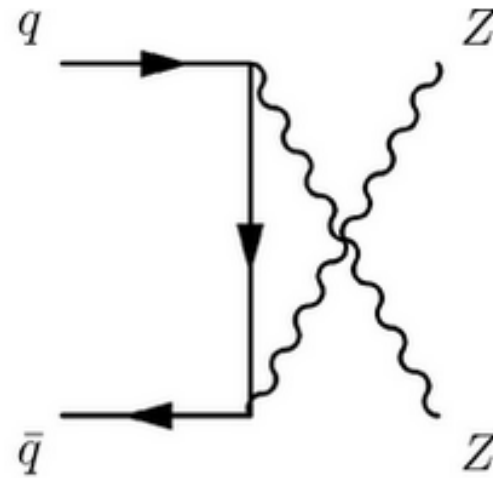
$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2$$

# Crossed Diagrams

- Different Process physically how to describe
- So in this slide we have t and u cases, but these processes also have the q's converting to Z's (which is also common for t channel processes – even QCD processes will change the color of the top and bottom quarks, e.g. if they are exchanging a gluon) (“q's scatter by emitting/absorbing a virtual q in this case, AND convert to Z's both”)



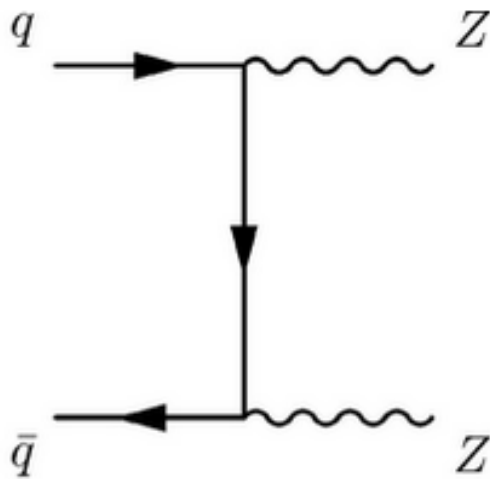
$$A + B \rightarrow \underset{Z_1}{C} + \underset{Z_2}{D}$$



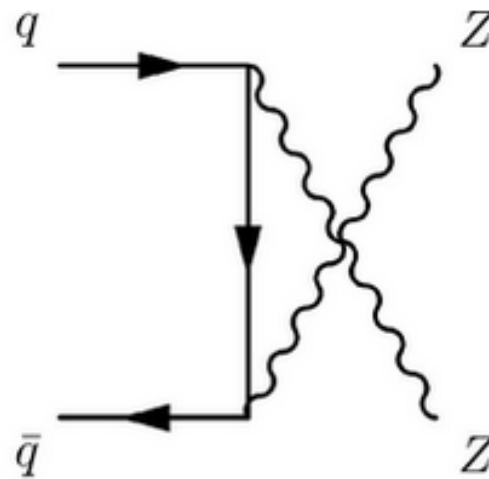
$$A + B \rightarrow \underset{Z_1}{D} + \underset{Z_2}{C}$$

# Crossed Diagrams

- Different Process physically how to describe
- So now to explain what the “crossed” diagram on right is compared to first/left:



$$A + B \rightarrow C_{Z_1} + D_{Z_2}$$



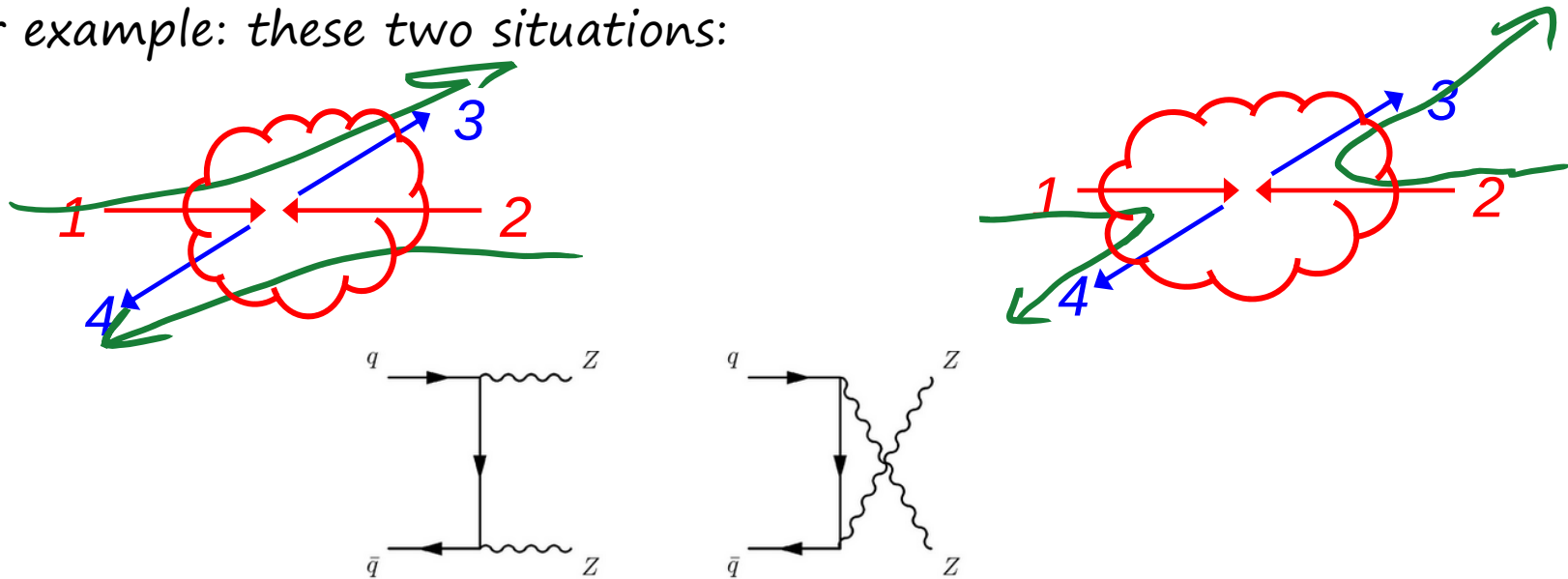
$$A + B \rightarrow D_{Z_1} + C_{Z_2}$$

- because aren't  $C$  and  $D$  arbitrary labels?
- The difference is a kinematic one, think of  $C + D$  vs  $D + C$  as assignments for two different kinematic situations



# Crossed Diagrams – Kine situations

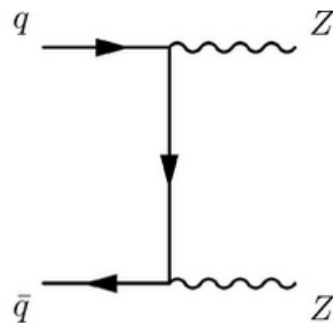
- ...for example: these two situations:



- So the two kinematic assignments  $C+D$  vs  $D+C$  are like left upper vs right upper cases –  $C+D$  is when  $C$  is essentially originally  $A$ , ( $1 \rightarrow 3$ )  $2 \rightarrow 4$  /  $B \rightarrow D$ 
  - “small deflection : momentum of  $C$  is almost same as  $A$ , etc. “ $A \rightarrow C$ ” “single” particle trajectory
- $D+C$  is case where  $A$  and  $D$  ( $1 \rightarrow 4$ ) are “single” trajectory, and there is large deflection, both  $A, B$  are turned completely around.
- The two kine situs are clearly different—even if labels are ambiguous
- And top assignments aren't actually unambiguously associated with  $t$  vs  $u$  being Left vs Right –  $t$  could be right and  $u$  left—but since it only makes sense to consider  $u$  (when possible to cross) in comparison to “ $t$ ”, so...
- So think of  $t$  and  $u$  together, for every contribution for  $t$  channel on left, there is always also a contribution from  $u$  like the one on the right.
- ALSO NOTE: SINCE ALWAYS TOGETHER, THESE TWO Amps INTERFERE QM'ically**

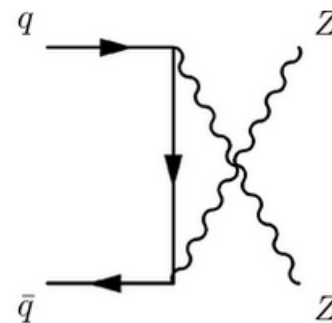
# Another thing - Interference

- Feynman diagrams mathematically typically are used for cross section measurements.
- They result in complex numbers and represent “matrix element” “ $M$ ”
- Therefore they are therefore squared (c.c.) for physical observables
- This results in



$M \propto$

$M1$



$M2$

$\sigma \propto$

$$|M1|^2 + |M2|^2 + 2 \operatorname{Re}[M1M2]$$

Interference

# Later from Griffiths

- Enlightening

Table 7.1 Catalog of basic quantum electrodynamic processes.

## Second-order processes

### Elastic



{ Electron-muon scattering ( $e + \mu \rightarrow e + \mu$ )  
(Mott scattering ( $M \gg m$ )  $\Rightarrow$  Rutherford scattering ( $v \ll c$ ))



{ Electron-electron scattering ( $e^- + e^- \rightarrow e^- + e^-$ )  
(Møller scattering)



{ Electron-positron scattering ( $e^- + e^+ \rightarrow e^- + e^+$ )  
(Bhabha scattering)



{ Compton scattering ( $\gamma + e^- \rightarrow \gamma + e^-$ )

### Inelastic



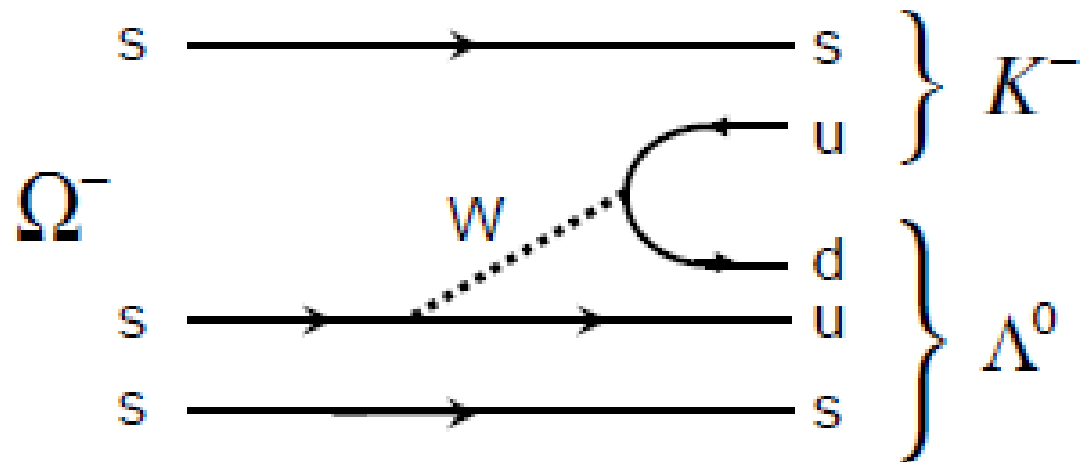
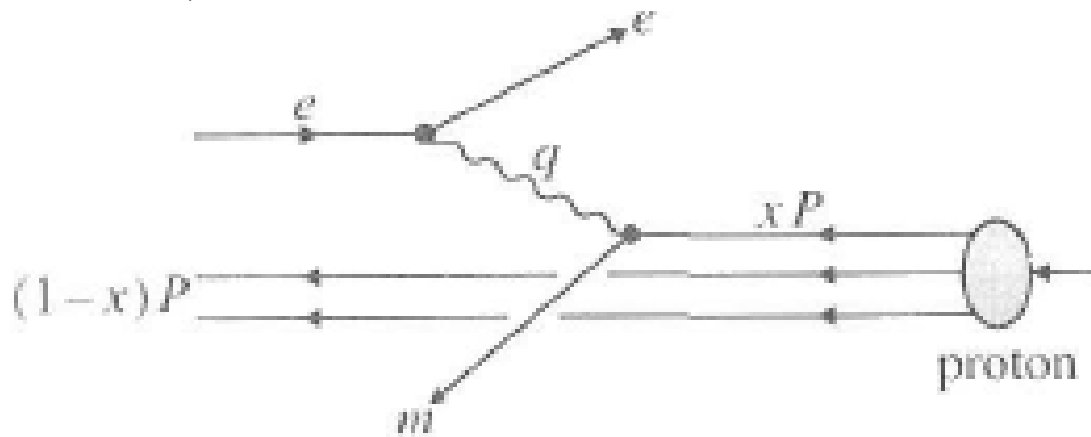
{ Pair annihilation ( $e^- + e^+ \rightarrow \gamma + \gamma$ )



{ Pair production ( $\gamma + \gamma \rightarrow e^- + e^+$ )

# Hadron Feynman's

- Other “spectator quarks”

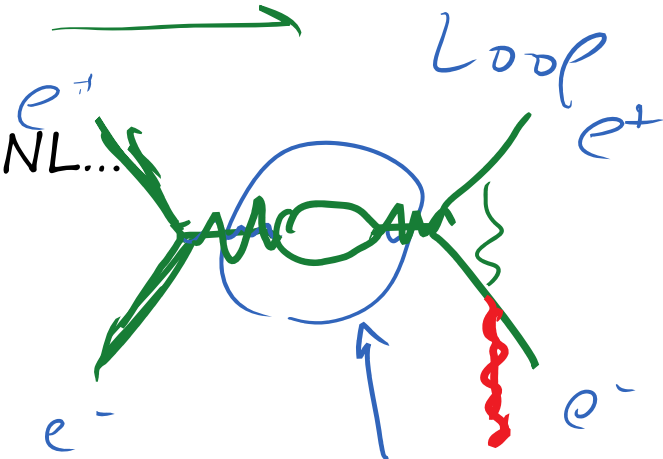
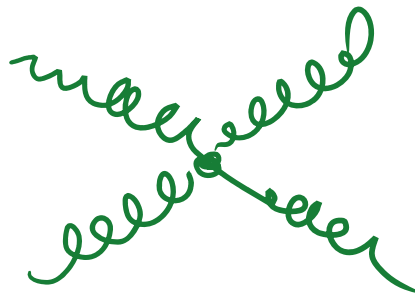
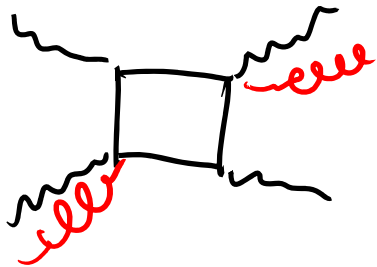


- Also for nuclei “spectator” nucleons

# Others → Beyond LO

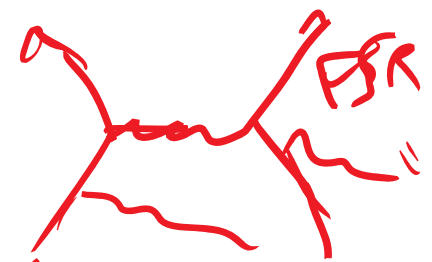
- Bare vertices
- Box diagrams
- Loops, Radiative Corrections NLO, NNL...
  - $\propto \alpha^2, \alpha^3 \dots$

## • Contact Interactions $\ell$



Divergent

Radiative



"initial" state

Radiation  
ISE

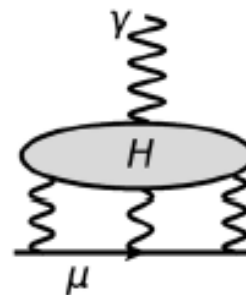
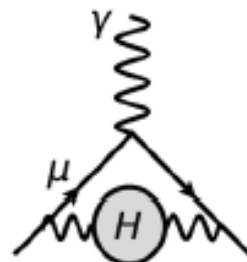
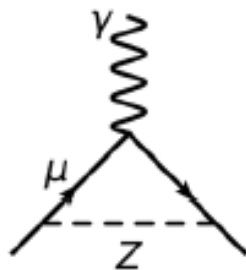
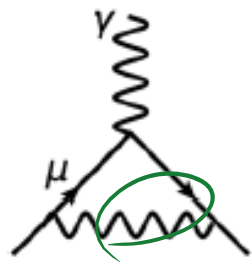
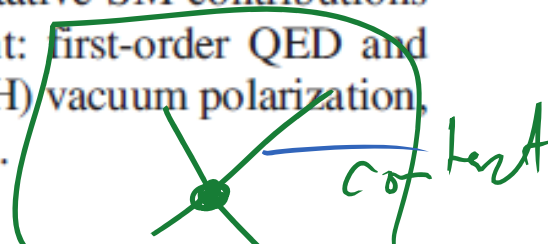


FIG. 1. Feynman diagrams of representative SM contributions to the muon anomaly. From left to right: first-order QED and weak processes, leading-order hadronic (H) vacuum polarization, and hadronic light-by-light contributions.



# QFT's – Renormalization

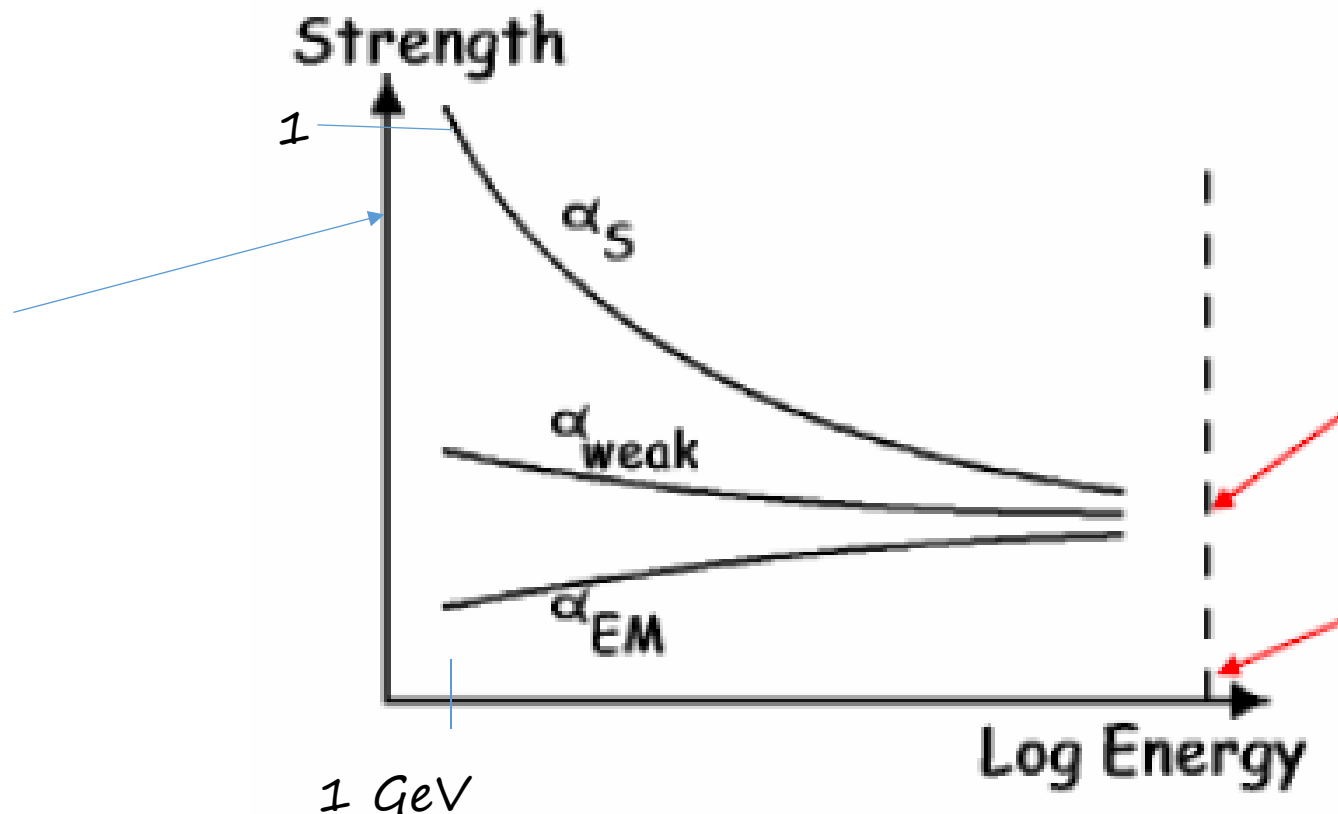
- These methods result in Pert. Theory expansion
  - “Tree level” leading order LO term + “next”/ 2nd order, NLO + NNLO + NNNLO + .....N<sup>th</sup>LO
- Successively more complicated Integrations for higher and higher order terms
  - Order parameter is  $\propto$  naturalized coupling  $g^2$  ( $\alpha = 1/137$  for EM)
- Issue: **ALL BUT LO TERMS HAVE DIVERGENT INTEGRALS!!!!**
- Solution:
  - “Renormalize”
    - a) introduce “arbitrary” cutoffs in integrals
    - b) Dimensional regularization  $\rightarrow$  make the integral in more dimensions, take a limit to 4 D (3+1)
    - c) Derive combinations of the terms that cancel (infinity – infinity)
- Renormilization “Scheme”
  - MS normalization
  - It is imposing this that causes the “running coupling” (charge screened by vacuum, length scale of interaction shorter reveals diff charge)

# Asymptotic Freedom/ Confinement

- Coupling  $\rightarrow 1$  “all” orders are needed -- perturbative theory becomes unusable

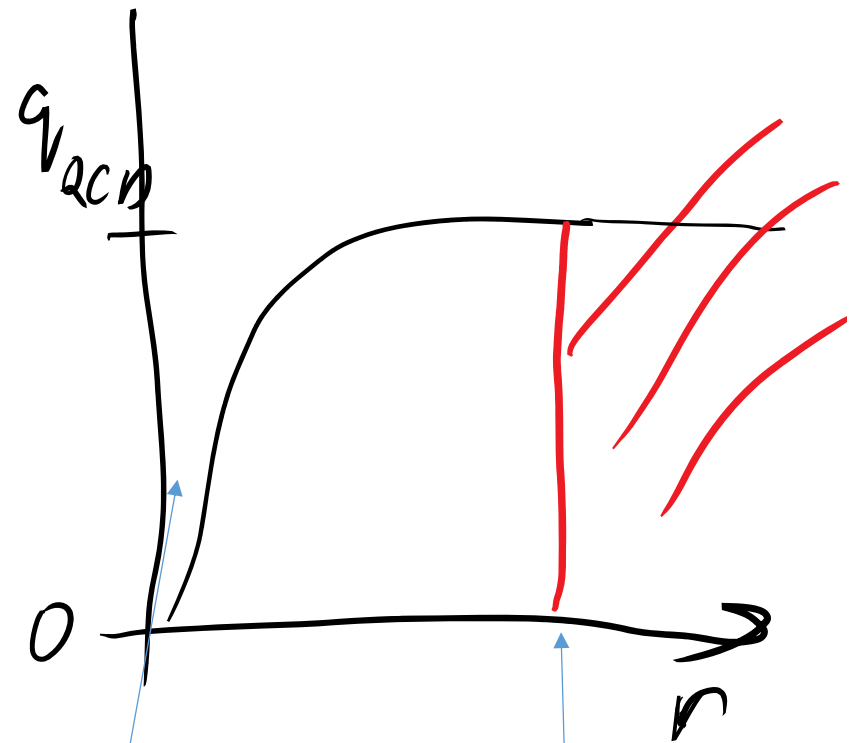
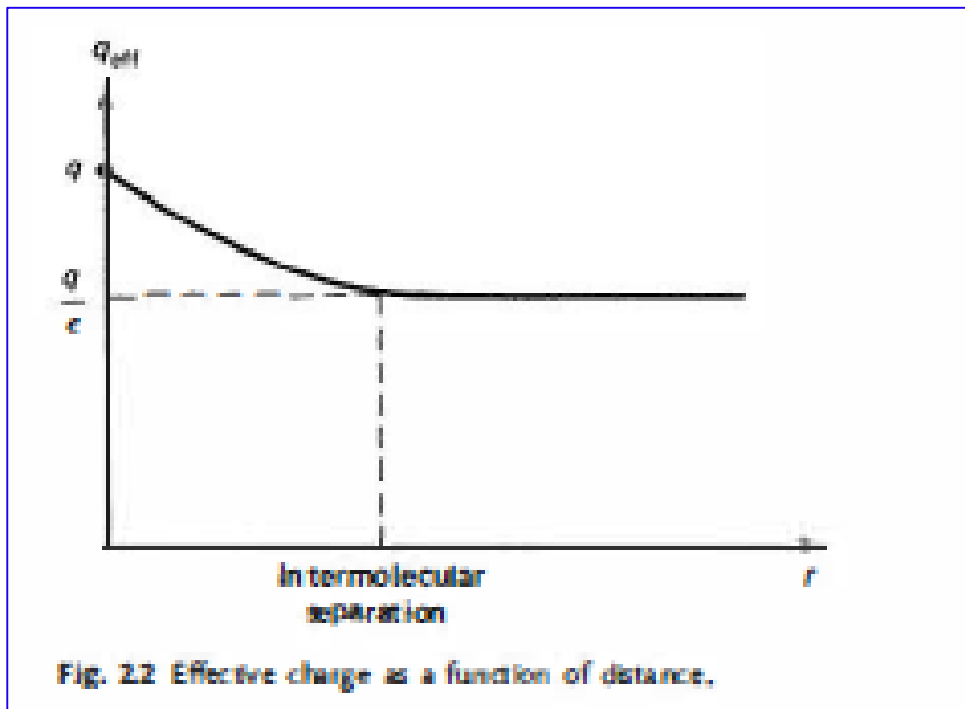
NON-PERTURBATIVE REGIME QCD

- Note that weak coupling also has asymptotic “free-er-dom”



# This plot only applies for EM

- For the others it's opposite! QCD asymptotic freedom  $q \rightarrow 0$



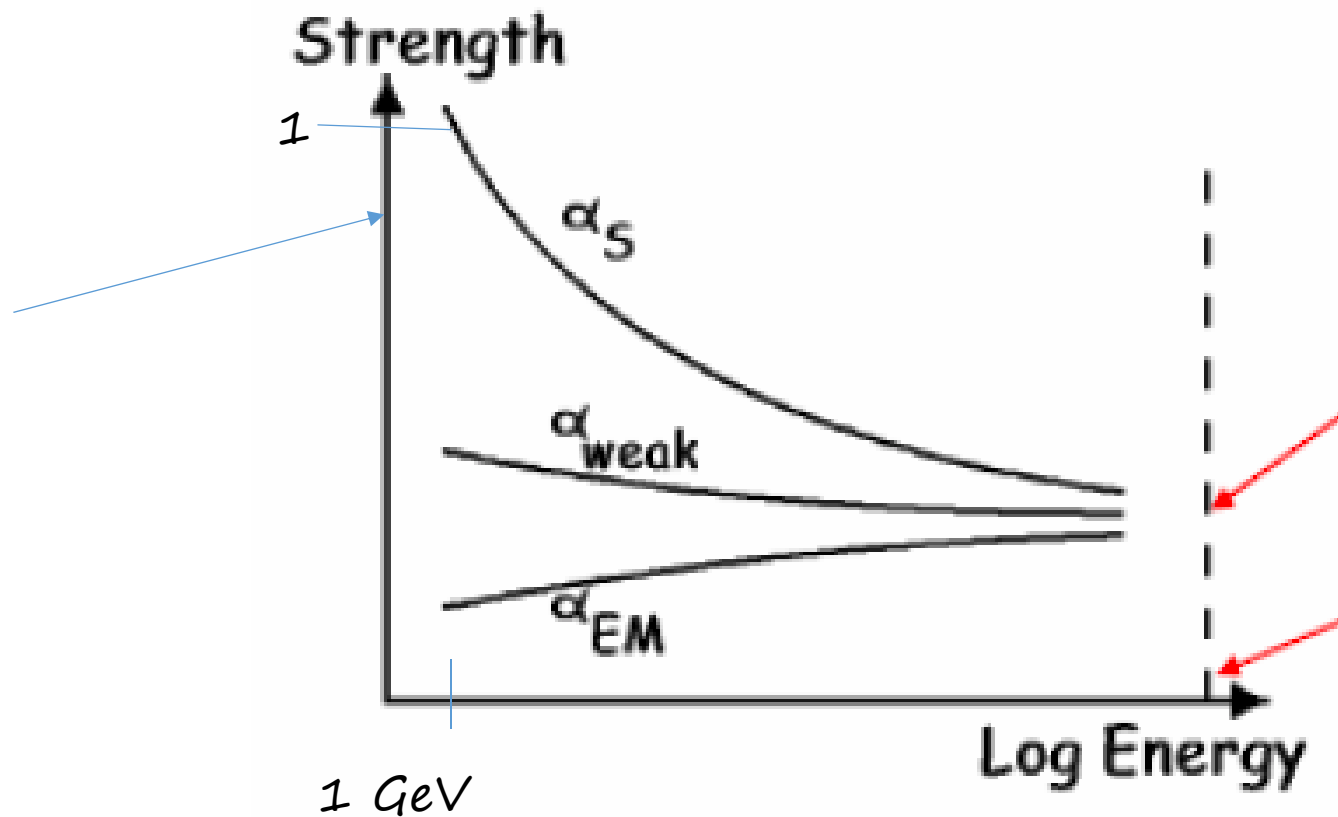
For QCD, the charge goes to 0 for small lengths/high energy, (screening works in weird way)

and technically confinement imposes length over which it doesn't operate (nuclear size  $\sim 10\text{fm}$ ) at all again



# DECAYS - Griff problems

- Coupling  $\rightarrow$  Higher faster more dominant
- IN decays or scattering



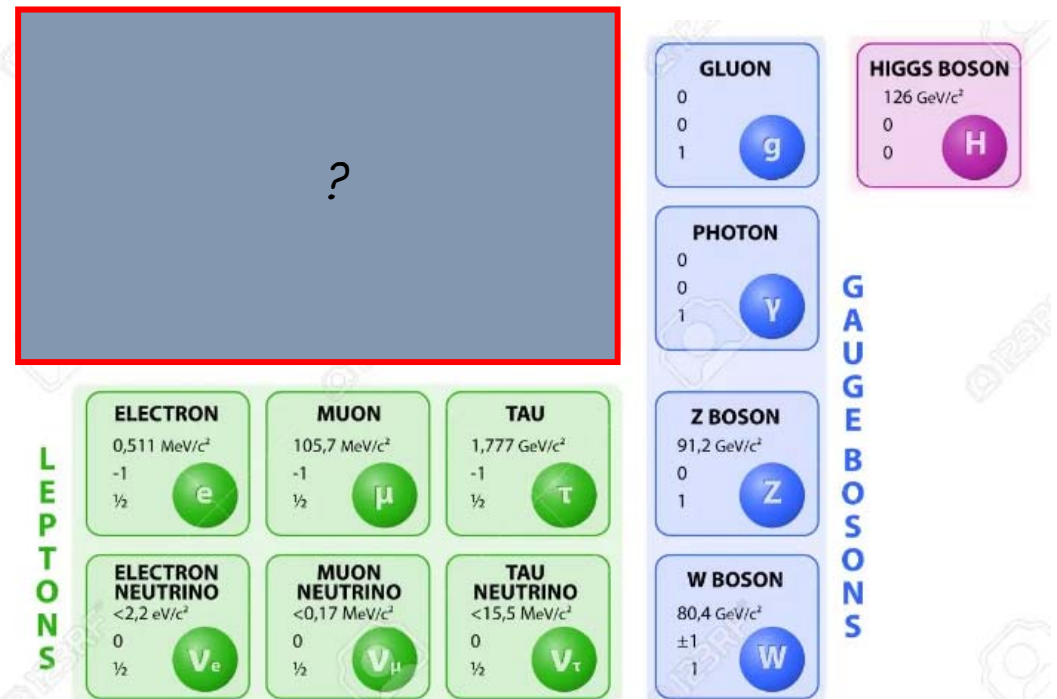
# Non-perturbative

- These methods result in *Pert. Theory* expansion
  - “Tree level” leading order LO term + “next”/ 2nd order, NLO + NNLO + NNNLO + .....N<sup>i</sup>LO
- Successively more complicated Integrations for higher and higher order terms
  - Order parameter is  $\propto$  naturalized coupling  $g^2$  ( $\alpha = 1/137$  for EM)
- Need other theory. “Lattice QCD” Use Quark Gluon DOF but LOTS AND LOTS OF OTHER “tricks”
- Different DOF’s better way to formulate? Another way to view altogether?
  - A type of Coupled Cluster theory (?) [Argonne ME theorist told me this]
- Lattice  $\chi$ EFT

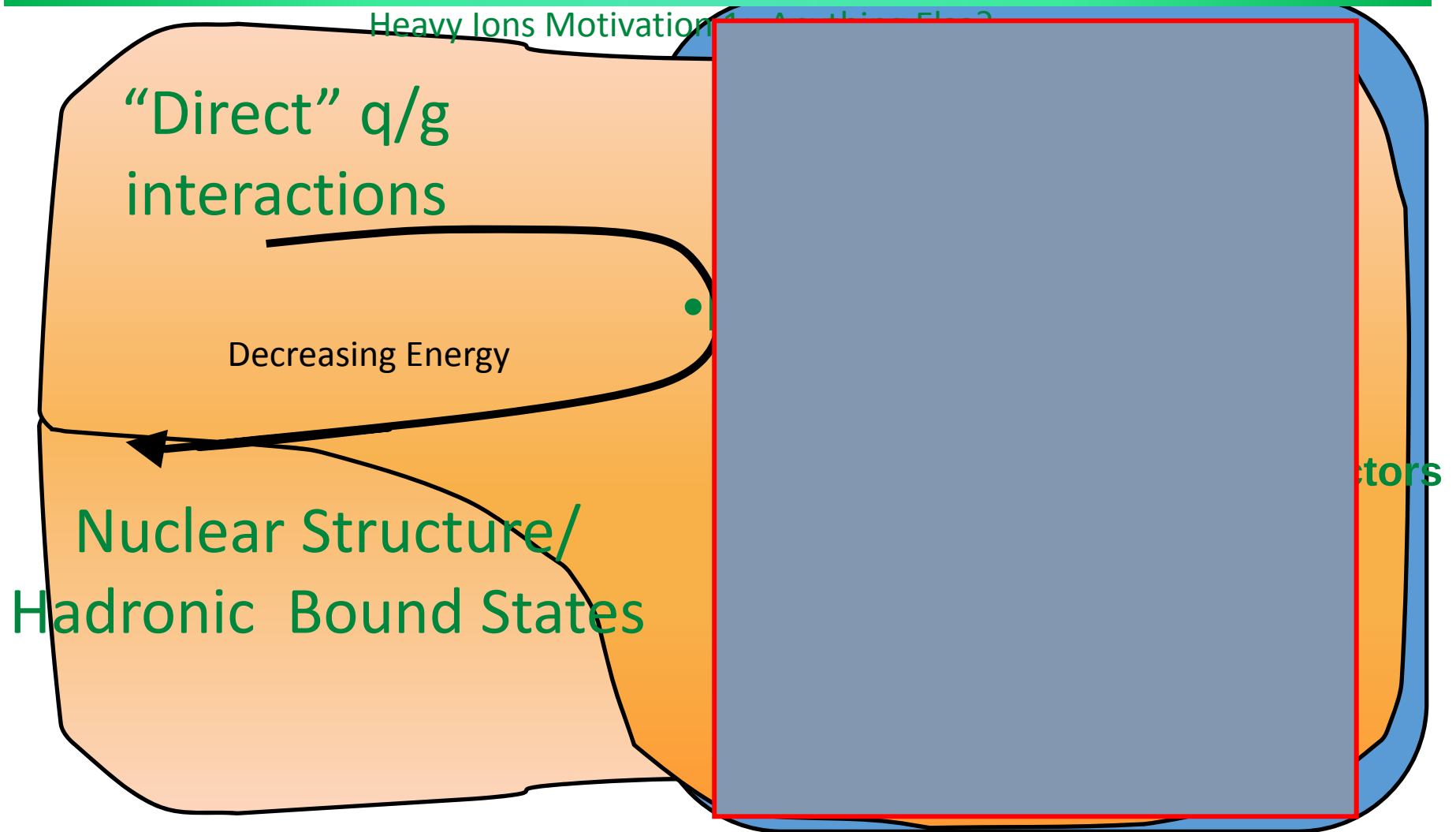
# LOW ENERGY Standard Model - Particles

- Since the SM is a collection of field theories
- And those field theories can't be used for low E QCD then we can't actually be sure there should be some different or additional "particles" at low E
- Lattice QCD is a way to pose the theory - *I don't know that it is yet unambiguously shown that using other DOF could also reproduce...*
  - After ~50 years LQCD dev...
  - Number of lattice sites-
  - Use of unphysical masses
  - Not always using all quark flavors/gluon ?

IF BEEN SHOWN  
ONLY RECENTLY  
(only very recently  
hadron masses  
are being predicted  
very accurately by LQCD

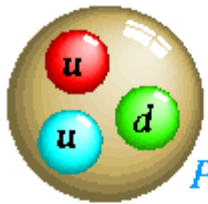


# Map of QCD Phenomena

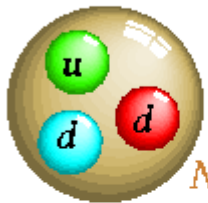


Practical upshot -- are quarks and gluons necessarily a valid or the only valid concept here !?!!?!?! Lattice results would argue yes

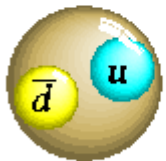
# QCD: Quarks, Gluons $\rightarrow$ Hadron, Nuclei



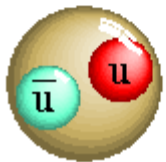
Proton



Neutron



$\pi^+$



$\pi^0$

- QCD: Quantum Chromo-Dynamics theory of the Strong Force!
- Color charged quarks interacting through gluons
- Quarks/Gluons observed two ways:
  - Hadrons (particles made of q/g) : mass hierarchy based on quarks
  - Collisions of hadrons reveals “parton” quark/gluon interactions
- Gluons are colored charged too! Short range effects
- Confinement ! Raw color charge always neutralized by vacuum fluctuations
- Everything from hadron collisions to q/g dynamics to fundamental QCD properties intrinsically related!

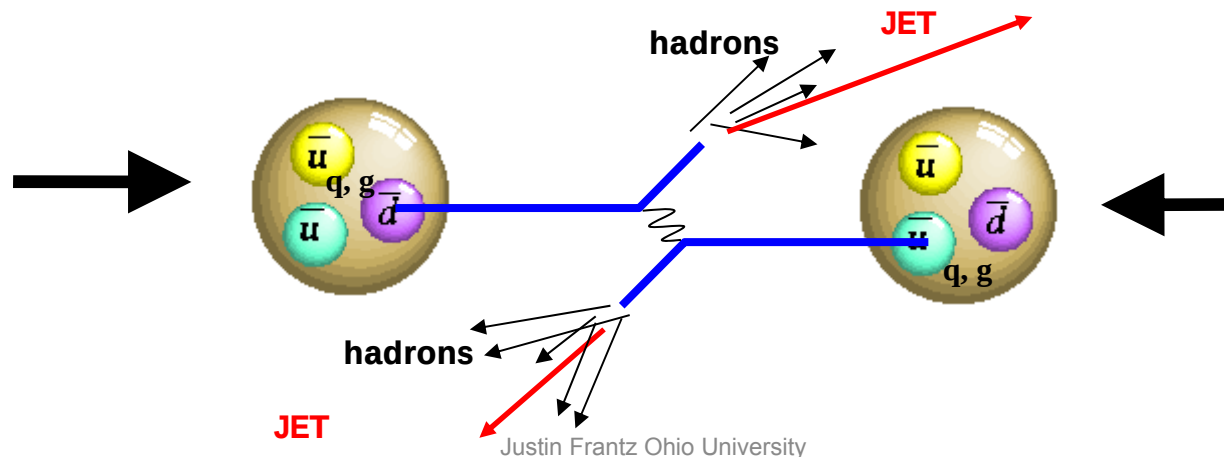
# Zoom-in on Medium Energy

## Heavy Ions Motivation 2: A tool

- **Hadron (Proton) Structure**
  - Steady State
  - Momentum Density
  - Mass Density
  - Spin Density
  - Quark Content
- **Fragmentation/ Hadronization**
  - $q/g$  Parton's "knocked out"/created--"hard" collisions
  - Particle "Jets" produced
  - Transient behavior

*Perturbative vs "Non Perturbative"*

*Medium Energy Nucl/Particle Physics Involves Mixture*



# Zoom-in on Medium Energy

## Heavy Ions Motivation 2: A tool

- **Hadron (Proton) Structure**
  - Steady State
  - Momentum Density
  - Mass Density
  - Spin Density
  - Quark Content
- **Fragmentation/ Hadronization**
  - $q/g$  Parton's "knocked out"/created -- "hard" collisions
  - Particle "Jets" produced
  - Transient behavior

Closely Related Theoretically: Usually phenomena are intertwined

Studied through scattering ( $e+e$ , hadrons)

Jefferson Lab (Hicks, Roche...) -- Spin program at RHIC



# Heavy Ions: Add Heat and Density!

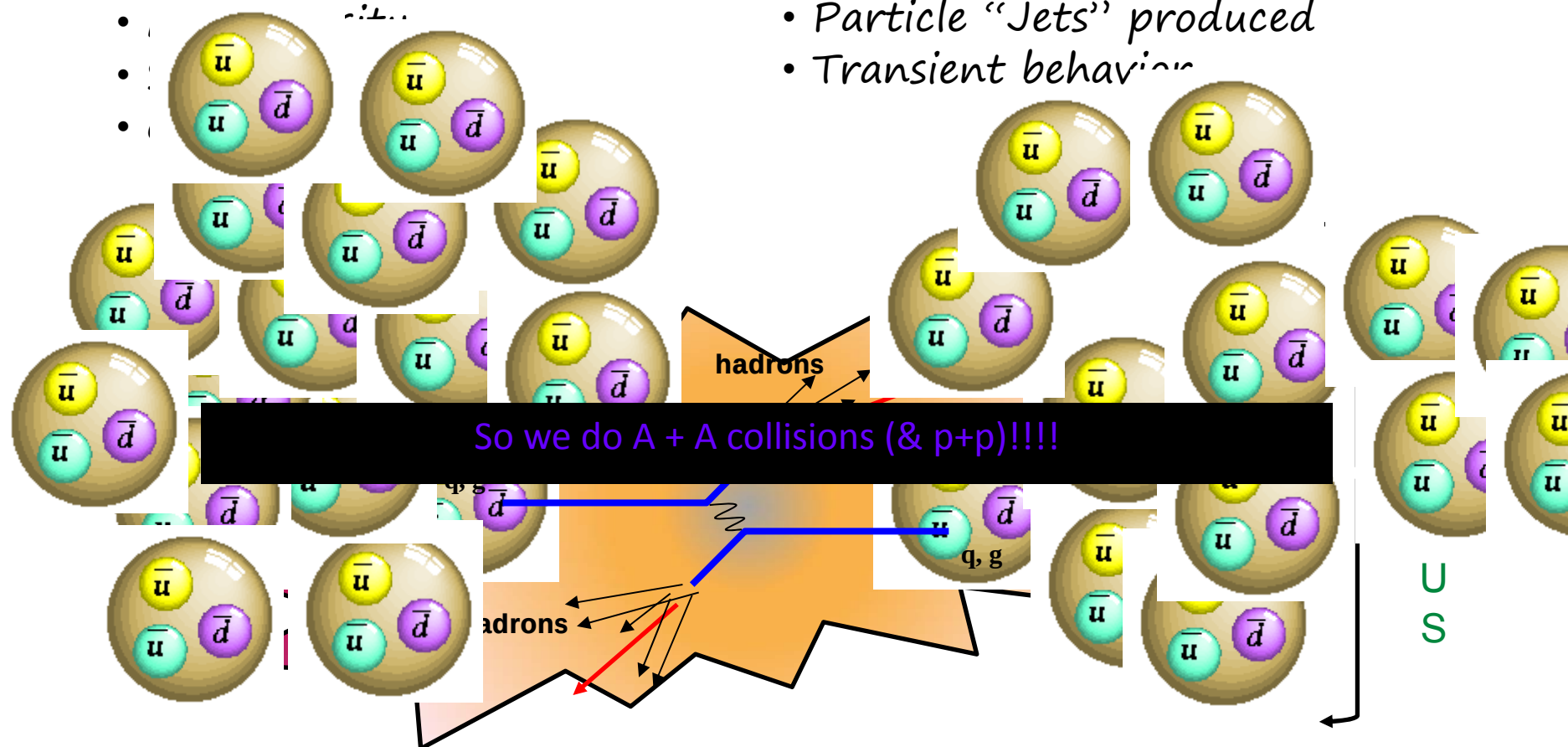
## Heavy Ions Motivation 2: A tool

- Hadron Structure

- Steady State
- Momentum Density

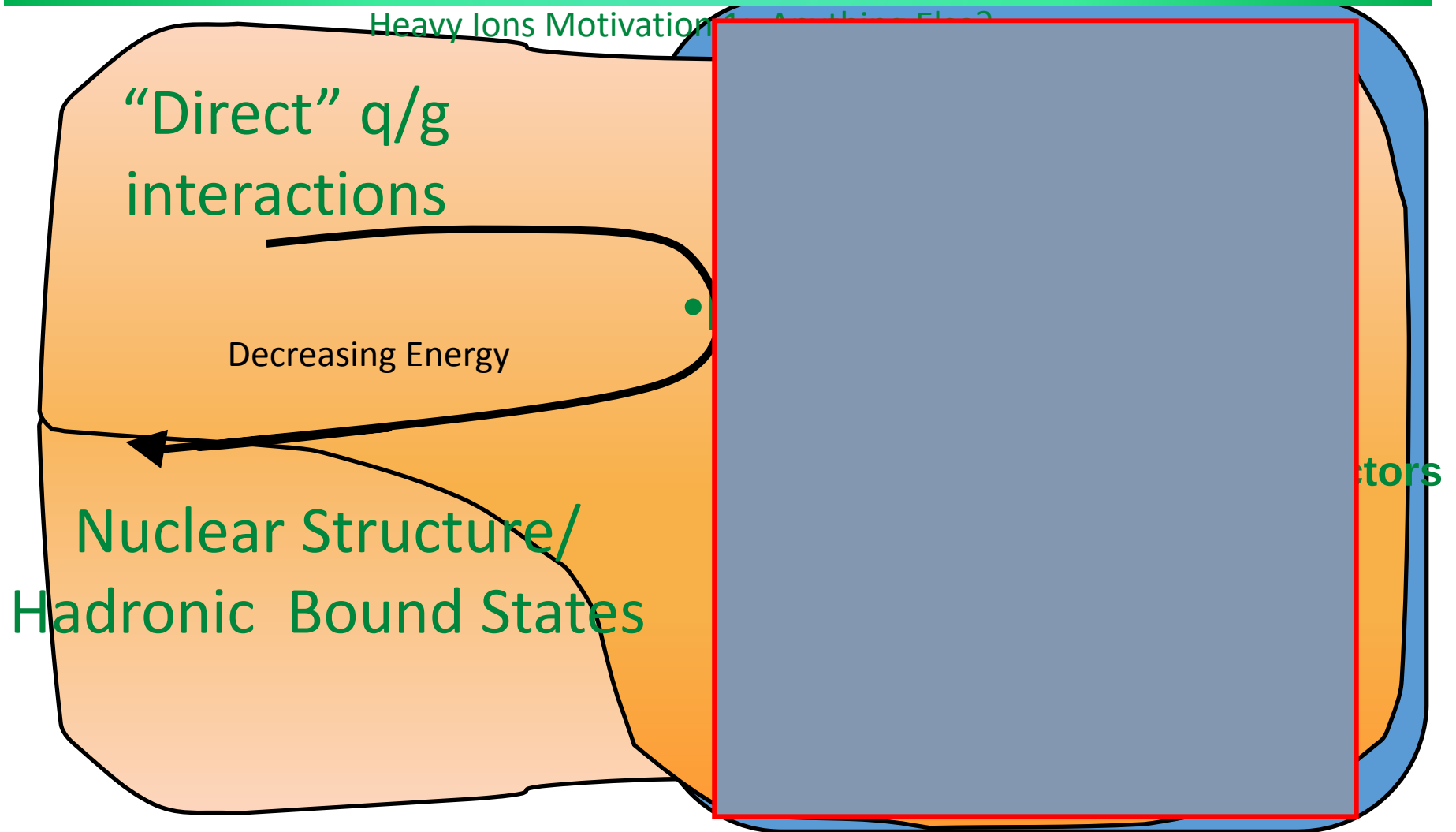
- Fragmentation/ Hadronization

- $q/g$  Parton's "knocked out"/created--"hard" collisions
- Particle "Jets" produced
- Transient behavior





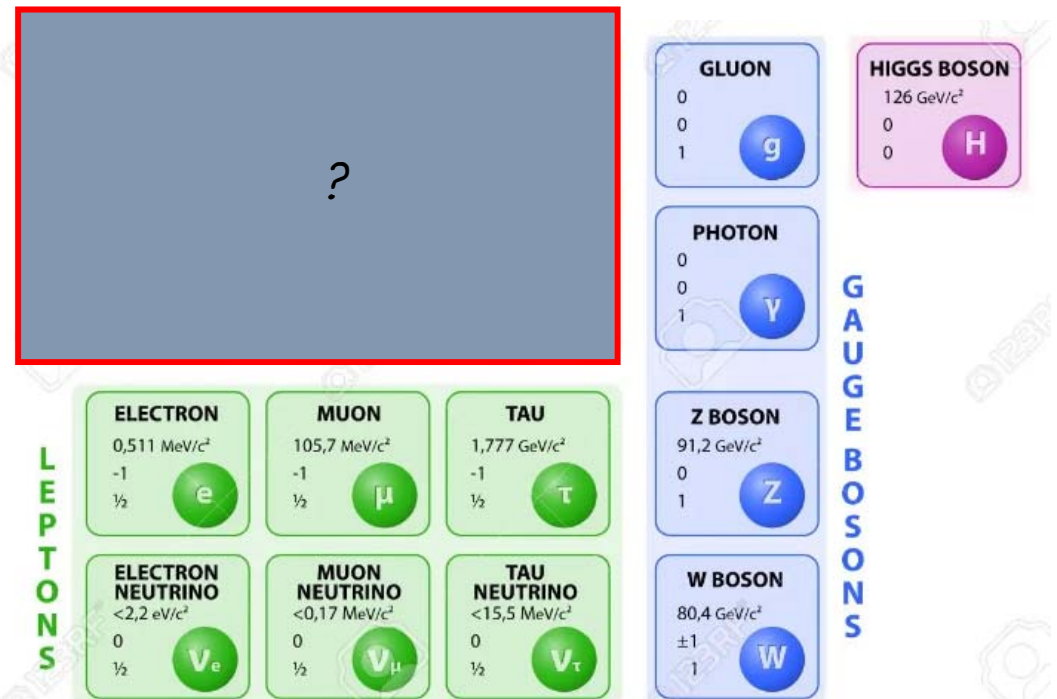
# Map of QCD Phenomena



Practical upshot -- are quarks and gluons necessarily a valid or the only valid concept here !?!!?!?! Lattice results would argue yes

# LOW ENERGY Standard Model - Particles

- Since the SM is a collection of field theories
- And those field theories can't be used for low E QCD then we can't actually be sure there should be some different or additional "particles" at low E
- Lattice QCD is a way to pose the theory



# Truly Elementary?

- However – because of confinement/coupling reaching 1 in QCD at low energy (as  $Q \rightarrow 0$ ) it actually not necessarily proven that quarks and gluons are 100% COMPLETE concepts
  - The “Non-perturbative” QCD regime – the SM is mostly based on using perturbation theory
  - This uncertainty even have ramifications for other parts of the SM since the 3 QFT's are not really fully separable
  - However, Lattice QCD ( $\sim$ fully rigorous NonPT QCD) has had tremendous success  $\rightarrow$  all indications are that the  $q, g$  concepts are good even there
  - They are still not necessarily the best description of matter in that regime!

# More Comments to Griff Ch 1,2,4 and Esp 5

- Quark combining in wave functions/ Ang momentum characterization of hadrons OK – obviously legit why to characterize
- Also predict decay properties
- ISOSPIN -- Quark Flavor up vs down
- BUT YOU CAN'T TAKE THE "WAVEFUNCTIONS" suggested as the ACTUAL WAVEFUNCTIONS
  - They are only approximations – Non-perturbative theory makes sometimes serious corrections to this way of thinking.

$$\psi = \psi(\text{space})\psi(\text{spin})\psi(\text{flavor})\psi(\text{color}) \quad (5.58)$$

$$\pi^0, \rho^0 = (u\bar{u} - d\bar{d})/\sqrt{2}$$

$$\psi(\text{baryon decuplet}) = \psi_s(\text{spin})\psi_f(\text{flavor}) \quad (5.61)$$

Example 5.1 Write down the wave function for the  $\Delta^+$ , in the spin state  $m_j = -\frac{1}{2}$  (neglect the space and color parts).

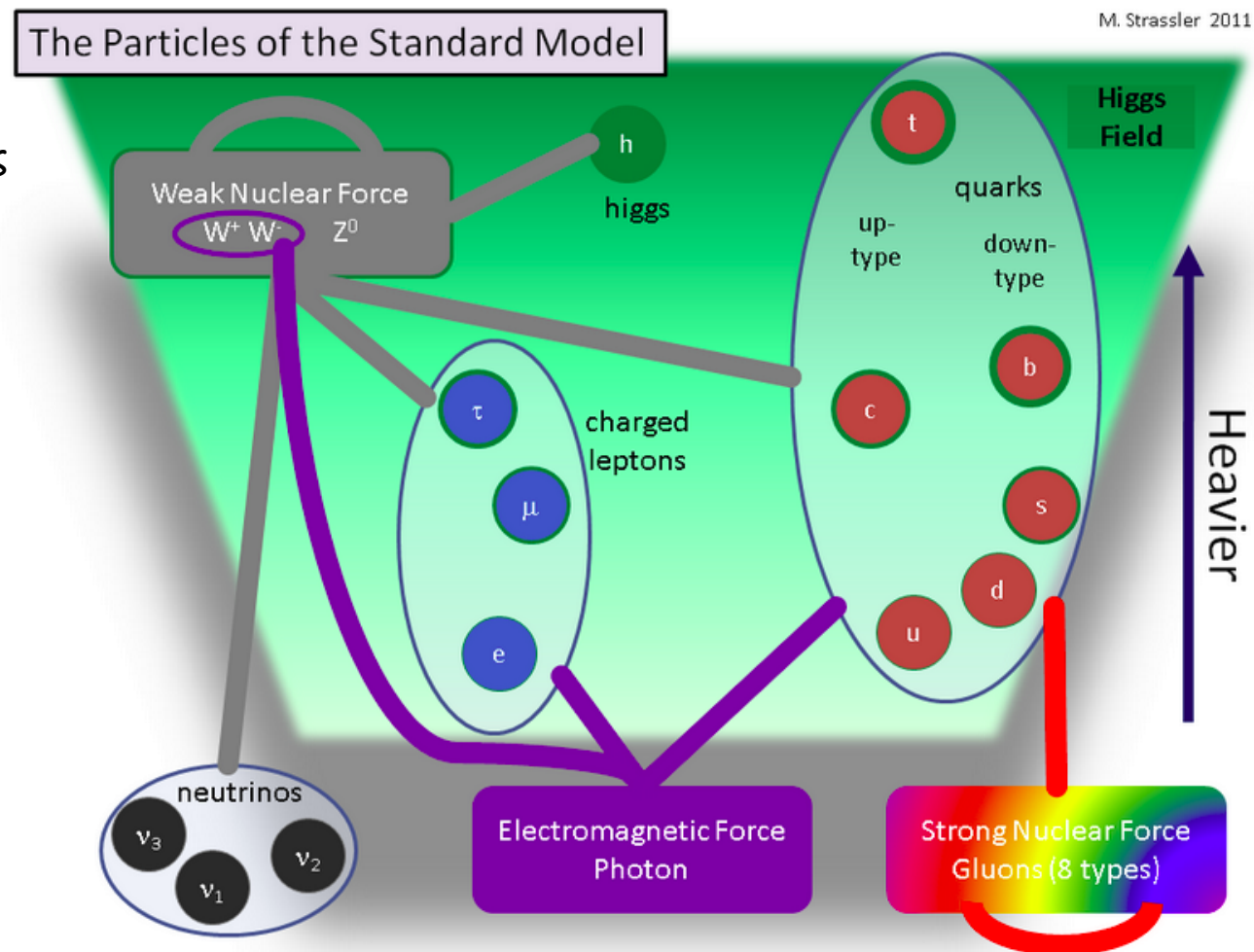
Solution:

$$\begin{aligned} |\Delta^+; \frac{3}{2}, -\frac{1}{2}\rangle &= \left[ (uud + udu + duu)/\sqrt{3} \right] \left[ (|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle + |\uparrow\uparrow\downarrow\rangle)/\sqrt{3} \right] \\ &= \left[ u(\downarrow)u(\downarrow)d(\uparrow) + u(\downarrow)u(\uparrow)d(\downarrow) + u(\uparrow)u(\downarrow)d(\downarrow) \right. \\ &\quad \left. + u(\downarrow)d(\downarrow)u(\uparrow) + u(\downarrow)d(\uparrow)u(\downarrow) + u(\uparrow)d(\downarrow)u(\downarrow) \right] \end{aligned}$$

# By Higg's Interaction ("Mass")

- Higgs Field all massive SM particles "bathed" in it, inertia generated by interactions with Higg's field (usual description) → but from gravity point of view, needs ENERGY →

- so more properly, all SM particles have Higg's charge/ emit a Higg's field



# Standard Model – Higgs Sector QFT

- The Higgs part of the SM is certainly an independent part of the rest of the SM
- It presumably could be another separate quantum field theory
- More likely just tacked on to the other three
  - Like an extra set of terms in all the Lagrangians
- Mostly it has only been applied to the Electroweak sector and “explained” the mass of the Weak bosons
- I’ve never seen a formulation of the Higgs field applied to quarks

|                   | LEPTONS  |     |                | QUARKS |      |                |
|-------------------|----------|-----|----------------|--------|------|----------------|
|                   |          | $q$ | $m/\text{GeV}$ |        | $q$  | $m/\text{GeV}$ |
| First Generation  | $e^-$    | -1  | 0.0005         | $d$    | -1/3 | 0.3            |
|                   | $\nu_1$  | 0   | $\approx 0$    | $u$    | +2/3 | 0.3            |
| Second Generation | $\mu^-$  | -1  | 0.106          | $s$    | -1/3 | 0.5            |
|                   | $\nu_2$  | 0   | $\approx 0$    | $c$    | +2/3 | 1.5            |
| Third Generation  | $\tau^-$ | -1  | 1.77           | $b$    | -1/3 | 4.5            |
|                   | $\nu_3$  | 0   | $\approx 0$    | $t$    | +2/3 | 175            |

Mass from Higgs

7.5 MeV

4.2 MeV

150 MeV

1.3 G

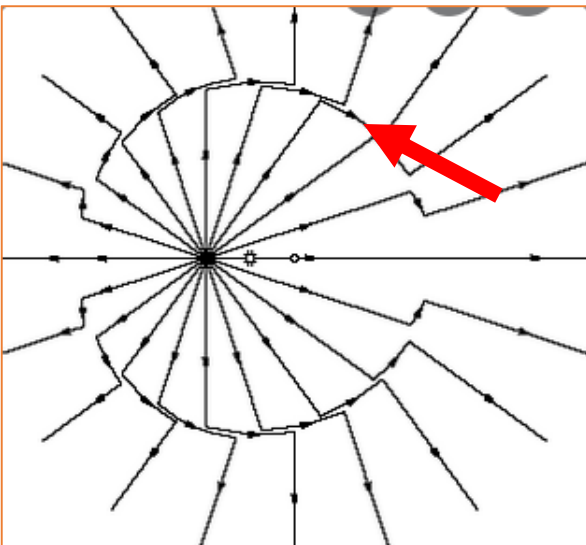
4.2

175

constituent mass

# Masses – Constituent vs Higgs

- First it's useful to remember: charge and associated field automatically implies “its own” inertia which is indistinguishable from mass
- Griffiths mentions this in comparison  $p$  vs  $n$  mass just from electric charge
- Even a charged particle with zero interaction with the Higgs particle (Zero Higgs Mass) would have a mass
- This is the case for the gluon –it is mostly quoted as having zero mass—this is strictly only in the sense of Higgs's mass



|                   | LEPTONS  |     |                | QUARKS |      |                |
|-------------------|----------|-----|----------------|--------|------|----------------|
|                   |          | $q$ | $m/\text{GeV}$ |        | $q$  | $m/\text{GeV}$ |
| First Generation  | $e^-$    | -1  | 0.0005         | $d$    | -1/3 | 0.3            |
|                   | $\nu_1$  | 0   | $\approx 0$    | $u$    | +2/3 | 0.3            |
| Second Generation | $\mu^-$  | -1  | 0.106          | $s$    | -1/3 | 0.5            |
|                   | $\nu_2$  | 0   | $\approx 0$    | $c$    | +2/3 | 1.5            |
| Third Generation  | $\tau^-$ | -1  | 1.77           | $b$    | -1/3 | 4.5            |
|                   | $\nu_3$  | 0   | $\approx 0$    | $t$    | +2/3 | 175            |

Mass from Higgs

7.5 MeV

4.2 MeV

150 MeV

1.3 G

4.2

175

constituent mass



# What? Gluons have mass?

- The non-Higgs mass I'm talking about here could be called "constituent mass" of the gluon
  - Example paper: "Are Gluons Massive?"  
<https://www.worldscientific.com/doi/abs/10.1142/S0217751X94001291?journalCode=ijmpa>
  - Abstract: The mass of the gluon is determined using asymptotic perturbative QCD in the renormalization scheme. The experimental determination of the couplant  $\bar{\alpha}_s^{(2)}$ , which occurs in the perturbation expansions for and together with an estimate of the on-shell couplant,  $\alpha_s = 0.096 \pm 0.016$  [or  $\alpha_s(0) = 0.30 \pm 0.05$ ], yields the results if typical current (constituent) masses are assigned to the quarks."
- Argument of charge-generating field automatically implying inertia is pretty rock solid - of course there is confinement.
- GLUEBALLS - definitely have valence gluons with constituent mass:
  - <https://physics.stackexchange.com/questions/267223/why-do-glueballs-have-mass-when-individual-gluons-are-massless>
  - That exchange people are generally avoiding our question
  - Glueballs haven't been unambiguously discovered but there are good signals that candidates - but could also be some kinds of regular hadrons  $f_0$ 's resonances esp  $f_0(1710)$
  - Gluex at Jlab designed to look for glueballs



## One last point about these mass questions – photon gravitational mass?

- If gluons have mass what about photons?
- YES photons also “only” have zero mass in the Higgs mass sense
  - Even without confinement effects to worry about
- See this stackexchange post: Do photons bend spacetime ?  
<https://physics.stackexchange.com/questions/481557/do-photons-bend-spacetime-or-not>
  - The winning argument from this and several posts is affirmative.
  - Whether you want to refer to this as mass may be a question of how you define “mass” vs energy
  - Post states that technically it’s a question that can only be answered 100% for sure in a true Quantum Gravity sense.
  - But most theorists believe they should—certainly the classical photon fields since they carry momentum and energy, should/do.
  - They are expected even to cause gravitational attraction to other masses (at least most theorists believe they can). [kugelblitz](#) Blackhole like object from only trapped photon radiation.
  - And there are mechanisms for them to interact with a graviton.

# Mass → Other Properties - Group Theory

- So we've discussed mass and its peculiarities
- At least when particles (even composite) are "isolated", they exist in definite states of "mass"
- Mass is the most fundamental determiner of a "state" for our particles
- Most of the other properties on the table are not so simple as they appear
- This is because most of the other properties seen there, even "up" vs down flavor, and most of the charges (except electric) are actually more complex than just a scalar property like electric charge
- Rather they more like elements (basis vectors) of a vector space and can be combined in that way. in the same way q.m. spin can be combined.

## Spin $\frac{1}{2}$

LEPTONS

|                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UP</b><br>mass 2,3 MeV/c <sup>2</sup><br>charge $\frac{2}{3}$<br>spin $\frac{1}{2}$<br> | <b>CHARM</b><br>1,275 GeV/c <sup>2</sup><br>$\frac{2}{3}$<br>$\frac{1}{2}$<br>  | <b>TOP</b><br>173,07 GeV/c <sup>2</sup><br>$\frac{2}{3}$<br>$\frac{1}{2}$<br>    |
| <b>DOWN</b><br>4,8 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br>              | <b>STRANGE</b><br>95 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br> | <b>BOTTOM</b><br>4,18 GeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br> |
| <b>ELECTRON</b><br>0,511 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br>                    | <b>MUON</b><br>105,7 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br>             | <b>TAU</b><br>1,777 GeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br>               |
| <b>ELECTRON NEUTRINO</b><br><2,2 eV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br>              | <b>MUON NEUTRINO</b><br><0,17 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br>     | <b>TAU NEUTRINO</b><br><15,5 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br>       |

## Integer Spin

GAUGE BOSONS

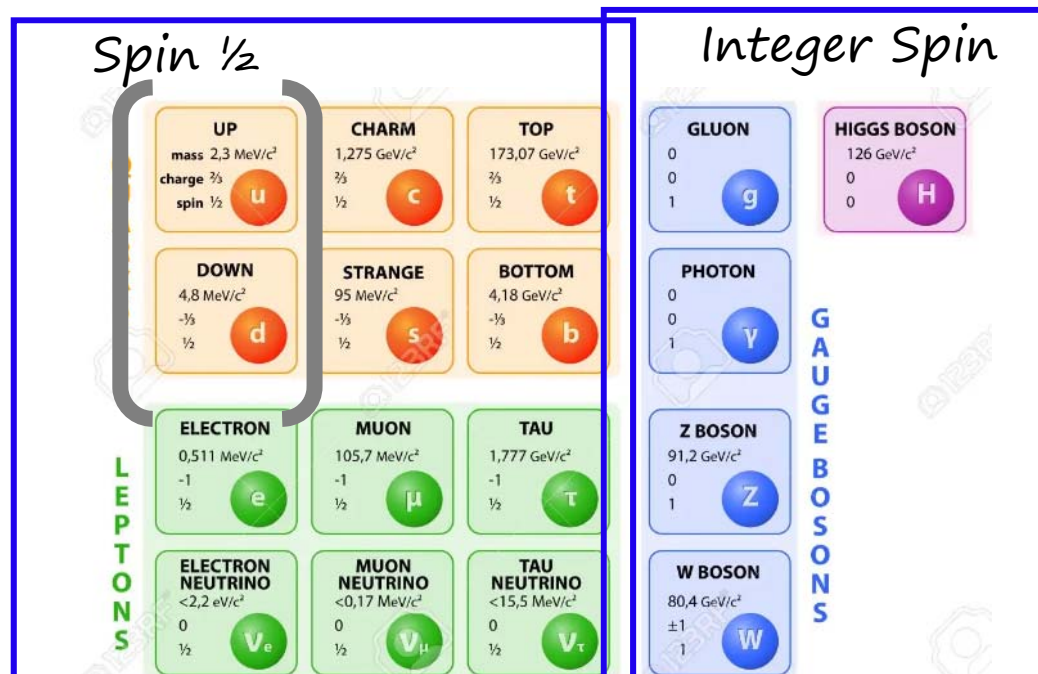
|                                                                                                                                                    |                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GLUON</b><br>0<br>0<br>1<br>                                | <b>HIGGS BOSON</b><br>126 GeV/c <sup>2</sup><br>0<br>0<br> |
| <b>PHOTON</b><br>0<br>0<br>1<br>                              |                                                                                                                                                |
| <b>Z BOSON</b><br>91,2 GeV/c <sup>2</sup><br>0<br>1<br>       |                                                                                                                                                |
| <b>W BOSON</b><br>80,4 GeV/c <sup>2</sup><br>$\pm 1$<br>1<br> |                                                                                                                                                |

# Mass → Other Properties Baryon Number Lepton Number

- Conservation of Baryon Number Lepton Number Angular momentum Charge, determines what decays are allowed

- Griffiths problems

-Eliminate half/most with easy things like that  
-If taking lots of time deep Feynman diagram ~half those is OK



# Mass → Other Properties – Group Theory

- The idea is captured by considering this:
- Electric charge has two types + / -
- How to represent this mathematically?
- Just the sign of a single number?
- Or how about making spin up + and spin down - ?
- The latter is how ~all other charges (except electric) and properties of the particles are described by the SM

| Spin $\frac{1}{2}$ |                                                                                                                                                                              |                                                                                                                                                                   | Integer Spin                                                                                                                                                       |                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| LEPTONS            | <b>UP</b><br>mass 2,3 MeV/c <sup>2</sup><br>charge $\frac{2}{3}$<br>spin $\frac{1}{2}$<br> | <b>CHARM</b><br>1,275 GeV/c <sup>2</sup><br>$\frac{2}{3}$<br>$\frac{1}{2}$<br>  | <b>TOP</b><br>173,07 GeV/c <sup>2</sup><br>$\frac{2}{3}$<br>$\frac{1}{2}$<br>    | <b>HIGGS BOSON</b><br>126 GeV/c <sup>2</sup><br>0<br>0<br>0<br> |
|                    | <b>DOWN</b><br>4,8 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br>              | <b>STRANGE</b><br>95 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br> | <b>BOTTOM</b><br>4,18 GeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br> |                                                                                                                                                    |
|                    | <b>ELECTRON</b><br>0,511 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br>                    | <b>MUON</b><br>105,7 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br>             | <b>TAU</b><br>1,777 GeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br>               | GAUGE BOSONS                                                                                                                                       |
|                    | <b>ELECTRON NEUTRINO</b><br><2,2 eV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br>              | <b>MUON NEUTRINO</b><br><0,17 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br>     | <b>TAU NEUTRINO</b><br><15,5 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br>       |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   |                                                                                                                                                                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   | <b>GLUON</b><br>0<br>0<br>0<br>1<br>                                             |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   | <b>PHOTON</b><br>0<br>0<br>0<br>1<br>                                           |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   | <b>Z BOSON</b><br>91,2 GeV/c <sup>2</sup><br>0<br>0<br>1<br>                    |                                                                                                                                                    |
|                    |                                                                                                                                                                              |                                                                                                                                                                   | <b>W BOSON</b><br>80,4 GeV/c <sup>2</sup><br>$\pm 1$<br>1<br>                   |                                                                                                                                                    |

So that is partially I'm sure why the column vector notation  $u/d$  ness are a property that behaves like spin up (up) and spin down (down).

Implication best captured by addition each quark in a "two quark state" can have "half up and half downness" not just one or the other

$$1/\sqrt{2} \left( \begin{pmatrix} u \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ d \end{pmatrix} \right)$$

# Charges & Chirality

- (QED) Electric charge 2 charges  $+$   $-$   $U(1)$  algebra
  - magnitude units  $|e| \rightarrow g_{em} \rightarrow \alpha$
- (QCD) Color charge 6 charges  $r, g, b, \bar{r}, \bar{g}, \bar{b}$   $SU(3)$ 
  - - neutralize two ways color-anticolor = "white" all three colors/anti's  $rgb = \text{white}$
  - "magnitude"  $g_s \rightarrow \alpha_s$
  - In QFT Represented by  $SU(3)$  matrices that satisfy this algebra (See below)
  - "Longer range": isospin "u vs d"-ness "p vs n"-ness "approximately also conserved"  $\rightarrow$  Chirality/Helicity ChPT  $\chi$ EFT
- Best to see by what is the "color" of the gluons:

 <https://slideplayer.com/slide/8741005/>

6 of the gluon fields are independent linear combinations of the simple gluon fields we enumerated

pg 285  
of  
Griffiths

$$\begin{aligned} G_\mu^1 &= (r\bar{g} + g\bar{r})/\sqrt{2} & G_\mu^4 &= (b\bar{g} + g\bar{b})/\sqrt{2} & G_\mu^6 &= (r\bar{b} + b\bar{r})/\sqrt{2} \\ G_\mu^2 &= i(r\bar{g} - g\bar{r})/\sqrt{2} & G_\mu^5 &= i(b\bar{g} - g\bar{b})/\sqrt{2} & G_\mu^7 &= i(r\bar{b} - b\bar{r})/\sqrt{2} \end{aligned}$$

# Charges & Chirality

- (QFD) Charge (?) "Hyper charge" "Weak charge" ?
  - It's complicated - Weak interactions always involve "mixing"
  - coupling parameter  $g$  separated into 2 parts:  $g_{\text{Axial}}$   $g_{\text{Vector}}$
  - Charge structure?
    - Spin-like SU(2)
    - Weak Isospin
    - Mixture of this and

### Left-handed fermions in the Standard Model<sup>[1]</sup>

| Generation 1                                                                                                                                           |         |                | Generation 2  |           |                | Generation 3 |            |                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|---------------|-----------|----------------|--------------|------------|----------------|
| Fermion                                                                                                                                                | Symbol  | Weak isospin   | Fermion       | Symbol    | Weak isospin   | Fermion      | Symbol     | Weak isospin   |
| Electron neutrino                                                                                                                                      | $\nu_e$ | $+\frac{1}{2}$ | Muon neutrino | $\nu_\mu$ | $+\frac{1}{2}$ | Tau neutrino | $\nu_\tau$ | $+\frac{1}{2}$ |
| Electron                                                                                                                                               | $e^-$   | $-\frac{1}{2}$ | Muon          | $\mu^-$   | $-\frac{1}{2}$ | Tau          | $\tau^-$   | $-\frac{1}{2}$ |
| Up quark                                                                                                                                               | $u$     | $+\frac{1}{2}$ | Charm quark   | $c$       | $+\frac{1}{2}$ | Top quark    | $t$        | $+\frac{1}{2}$ |
| Down quark                                                                                                                                             | $d$     | $-\frac{1}{2}$ | Strange quark | $s$       | $-\frac{1}{2}$ | Bottom quark | $b$        | $-\frac{1}{2}$ |
| All of the above left-handed ( <i>regular</i> ) particles have corresponding right-handed <i>anti</i> -particles with equal and opposite weak isospin. |         |                |               |           |                |              |            |                |
| All right-handed ( <i>regular</i> ) particles and left-handed anti-particles have weak isospin of 0.                                                   |         |                |               |           |                |              |            |                |

# Break Exercise

- Find links on the web of highest number of  $N$ 's in NN..NLO
  - or non-qcd?
  - anything that looks interesting
- Put into tophat: ( title first author )



# Particles and Nuclei 2

- Third Day 1/22/26
- Announcements:
- Canvas
- HW 1: Due ~Tuesday in Class
- **READING Chapter 3! by Tuesday** → Lancaster & Blundell (L&B) Ch 3, 4, [handout pdf, slightly less than that]:

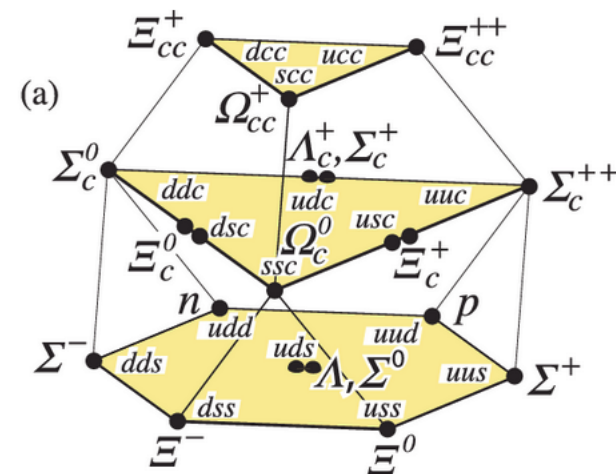
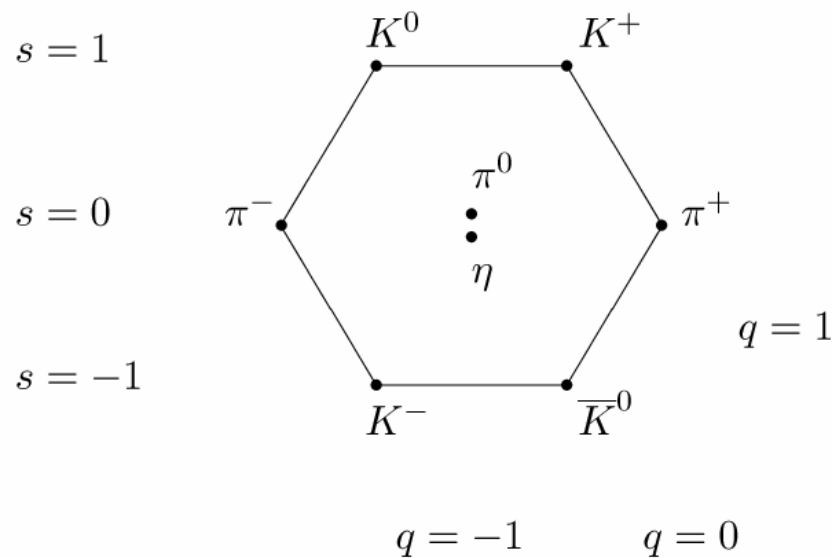


# Roadmap

- BASICS (Ch1,2,4) [Today Finish → Some Leftover topics]
  - Particles (Properties : mass, ctau, charges1, ..., spin,...)
  - Interactions
  - QCD Complications finish some more
  - Particles2 (Properties : mass, ctau, charges1, ..., spin,...)
    - Some Weak Phenomenology
    - Experimental Point of View
  - Some Experimental
- Relativistic Kinematics (Ch3 + Rapidity + )

# Quark Model and Gell-Mann Multiplets

- Hadron "Periodic Table" : 8-fold Way  $\rightarrow$  Multiplets  
Supermultiplets
- Not used in practice much, except e.g. when discovering corners that hadn't been confirmed
- Iso spin here



# Heavy and Light Flavors - QUARKS

- Commonly used

|                   | LEPTONS  |     |                | QUARKS |      |                |
|-------------------|----------|-----|----------------|--------|------|----------------|
|                   |          | $q$ | $m/\text{GeV}$ |        | $q$  | $m/\text{GeV}$ |
| First Generation  | $e^-$    | -1  | 0.0005         | $d$    | -1/3 | 0.3            |
|                   | $\nu_1$  | 0   | $\approx 0$    | $u$    | +2/3 | 0.3            |
| Second Generation | $\mu^-$  | -1  | 0.106          | $s$    | -1/3 | 0.5            |
|                   | $\nu_2$  | 0   | $\approx 0$    | $c$    | +2/3 | 1.5            |
| Third Generation  | $\tau^-$ | -1  | 1.77           | $b$    | -1/3 | 4.5            |
|                   | $\nu_3$  | 0   | $\approx 0$    | $t$    | +2/3 | 175            |

Light quarks  
(even  $s$  usually,  
sometimes not)

Heavy Flavor  
Quarks,  $c, b$

Top Quark: Discovered 1995 (last SM discovery before Higgs)

-VERY HEAVY  $\rightarrow$  Why? (Higgs)

- ( $Z, W \sim 90 \text{ GeV}$ )

- These are the Q's driving LHC "3" and future things discussed before from p5 like Higgs' factory

Top observed recently in LHC Pb+Pb, good tool there

top not considered HF because lifetime too small to form bound states

# Heavy Baryons

- Surprisingly many of the heavy baryons in these supermultiplets still are to be discovered

| Omega                        |                              |                 |                                   |                   |         |     |     |      |                                   |                                                                                                                                          |
|------------------------------|------------------------------|-----------------|-----------------------------------|-------------------|---------|-----|-----|------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Particle                     | Symbol ⇄                     | Quark content ⇄ | Rest mass (MeV/c <sup>2</sup> ) ⇄ | J <sup>P</sup> ⇄  | Q (e) ⇄ | S ⇄ | C ⇄ | B' ⇄ | Mean lifetime (s) ⇄               | Decays to                                                                                                                                |
| Omega <sup>[6]</sup>         | Ω <sup>−</sup>               | sss             | 1 672.45 ± 0.29                   | $\frac{3}{2}^{+}$ | −1      | −3  | 0   | 0    | (8.21 ± 0.11) × 10 <sup>−11</sup> | Λ <sup>0</sup> + K <sup>−</sup><br>or<br>Ξ <sup>0</sup> + π <sup>−</sup><br>or<br>Ξ <sup>−</sup> + π <sup>0</sup>                        |
| Charmed omega <sup>[7]</sup> | Ω <sub>c</sub> <sup>0</sup>  | ssc             | 2 697.5 ± 2.6                     | $\frac{1}{2}^{+}$ | 0       | −2  | +1  | 0    | (6.9 ± 1.2) × 10 <sup>−14</sup>   | See<br>Ω <sub>c</sub> <sup>0</sup><br>Decay<br>Modes  |
| Bottom omega <sup>[8]</sup>  | Ω <sub>b</sub> <sup>−</sup>  | ssb             | 6 054.4 ± 6.8                     | $\frac{1}{2}^{+}$ | −1      | −2  | 0   | −1   | (1.13 ± 0.53) × 10 <sup>−12</sup> | Ω <sup>−</sup> + J/ψ<br>(seen)                                                                                                           |
| Double charmed omega†        | Ω <sub>cc</sub> <sup>+</sup> | scc             |                                   | $\frac{1}{2}^{+}$ | +1      | −1  | +2  | 0    |                                   |                                                                                                                                          |
| Charmed bottom omega†        | Ω <sub>cb</sub> <sup>0</sup> | scb             |                                   | $\frac{1}{2}^{+}$ | 0       | −1  | +1  | −1   |                                   |                                                                                                                                          |
| Double bottom omega†         | Ω <sub>bb</sub> <sup>−</sup> | sbb             |                                   | $\frac{1}{2}^{+}$ | −1      | −1  | 0   | −2   |                                   |                                                                                                                                          |
| Triple charmed               | Ω <sup>++</sup>              | ccc             |                                   | $\frac{3}{2}^{+}$ | +2      | 0   | +3  | 0    |                                   |                                                                                                                                          |

There was a big amount of work on this at a small e+e- facility at Cornell University

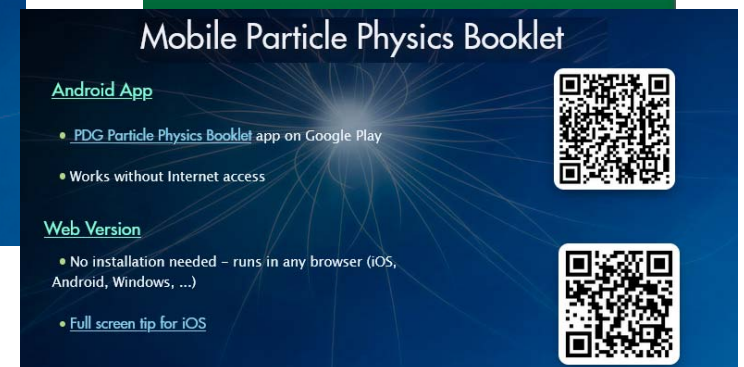
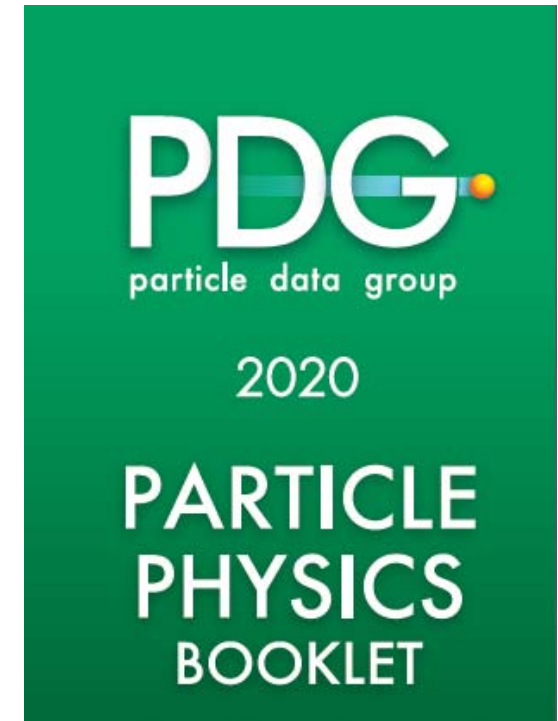
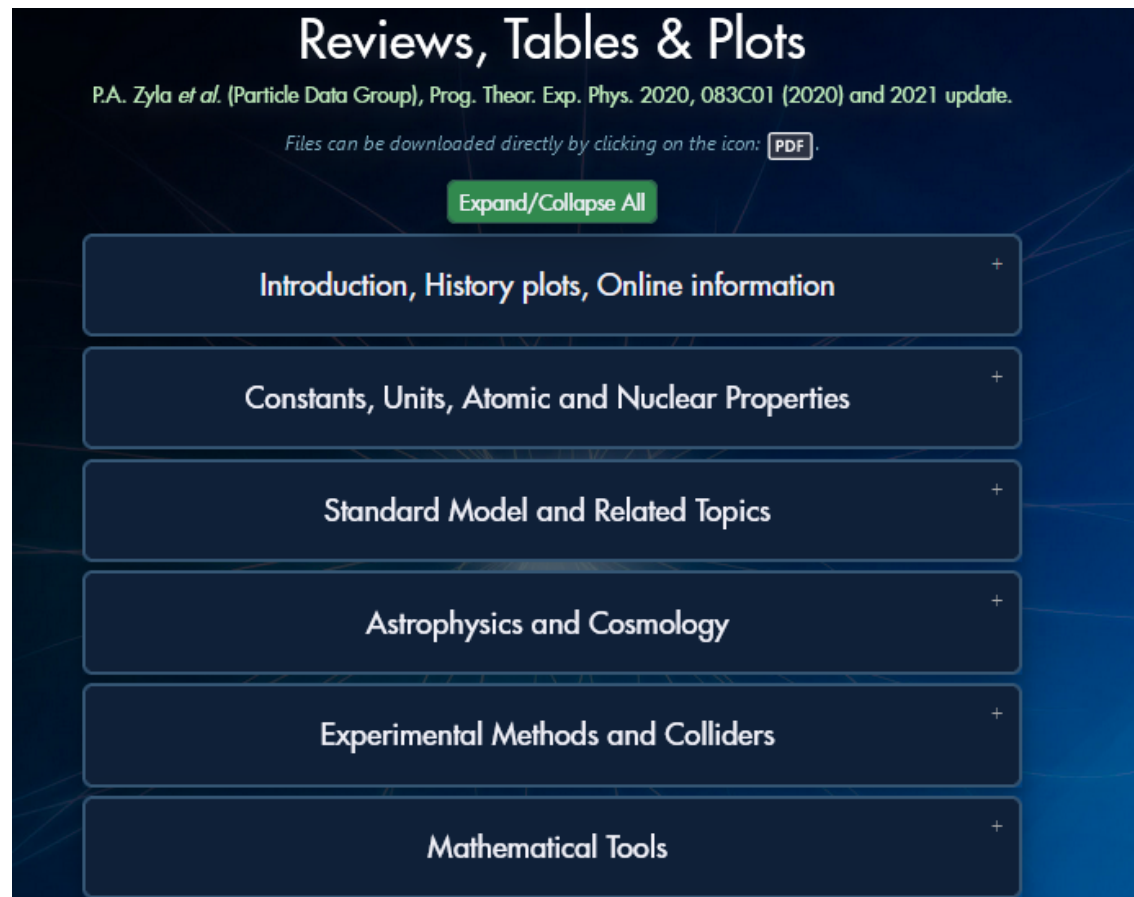
CESR : Early 2000's  
Babar last thing at SLAC

Now LHC



# The Ultimate Resource – PDG

- Particle Data Group
- First place to go – e.g. to understand detectors especially

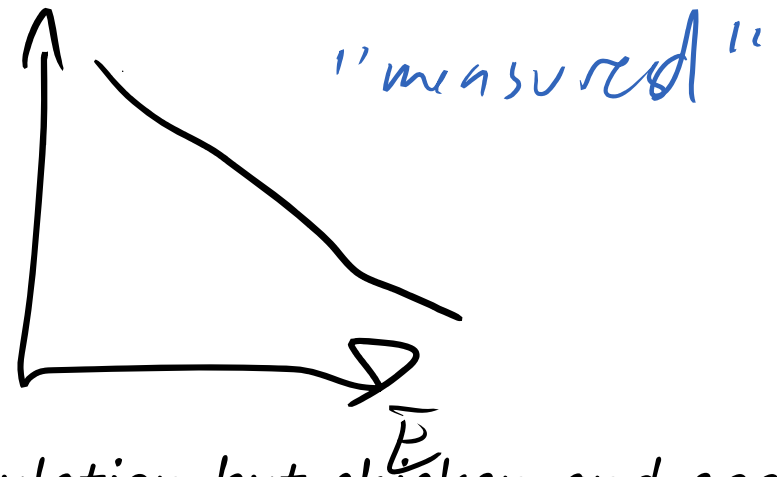


# Break/Exercise

- **EXERCISE** : From the PDG Review of Machine Learning:
  - a) How many sections are there under sub section Neural Networks
  - b) Name any other “flavor” of machine learning defined by another subsection next to the section on NN
  - c) What is word is defined in the second paragraph of the short section 41.3.5.2

# Break/Exercise

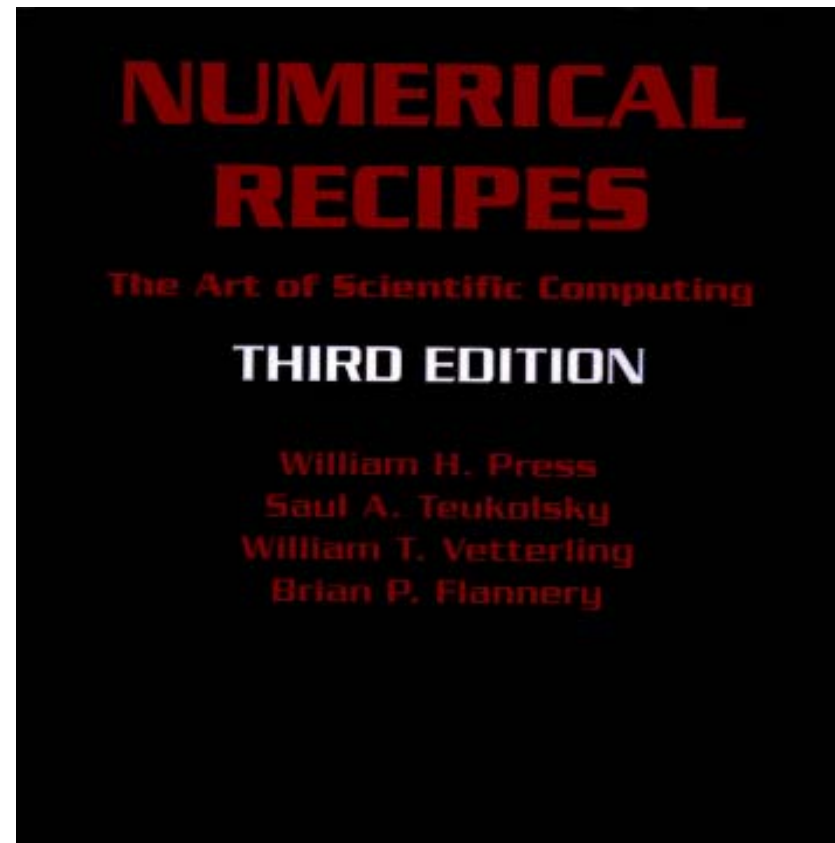
- PDG Review of Machine Learning:
  - a) How many sections are there under sub section Neural Networks
  - b) Name any other "flavor" of machine learning defined by another subsection next to the section on NN
  - c) What is word is defined in the second paragraph of the short section 41.3.5.2
- "Unfolding" → the general problem of undoing measurement effects (called "smearing") for a measured distribution
  - There could almost be a section on itself – maybe in one of the others



- Transformation function from simulation but chicken and egg problem because you don't know underlying true distribution to use for simulation -- e.g. why you are measuring it. Involves iterating

# Another historically common reference:

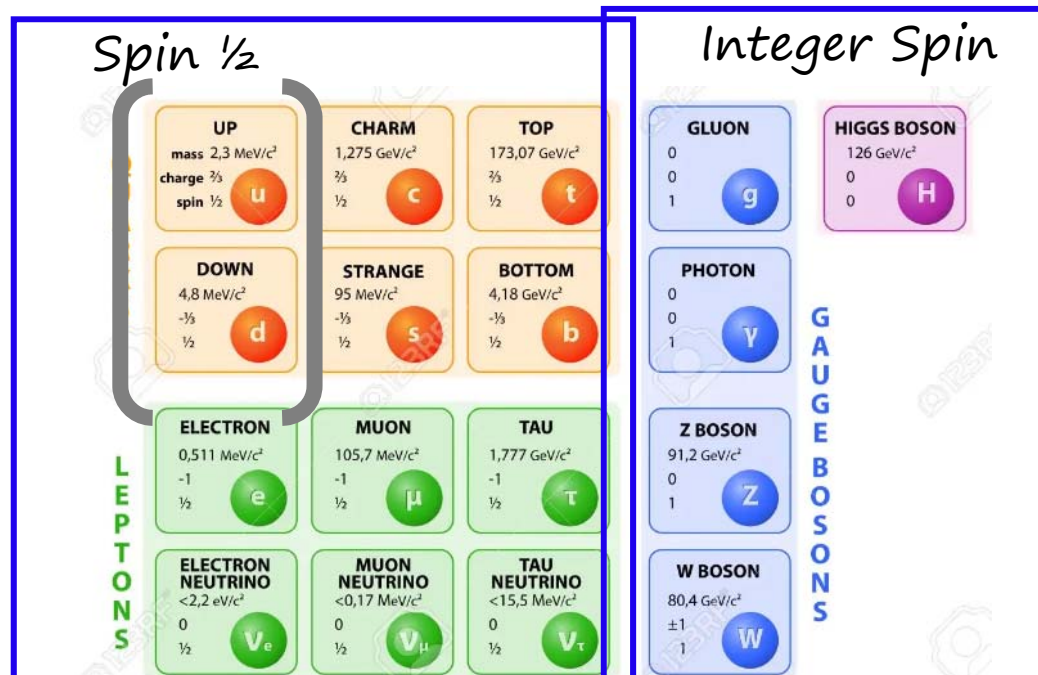
- Numerical Recipes – good read, especially the section on fitting functions
- C++ examples, though criticized... still good basis for concepts even 1<sup>st</sup> fortran version





## Other Properties - Group Theory

- So we've discussed mass and its peculiarities
- At least when particles (even composite) are "isolated", they exist in definite states of "mass"
- Mass is the most fundamental determiner of a "state" for our particles
- Most of the other properties on the table are not so simple as they appear
- This is because most of the other properties seen there, even "up" vs down flavor, and most of the charges (except electric) are actually more complex than just a scalar property like electric charge
- Rather they more like elements (basis vectors) of a vector space and can be combined in that way. in the same way q.m. spin can be combined.



# Mass $\rightarrow$ Other Properties - Group Theory

- The idea is captured by considering this:
- Electric charge has two types + / -
- How to represent this mathematically?
- Just the sign of a single number?
- Or how about making spin up + and spin down - ?
- The latter is how ~all other charges (except electric) and properties of the particles are described by the SM

## Spin $\frac{1}{2}$

**UP**  
mass 2,3 MeV/c<sup>2</sup>  
charge  $\frac{2}{3}$   
spin  $\frac{1}{2}$  

**CHARM**  
1,275 GeV/c<sup>2</sup>  
 $\frac{2}{3}$   
 $\frac{1}{2}$  

**TOP**  
173,07 GeV/c<sup>2</sup>  
 $\frac{2}{3}$   
 $\frac{1}{2}$  

**DOWN**  
4,8 MeV/c<sup>2</sup>  
 $-\frac{1}{3}$   
 $\frac{1}{2}$  

**STRANGE**  
95 MeV/c<sup>2</sup>  
 $-\frac{1}{3}$   
 $\frac{1}{2}$  

**BOTTOM**  
4,18 GeV/c<sup>2</sup>  
 $-\frac{1}{3}$   
 $\frac{1}{2}$  

**ELECTRON**  
0,511 MeV/c<sup>2</sup>  
-1  
 $\frac{1}{2}$  

**MUON**  
105,7 MeV/c<sup>2</sup>  
-1  
 $\frac{1}{2}$  

**TAU**  
1,777 GeV/c<sup>2</sup>  
-1  
 $\frac{1}{2}$  

**ELECTRON NEUTRINO**  
<2,2 eV/c<sup>2</sup>  
0  
 $\frac{1}{2}$  

**MUON NEUTRINO**  
<0,17 MeV/c<sup>2</sup>  
0  
 $\frac{1}{2}$  

**TAU NEUTRINO**  
<15,5 MeV/c<sup>2</sup>  
0  
 $\frac{1}{2}$  

LEPTONS

## Integer Spin

**GLUON**  
0  
0  
1 

**HIGGS BOSON**  
126 GeV/c<sup>2</sup>  
0  
0 

**PHOTON**  
0  
0  
1 

**Z BOSON**  
91,2 GeV/c<sup>2</sup>  
0  
1 

**W BOSON**  
80,4 GeV/c<sup>2</sup>  
 $\pm 1$   
1 

GAUGE BOSONS

So that is partially I'm sure why the column vector notation u/d ness are a property that behaves like spin up (up) and spin down (down).

Implication best captured by addition each quark in a "two quark state" can have "half up and half downness" not just one or the other

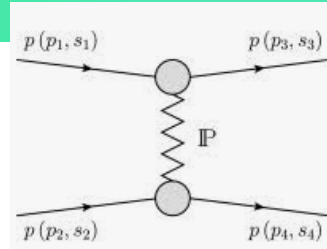
$$\frac{1}{\sqrt{2}} \left( \begin{pmatrix} u \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ d \end{pmatrix} \right)$$

# QCD Particles

- Hadrons
  - Bound states of quarks/gluons
  - For HEP this is distinguisher that causes them to exclude hadronic physics → that's all “boring” NP
- Mesons
  - quark – antiquark “pairs” ( $u\bar{d}$  or  $\frac{1}{\sqrt{2}}(u\bar{u} + d\bar{d})$ )
  - color anti-color
- Baryons
  - $qqq$  or  $\bar{q}\bar{q}\bar{q}$  color anti-color
- **NEW:** Tetraquarks and Pentaquarks (confirmed lately after an initial false positive slew of results by )
  - Still some ambiguity of whether they are “LIKE” bound states of 2 mesons (tetra quark) or Meson-Baryon (pentaquarks)

# Case Study: Other Particles – Pomeron

P

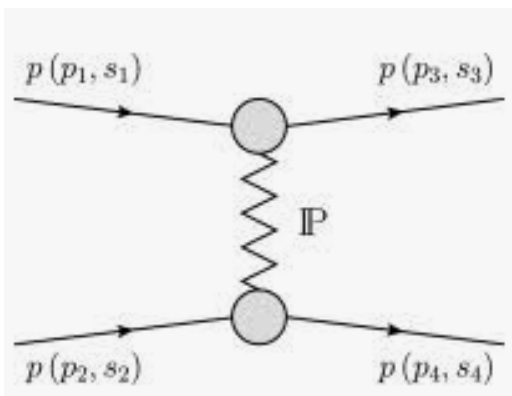


(I forgot to mention in class that one reason I wanted to cover this was for seminar prep for Brandenburg seminar)

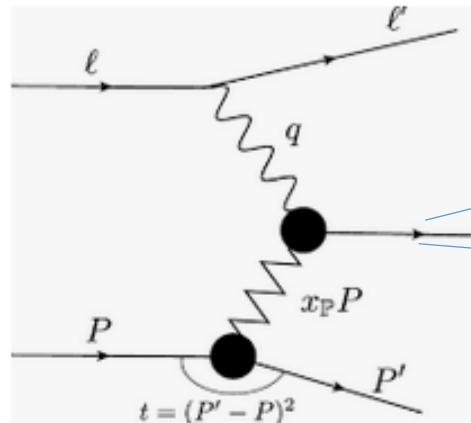
- Pomeron / Odderon / Regge-ons – Diffractive Physics – very odd “states”
- These are multigluon “states” that ~only act as force mediators of a sort, somewhat like pion is an effective force mediator for nucleons
- Usually these are not considered among the glueballs themselves e.g. predicted by the SM/Lattice QCD – more of an empirical data explainer
  - and like virtual force mediators particles only
  - though Wikipedia connects the recent odderon observation in Tevatron to these <https://en.wikipedia.org/wiki/Glueball>
- The scattering process involving pomeron exchange are responsible for a large percentage (majority at some energies) of the total high energy nuclear scattering **total cross section (x.s.) – IMPORTANT!**
  - But a little like low E photon minor scat/Bremst. leading to  $\infty$  x.s. for in QED
- An early ( $\leq 70$ 's) general classification of hadrons in scattering grouped them in a classifier type called Regge-ons  $\rightarrow$  a theory that successfully explained the mysteriously rising x.s. energy dependence through exchange of a Reggeon not observed elsewhere that had “vacuum quantum numbers” e.g. spin 0
  - This x.s. rise is a non-perturbative QCD phenomenon

# “Diffractive” Scattering – Pomeron

- More importantly these processes are “elastic” or pseudo elastic  $\rightarrow$  despite high energy, all incoming hadrons are left intact by the scattering.
  - But ~usually, additional particles are produced
- The pomeron attains seemingly very real nature in some cases, because some of these processes involve e.g. jet production where it is thought that the quarks emerge from the breakup of the pomerons themselves.
- The EIC will have this kind of scattering as a major focus



Elastic most common

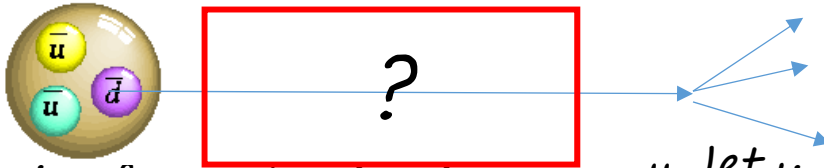


breakup -  
but not of  
incoming  
particles -  
Pom is  
dissected

Inelastic - more rare

# Other QCD Particle States

- Jets – Partons (general term for quarks or gluons) knocked out of hadrons ... “temporarily” deconfined (?)



- This timeframe/region is not well studied experimentally
- Best data? RHIC: put QGP plasma there – yes, has color– theories ?
- Pomeron/ Odd ball
  - DOMINATES OVERALL HADRON-HADRON SCATTERING RATE!
- Glueballs



# Some Lifetimes

TABLE 32-2 Particles (selected)<sup>†</sup>

| Category                      | Forces involved    | Particle name      | Symbol                      | Anti-particle     | Spin          | Mass (MeV/c <sup>2</sup> )                | B             | L <sub>e</sub> | L <sub>μ</sub> | L <sub>τ</sub> | S  | Mean life (s)             | Principal Decay Modes                                                                                                  |
|-------------------------------|--------------------|--------------------|-----------------------------|-------------------|---------------|-------------------------------------------|---------------|----------------|----------------|----------------|----|---------------------------|------------------------------------------------------------------------------------------------------------------------|
| Fundamental                   |                    |                    |                             |                   |               |                                           |               |                |                |                |    |                           |                                                                                                                        |
| Gauge bosons (force carriers) | str                | Gluons             | g                           | Self              | 1             | 0                                         | 0             | 0              | 0              | 0              | 0  | Stable                    |                                                                                                                        |
|                               | em                 | Photon             | γ                           | Self              | 1             | 0                                         | 0             | 0              | 0              | 0              | 0  | Stable                    |                                                                                                                        |
|                               | w, em              | W                  | W <sup>+</sup>              | W <sup>-</sup>    | 1             | 80.385 × 10 <sup>3</sup>                  | 0             | 0              | 0              | 0              | 0  | 3 × 10 <sup>-25</sup>     | eν <sub>e</sub> , μν <sub>μ</sub> , τν <sub>τ</sub> , hadrons                                                          |
|                               | w                  | Z                  | Z <sup>0</sup>              | Self              | 1             | 91.19 × 10 <sup>3</sup>                   | 0             | 0              | 0              | 0              | 0  | 3 × 10 <sup>-25</sup>     | e <sup>+</sup> e <sup>-</sup> , μ <sup>+</sup> μ <sup>-</sup> , τ <sup>+</sup> τ <sup>-</sup> , hadrons                |
| Higgs boson                   | w, str             | Higgs              | H <sup>0</sup>              | Self              | 0             | 125 × 10 <sup>3</sup>                     | 0             | 0              | 0              | 0              | 0  | 1.6 × 10 <sup>-22</sup>   | b $\bar{b}$ , Z <sup>0</sup> Z <sup>0</sup> , W <sup>+</sup> W <sup>-</sup> , gg, τ $\bar{\tau}$ , γγ                  |
| Leptons                       | w, em <sup>‡</sup> | Electron           | e <sup>-</sup>              | e <sup>+</sup>    | $\frac{1}{2}$ | 0.511                                     | 0             | +1             | 0              | 0              | 0  | Stable                    |                                                                                                                        |
|                               |                    | Neutrino (e)       | ν <sub>e</sub>              | $\bar{\nu}_e$     | $\frac{1}{2}$ | 0 (<0.14 eV/c <sup>2</sup> ) <sup>‡</sup> | 0             | +1             | 0              | 0              | 0  | Stable                    |                                                                                                                        |
|                               |                    | Muon               | μ <sup>-</sup>              | μ <sup>+</sup>    | $\frac{1}{2}$ | 105.7                                     | 0             | 0              | +1             | 0              | 0  | 2.20 × 10 <sup>-6</sup>   | e <sup>-</sup> $\bar{\nu}_e$ ν <sub>μ</sub>                                                                            |
|                               |                    | Neutrino (μ)       | ν <sub>μ</sub>              | $\bar{\nu}_\mu$   | $\frac{1}{2}$ | 0 (<0.14 eV/c <sup>2</sup> ) <sup>‡</sup> | 0             | 0              | +1             | 0              | 0  | Stable                    |                                                                                                                        |
|                               |                    | Tau                | τ <sup>-</sup>              | τ <sup>+</sup>    | $\frac{1}{2}$ | 1777                                      | 0             | 0              | 0              | +1             | 0  | 2.91 × 10 <sup>-13</sup>  | μ <sup>-</sup> $\bar{\nu}_\mu$ ν <sub>τ</sub> , e <sup>-</sup> $\bar{\nu}_e$ ν <sub>τ</sub> , hadrons + ν <sub>τ</sub> |
|                               |                    | Neutrino (τ)       | ν <sub>τ</sub>              | $\bar{\nu}_\tau$  | $\frac{1}{2}$ | 0 (<0.14 eV/c <sup>2</sup> ) <sup>‡</sup> | 0             | 0              | 0              | +1             | 0  | Stable                    |                                                                                                                        |
| Quarks                        | w, em, str         | (see Table 32-3)   |                             |                   |               |                                           |               |                |                |                |    |                           |                                                                                                                        |
| Hadrons (composite), selected |                    |                    |                             |                   |               |                                           |               |                |                |                |    |                           |                                                                                                                        |
| Mesons (quark-antiquark)      | str, em, w         | Pion               | π <sup>+</sup>              | π <sup>-</sup>    | 0             | 139.6                                     | 0             | 0              | 0              | 0              | 0  | 2.60 × 10 <sup>-8</sup>   | μ <sup>+</sup> ν <sub>μ</sub>                                                                                          |
|                               |                    |                    | π <sup>0</sup>              | Self              | 0             | 135.0                                     | 0             | 0              | 0              | 0              | 0  | 0.85 × 10 <sup>-16</sup>  | 2γ                                                                                                                     |
|                               |                    | Kaon               | K <sup>+</sup>              | K <sup>-</sup>    | 0             | 493.7                                     | 0             | 0              | 0              | 0              | +1 | 1.24 × 10 <sup>-8</sup>   | μ <sup>+</sup> ν <sub>μ</sub> , π <sup>+</sup> π <sup>0</sup>                                                          |
|                               |                    |                    | K <sub>S</sub> <sup>0</sup> | $\bar{K}_S^0$     | 0             | 497.6                                     | 0             | 0              | 0              | 0              | +1 | 0.895 × 10 <sup>-10</sup> | π <sup>+</sup> π <sup>-</sup> , 2π <sup>0</sup>                                                                        |
|                               |                    |                    | K <sub>L</sub> <sup>0</sup> | $\bar{K}_L^0$     | 0             | 497.6                                     | 0             | 0              | 0              | 0              | +1 | 5.12 × 10 <sup>-8</sup>   | π <sup>+</sup> e <sup>-</sup> $\bar{\nu}_e$ , π <sup>+</sup> μ <sup>-</sup> $\bar{\nu}_\mu$ , 3π                       |
|                               |                    |                    | η <sup>0</sup>              | Self              | 0             | 547.9                                     | 0             | 0              | 0              | 0              | 0  | 5.1 × 10 <sup>-19</sup>   | 2γ, 3π <sup>0</sup> , π <sup>+</sup> π <sup>-</sup> π <sup>0</sup>                                                     |
|                               |                    | Rho                | ρ <sup>0</sup>              | Self              | 1             | 775                                       | 0             | 0              | 0              | 0              | 0  | 4.4 × 10 <sup>-24</sup>   | π <sup>+</sup> π <sup>-</sup> , 2π <sup>0</sup>                                                                        |
|                               |                    |                    | ρ <sup>+</sup>              | ρ <sup>-</sup>    | 1             | 775                                       | 0             | 0              | 0              | 0              | 0  | 4.4 × 10 <sup>-24</sup>   | π <sup>+</sup> π <sup>0</sup>                                                                                          |
|                               |                    |                    | and others                  |                   |               |                                           |               |                |                |                |    |                           |                                                                                                                        |
|                               |                    | Baryons (3 quarks) | str, em, w                  | Proton            | p             | $\bar{p}$                                 | $\frac{1}{2}$ | 938.3          | +1             | 0              | 0  | 0                         | 0                                                                                                                      |
| Neutron                       | n                  |                    |                             | $\bar{n}$         | $\frac{1}{2}$ | 939.6                                     | +1            | 0              | 0              | 0              | 0  | 882                       | p e <sup>-</sup> $\bar{\nu}_e$                                                                                         |
| Lambda                        | Λ <sup>0</sup>     |                    |                             | $\bar{\Lambda}^0$ | $\frac{1}{2}$ | 1115.7                                    | +1            | 0              | 0              | 0              | -1 | 2.63 × 10 <sup>-10</sup>  | p π <sup>-</sup> , n π <sup>0</sup>                                                                                    |
| Sigma                         | Σ <sup>+</sup>     |                    |                             | $\bar{\Sigma}^-$  | $\frac{1}{2}$ | 1189.4                                    | +1            | 0              | 0              | 0              | -1 | 0.80 × 10 <sup>-10</sup>  | p π <sup>0</sup> , n π <sup>+</sup>                                                                                    |
|                               | Σ <sup>0</sup>     |                    |                             | $\bar{\Sigma}^0$  | $\frac{1}{2}$ | 1192.6                                    | +1            | 0              | 0              | 0              | -1 | 7.4 × 10 <sup>-20</sup>   | Λ <sup>0</sup> γ                                                                                                       |
|                               | Σ <sup>-</sup>     |                    |                             | $\bar{\Sigma}^+$  | $\frac{1}{2}$ | 1197.4                                    | +1            | 0              | 0              | 0              | -1 | 1.48 × 10 <sup>-10</sup>  | n π <sup>-</sup>                                                                                                       |
|                               | Ξ <sup>0</sup>     |                    |                             | $\bar{\Xi}^0$     | $\frac{1}{2}$ | 1314.9                                    | +1            | 0              | 0              | 0              | -2 | 2.90 × 10 <sup>-10</sup>  | Λ <sup>0</sup> π <sup>0</sup>                                                                                          |
| Xi                            | Ξ <sup>-</sup>     |                    |                             | $\bar{\Xi}^+$     | $\frac{1}{2}$ | 1321.7                                    | +1            | 0              | 0              | 0              | -2 | 1.64 × 10 <sup>-10</sup>  | Λ <sup>0</sup> π <sup>-</sup>                                                                                          |
|                               | Ω <sup>-</sup>     |                    |                             | Ω <sup>+</sup>    | $\frac{1}{2}$ | 1672.5                                    | +1            | 0              | 0              | 0              | -3 | 0.82 × 10 <sup>-10</sup>  | Ξ <sup>0</sup> π <sup>-</sup> , Λ <sup>0</sup> K <sup>-</sup> , Ξ <sup>-</sup> π <sup>0</sup>                          |
| and others                    |                    |                    |                             |                   |               |                                           |               |                |                |                |    |                           |                                                                                                                        |

top 1e-25 s

“reso  
nance”

top 1e-25 s

“resonances”

## Fundamental

|                                  |                    |                     |            |                  |               |                              |   |    |    |    |   |                        |                                 |
|----------------------------------|--------------------|---------------------|------------|------------------|---------------|------------------------------|---|----|----|----|---|------------------------|---------------------------------|
| Gauge bosons<br>(force carriers) | str                | Gluons              | g          | Self             | 1             | 0                            | 0 | 0  | 0  | 0  | 0 | Stable                 |                                 |
|                                  | em                 | Photon              | $\gamma$   | Self             | 1             | 0                            | 0 | 0  | 0  | 0  | 0 | Stable                 |                                 |
|                                  | w,em               | W                   | $W^+$      | $W^-$            | 1             | $80.385 \times 10^3$         | 0 | 0  | 0  | 0  | 0 | $3 \times 10^{-25}$    | $e\nu_e, \mu\nu_\mu,$           |
|                                  | w                  | Z                   | $Z^0$      | Self             | 1             | $91.19 \times 10^3$          | 0 | 0  | 0  | 0  | 0 | $3 \times 10^{-25}$    | $e^+e^-, \mu^+\mu^-$            |
| Higgs boson                      | w, str             | Higgs               | $H^0$      | Self             | 0             | $125 \times 10^3$            | 0 | 0  | 0  | 0  | 0 | $1.6 \times 10^{-22}$  | $b\bar{b}, Z^0Z^0,$             |
| Leptons                          | w, em <sup>1</sup> | Electron            | $e^-$      | $e^+$            | $\frac{1}{2}$ | 0.511                        | 0 | +1 | 0  | 0  | 0 | Stable                 |                                 |
|                                  |                    | Neutrino (e)        | $\nu_e$    | $\bar{\nu}_e$    | $\frac{1}{2}$ | $0 (<0.14 \text{ eV}/c^2)^1$ | 0 | +1 | 0  | 0  | 0 | Stable                 |                                 |
|                                  |                    | Muon                | $\mu^-$    | $\mu^+$          | $\frac{1}{2}$ | 105.7                        | 0 | 0  | +1 | 0  | 0 | $2.20 \times 10^{-6}$  | $e^-\bar{\nu}_e\nu_\mu$         |
|                                  |                    | Neutrino ( $\mu$ )  | $\nu_\mu$  | $\bar{\nu}_\mu$  | $\frac{1}{2}$ | $0 (<0.14 \text{ eV}/c^2)^1$ | 0 | 0  | +1 | 0  | 0 | Stable                 |                                 |
|                                  |                    | Tau                 | $\tau^-$   | $\tau^+$         | $\frac{1}{2}$ | 1777                         | 0 | 0  | 0  | +1 | 0 | $2.91 \times 10^{-13}$ | $\mu^-\bar{\nu}_\mu\nu_\tau, e$ |
|                                  |                    | Neutrino ( $\tau$ ) | $\nu_\tau$ | $\bar{\nu}_\tau$ | $\frac{1}{2}$ | $0 (<0.14 \text{ eV}/c^2)^1$ | 0 | 0  | 0  | +1 | 0 | Stable                 |                                 |

Quarks w, em, str (see Table 32-3)

## Hadrons (composite), selected

|                  |             |            |          |                   |               |           |               |       |    |   |    |                         |                          |        |
|------------------|-------------|------------|----------|-------------------|---------------|-----------|---------------|-------|----|---|----|-------------------------|--------------------------|--------|
| Mesons           | str, em, w  | Pion       | $\pi^+$  | $\pi^-$           | 0             | 139.6     | 0             | 0     | 0  | 0 | 0  | $2.60 \times 10^{-8}$   | $\mu^+ \nu_\mu$          |        |
|                  |             |            | $\pi^0$  | Self              | 0             | 135.0     | 0             | 0     | 0  | 0 | 0  | $0.85 \times 10^{-16}$  | $2\gamma$                |        |
|                  |             | Kaon       | $K^+$    | $K^-$             | 0             | 493.7     | 0             | 0     | 0  | 0 | +1 | $1.24 \times 10^{-8}$   | $\mu^+ \nu_\mu, \pi^+$   |        |
|                  |             |            | $K_S^0$  | $\bar{K}_S^0$     | 0             | 497.6     | 0             | 0     | 0  | 0 | +1 | $0.895 \times 10^{-10}$ | $\pi^+ \pi^-, 2\pi$      |        |
|                  |             |            | $K_L^0$  | $\bar{K}_L^0$     | 0             | 497.6     | 0             | 0     | 0  | 0 | +1 | $5.12 \times 10^{-8}$   | $\pi^+ e^+ \bar{\nu}_e,$ |        |
|                  |             | Eta        | $\eta^0$ | Self              | 0             | 547.9     | 0             | 0     | 0  | 0 | 0  | $5.1 \times 10^{-19}$   | $2\gamma, 3\pi^0,$       |        |
|                  |             | Rho        | $\rho^0$ | Self              | 1             | 775       | 0             | 0     | 0  | 0 | 0  | $4.4 \times 10^{-24}$   | $\pi^+ \pi^-, 2\pi$      |        |
|                  |             |            | $\rho^+$ | $\rho^-$          | 1             | 775       | 0             | 0     | 0  | 0 | 0  | $4.4 \times 10^{-24}$   | $\pi^+ \pi^0$            |        |
|                  |             | and others |          |                   |               |           |               |       |    |   |    |                         |                          |        |
|                  |             | w          |          | Proton            | p             | $\bar{p}$ | $\frac{1}{2}$ | 938.3 | +1 | 0 | 0  | 0                       | 0                        | Stable |
| Neutron          | n           |            |          | $\bar{n}$         | $\frac{1}{2}$ | 939.6     | +1            | 0     | 0  | 0 | 0  | 882                     | $pe^+ \bar{\nu}_e$       |        |
| Lambda           | $\Lambda^0$ |            |          | $\bar{\Lambda}^0$ | $\frac{1}{2}$ | 1115.7    | +1            | 0     | 0  | 0 | -1 | $2.63 \times 10^{-10}$  | $p\pi^-, n\pi^+$         |        |
| Sigma            | $\Sigma^+$  |            |          | $\bar{\Sigma}^-$  | $\frac{1}{2}$ | 1189.4    | +1            | 0     | 0  | 0 | -1 | $0.80 \times 10^{-10}$  | $p\pi^0, n\pi^+$         |        |
|                  | $\Sigma^0$  |            |          | $\bar{\Sigma}^0$  | $\frac{1}{2}$ | 1192.6    | +1            | 0     | 0  | 0 | -1 | $7.4 \times 10^{-20}$   | $\Lambda^0 \gamma$       |        |
|                  | $\Sigma^-$  |            |          | $\bar{\Sigma}^+$  | $\frac{1}{2}$ | 1197.4    | +1            | 0     | 0  | 0 | -1 | $1.48 \times 10^{-10}$  | $n\pi^-$                 |        |
| Xi               | $\Xi^0$     |            |          | $\bar{\Xi}^0$     | $\frac{1}{2}$ | 1314.9    | +1            | 0     | 0  | 0 | -2 | $2.90 \times 10^{-10}$  | $\Lambda^0 \pi^0$        |        |
|                  | $\Xi^-$     |            |          | $\bar{\Xi}^+$     | $\frac{1}{2}$ | 1321.7    | +1            | 0     | 0  | 0 | -2 | $1.64 \times 10^{-10}$  | $\Lambda^0 \pi^-$        |        |
| Omega and others | $\Omega^-$  |            |          | $\bar{\Omega}^+$  | $\frac{1}{2}$ | 1672.5    | +1            | 0     | 0  | 0 | -3 | $0.82 \times 10^{-10}$  | $\Xi^0 \pi^-, \Lambda^+$ |        |

$\gamma c\tau \rightarrow 10^{-10}s$   
~1 m

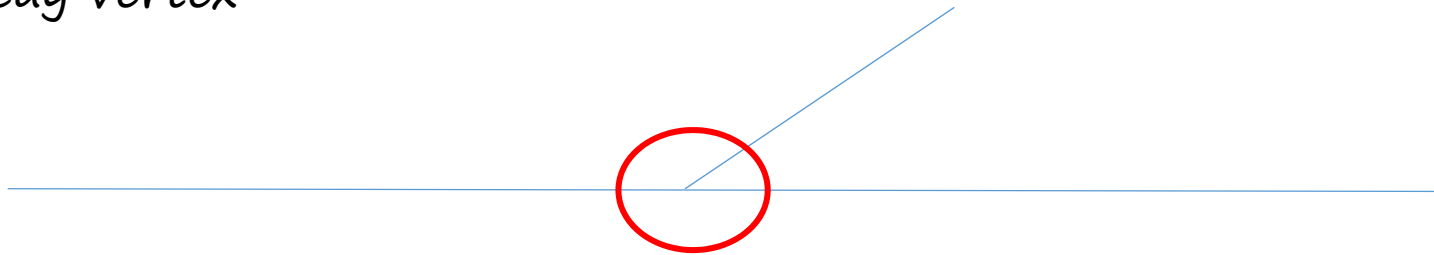
Usually inside  
detector  
volumes

ID particles  
by track  
vertex  
topology



# Experimental Rundown of Particles/Hadrons

- Pion: lightest hadron – In almost any collisions at energy/momentum higher than a  $\sim$ half a GeV lightest hadron pion will be produced
  - Speed of light 1 m in  $\sim 3$  nano s
- Charged Pions –  $\sim 3 \times 10^{-8}$  s = 30 ns lifetime (but also time-dilated with a Lorentz gamma factor of  $E/m \geq 3$ ) decay length ( $\gamma c \tau$ )  $\geq 10$  meters
  - Effectively Stable
- Neutral Pion –  $\sim 1 \times 10^{-16}$  s  $\rightarrow \geq \sim 0.1 \mu\text{m}$  so macroscopically far away from collision “vertex”
  - vertex not in Feynman diagram sense, rather in particle “track”/trajectories
  - generic term used lots in Particle Physics – either event vertex or decay vertex



# Experimental Rundown of Particles/Hadrons

- Here are some important experimental aspects of these particles based on their detection from Medium/High E scattering
- **Photons** – many commonly produced hadrons decay to photons especially the  $\pi^0$  – also can be produced directly (“direct photons”) These photons are much higher E than typically called “Gamma Rays”
- **Electrons** – produced very small quantities in comparison to hadrons (%) – mostly from “photon conversion”, hadron decay  $\gamma$ 's
  - almost never measured from direct production at  $\geq$  Medium E
  - Heavy Flavor hadrons commonly decay (weakly) to these so they do good way to detect
- **Muons** – same purpose, but cleaner but rarer (though not much rarer) than electron – generally “easy” to detect but require far detectors/matter to absorb all other particles
- **Neutrinos** – are undetectable [except in dedicated huge detectors], but for simple (e.g. elastic) reactions can be reconstructed via “missing mass” ... missing momentum, missing energy, “invisible”  $E/p$ , missing transverse energy  $E_T$  (MET)
- **Missing E** is also a common technique for reconstructing any particles out of detector acceptance in processes involving small numbers of particles such as “pseudo”-elastic [initial state particles scatter intact] or “exclusive” channels

# Some Lifetimes

TABLE 32-2 Particles (selected)<sup>†</sup>

| Category                      | Forces involved    | Particle name      | Symbol                      | Anti-particle     | Spin          | Mass (MeV/c <sup>2</sup> )                | B             | L <sub>e</sub> | L <sub>μ</sub> | L <sub>τ</sub>           | S                                                                                             | Mean life (s)             | Principal Decay Modes                                                                                                  |
|-------------------------------|--------------------|--------------------|-----------------------------|-------------------|---------------|-------------------------------------------|---------------|----------------|----------------|--------------------------|-----------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------|
| Fundamental                   |                    |                    |                             |                   |               |                                           |               |                |                |                          |                                                                                               |                           |                                                                                                                        |
| Gauge bosons (force carriers) | str                | Gluons             | g                           | Self              | 1             | 0                                         | 0             | 0              | 0              | 0                        | 0                                                                                             | Stable                    |                                                                                                                        |
|                               | em                 | Photon             | γ                           | Self              | 1             | 0                                         | 0             | 0              | 0              | 0                        | 0                                                                                             | Stable                    |                                                                                                                        |
|                               | w, em              | W                  | W <sup>+</sup>              | W <sup>-</sup>    | 1             | 80.385 × 10 <sup>3</sup>                  | 0             | 0              | 0              | 0                        | 0                                                                                             | 3 × 10 <sup>-25</sup>     | eν <sub>e</sub> , μν <sub>μ</sub> , τν <sub>τ</sub> , hadrons                                                          |
|                               | w                  | Z                  | Z <sup>0</sup>              | Self              | 1             | 91.19 × 10 <sup>3</sup>                   | 0             | 0              | 0              | 0                        | 0                                                                                             | 3 × 10 <sup>-25</sup>     | e <sup>+</sup> e <sup>-</sup> , μ <sup>+</sup> μ <sup>-</sup> , τ <sup>+</sup> τ <sup>-</sup> , hadrons                |
| Higgs boson                   | w, str             | Higgs              | H <sup>0</sup>              | Self              | 0             | 125 × 10 <sup>3</sup>                     | 0             | 0              | 0              | 0                        | 0                                                                                             | 1.6 × 10 <sup>-22</sup>   | b $\bar{b}$ , Z <sup>0</sup> Z <sup>0</sup> , W <sup>+</sup> W <sup>-</sup> , gg, τ $\bar{\tau}$ , γγ                  |
| Leptons                       | w, em <sup>‡</sup> | Electron           | e <sup>-</sup>              | e <sup>+</sup>    | $\frac{1}{2}$ | 0.511                                     | 0             | +1             | 0              | 0                        | 0                                                                                             | Stable                    |                                                                                                                        |
|                               |                    | Neutrino (e)       | ν <sub>e</sub>              | $\bar{\nu}_e$     | $\frac{1}{2}$ | 0 (<0.14 eV/c <sup>2</sup> ) <sup>‡</sup> | 0             | +1             | 0              | 0                        | 0                                                                                             | Stable                    |                                                                                                                        |
|                               |                    | Muon               | μ <sup>-</sup>              | μ <sup>+</sup>    | $\frac{1}{2}$ | 105.7                                     | 0             | 0              | +1             | 0                        | 0                                                                                             | 2.20 × 10 <sup>-6</sup>   | e <sup>-</sup> $\bar{\nu}_e$ ν <sub>μ</sub>                                                                            |
|                               |                    | Neutrino (μ)       | ν <sub>μ</sub>              | $\bar{\nu}_\mu$   | $\frac{1}{2}$ | 0 (<0.14 eV/c <sup>2</sup> ) <sup>‡</sup> | 0             | 0              | +1             | 0                        | 0                                                                                             | Stable                    |                                                                                                                        |
|                               |                    | Tau                | τ <sup>-</sup>              | τ <sup>+</sup>    | $\frac{1}{2}$ | 1777                                      | 0             | 0              | 0              | +1                       | 0                                                                                             | 2.91 × 10 <sup>-13</sup>  | μ <sup>-</sup> $\bar{\nu}_\mu$ ν <sub>τ</sub> , e <sup>-</sup> $\bar{\nu}_e$ ν <sub>τ</sub> , hadrons + ν <sub>τ</sub> |
|                               |                    | Neutrino (τ)       | ν <sub>τ</sub>              | $\bar{\nu}_\tau$  | $\frac{1}{2}$ | 0 (<0.14 eV/c <sup>2</sup> ) <sup>‡</sup> | 0             | 0              | 0              | +1                       | 0                                                                                             | Stable                    |                                                                                                                        |
| Quarks                        | w, em, str         | (see Table 32-3)   |                             |                   |               |                                           |               |                |                |                          |                                                                                               |                           |                                                                                                                        |
| Hadrons (composite), selected |                    |                    |                             |                   |               |                                           |               |                |                |                          |                                                                                               |                           |                                                                                                                        |
| Mesons (quark-antiquark)      | str, em, w         | Pion               | π <sup>+</sup>              | π <sup>-</sup>    | 0             | 139.6                                     | 0             | 0              | 0              | 0                        | 0                                                                                             | 2.60 × 10 <sup>-8</sup>   | μ <sup>+</sup> ν <sub>μ</sub>                                                                                          |
|                               |                    |                    | π <sup>0</sup>              | Self              | 0             | 135.0                                     | 0             | 0              | 0              | 0                        | 0                                                                                             | 0.85 × 10 <sup>-16</sup>  | 2γ                                                                                                                     |
|                               |                    | Kaon               | K <sup>+</sup>              | K <sup>-</sup>    | 0             | 493.7                                     | 0             | 0              | 0              | 0                        | +1                                                                                            | 1.24 × 10 <sup>-8</sup>   | μ <sup>+</sup> ν <sub>μ</sub> , π <sup>+</sup> π <sup>0</sup>                                                          |
|                               |                    |                    | K <sub>S</sub> <sup>0</sup> | $\bar{K}_S^0$     | 0             | 497.6                                     | 0             | 0              | 0              | 0                        | +1                                                                                            | 0.895 × 10 <sup>-10</sup> | π <sup>+</sup> π <sup>-</sup> , 2π <sup>0</sup>                                                                        |
|                               |                    |                    | K <sub>L</sub> <sup>0</sup> | $\bar{K}_L^0$     | 0             | 497.6                                     | 0             | 0              | 0              | 0                        | +1                                                                                            | 5.12 × 10 <sup>-8</sup>   | π <sup>+</sup> e <sup>-</sup> $\bar{\nu}_e$ , π <sup>+</sup> μ <sup>-</sup> $\bar{\nu}_\mu$ , 3π                       |
|                               |                    |                    | η <sup>0</sup>              | Self              | 0             | 547.9                                     | 0             | 0              | 0              | 0                        | 0                                                                                             | 5.1 × 10 <sup>-19</sup>   | 2γ, 3π <sup>0</sup> , π <sup>+</sup> π <sup>-</sup> π <sup>0</sup>                                                     |
|                               |                    | Rho                | ρ <sup>0</sup>              | Self              | 1             | 775                                       | 0             | 0              | 0              | 0                        | 0                                                                                             | 4.4 × 10 <sup>-24</sup>   | π <sup>+</sup> π <sup>-</sup> , 2π <sup>0</sup>                                                                        |
|                               |                    |                    | ρ <sup>+</sup>              | ρ <sup>-</sup>    | 1             | 775                                       | 0             | 0              | 0              | 0                        | 0                                                                                             | 4.4 × 10 <sup>-24</sup>   | π <sup>+</sup> π <sup>0</sup>                                                                                          |
|                               |                    |                    | and others                  |                   |               |                                           |               |                |                |                          |                                                                                               |                           |                                                                                                                        |
|                               |                    | Baryons (3 quarks) | str, em, w                  | Proton            | p             | $\bar{p}$                                 | $\frac{1}{2}$ | 938.3          | +1             | 0                        | 0                                                                                             | 0                         | 0                                                                                                                      |
| Neutron                       | n                  |                    |                             | $\bar{n}$         | $\frac{1}{2}$ | 939.6                                     | +1            | 0              | 0              | 0                        | 0                                                                                             | 882                       | p e <sup>-</sup> $\bar{\nu}_e$                                                                                         |
| Lambda                        | Λ <sup>0</sup>     |                    |                             | $\bar{\Lambda}^0$ | $\frac{1}{2}$ | 1115.7                                    | +1            | 0              | 0              | 0                        | -1                                                                                            | 2.63 × 10 <sup>-10</sup>  | p π <sup>-</sup> , n π <sup>0</sup>                                                                                    |
| Sigma                         | Σ <sup>+</sup>     |                    |                             | $\bar{\Sigma}^-$  | $\frac{1}{2}$ | 1189.4                                    | +1            | 0              | 0              | 0                        | -1                                                                                            | 0.80 × 10 <sup>-10</sup>  | p π <sup>0</sup> , n π <sup>+</sup>                                                                                    |
|                               | Σ <sup>0</sup>     |                    |                             | $\bar{\Sigma}^0$  | $\frac{1}{2}$ | 1192.6                                    | +1            | 0              | 0              | 0                        | -1                                                                                            | 7.4 × 10 <sup>-20</sup>   | Λ <sup>0</sup> γ                                                                                                       |
|                               | Σ <sup>-</sup>     |                    |                             | $\bar{\Sigma}^+$  | $\frac{1}{2}$ | 1197.4                                    | +1            | 0              | 0              | 0                        | -1                                                                                            | 1.48 × 10 <sup>-10</sup>  | n π <sup>-</sup>                                                                                                       |
| Xi                            | Ξ <sup>0</sup>     |                    |                             | $\bar{\Xi}^0$     | $\frac{1}{2}$ | 1314.9                                    | +1            | 0              | 0              | 0                        | -2                                                                                            | 2.90 × 10 <sup>-10</sup>  | Λ <sup>0</sup> π <sup>0</sup>                                                                                          |
|                               | Ξ <sup>-</sup>     |                    |                             | $\bar{\Xi}^+$     | $\frac{1}{2}$ | 1321.7                                    | +1            | 0              | 0              | 0                        | -2                                                                                            | 1.64 × 10 <sup>-10</sup>  | Λ <sup>0</sup> π <sup>-</sup>                                                                                          |
| Omega and others              | Ω <sup>-</sup>     | Ω <sup>+</sup>     | $\frac{1}{2}$               | 1672.5            | +1            | 0                                         | 0             | 0              | -3             | 0.82 × 10 <sup>-10</sup> | Ξ <sup>0</sup> π <sup>-</sup> , Λ <sup>0</sup> K <sup>-</sup> , Ξ <sup>-</sup> π <sup>0</sup> |                           |                                                                                                                        |

top 1e-25 s

“resonances”

## Fundamental

|                                  |                    |                     |            |                  |               |                              |   |    |    |    |   |                        |                                 |
|----------------------------------|--------------------|---------------------|------------|------------------|---------------|------------------------------|---|----|----|----|---|------------------------|---------------------------------|
| Gauge bosons<br>(force carriers) | str                | Gluons              | g          | Self             | 1             | 0                            | 0 | 0  | 0  | 0  | 0 | Stable                 |                                 |
|                                  | em                 | Photon              | $\gamma$   | Self             | 1             | 0                            | 0 | 0  | 0  | 0  | 0 | Stable                 |                                 |
|                                  | w,em               | W                   | $W^+$      | $W^-$            | 1             | $80.385 \times 10^3$         | 0 | 0  | 0  | 0  | 0 | $3 \times 10^{-25}$    | $e\nu_e, \mu\nu_\mu,$           |
|                                  | w                  | Z                   | $Z^0$      | Self             | 1             | $91.19 \times 10^3$          | 0 | 0  | 0  | 0  | 0 | $3 \times 10^{-25}$    | $e^+e^-, \mu^+\mu^-$            |
| Higgs boson                      | w, str             | Higgs               | $H^0$      | Self             | 0             | $125 \times 10^3$            | 0 | 0  | 0  | 0  | 0 | $1.6 \times 10^{-22}$  | $b\bar{b}, Z^0Z^0,$             |
| Leptons                          | w, em <sup>1</sup> | Electron            | $e^-$      | $e^+$            | $\frac{1}{2}$ | 0.511                        | 0 | +1 | 0  | 0  | 0 | Stable                 |                                 |
|                                  |                    | Neutrino (e)        | $\nu_e$    | $\bar{\nu}_e$    | $\frac{1}{2}$ | $0 (<0.14 \text{ eV}/c^2)^1$ | 0 | +1 | 0  | 0  | 0 | Stable                 |                                 |
|                                  |                    | Muon                | $\mu^-$    | $\mu^+$          | $\frac{1}{2}$ | 105.7                        | 0 | 0  | +1 | 0  | 0 | $2.20 \times 10^{-6}$  | $e^-\bar{\nu}_e\nu_\mu$         |
|                                  |                    | Neutrino ( $\mu$ )  | $\nu_\mu$  | $\bar{\nu}_\mu$  | $\frac{1}{2}$ | $0 (<0.14 \text{ eV}/c^2)^1$ | 0 | 0  | +1 | 0  | 0 | Stable                 |                                 |
|                                  |                    | Tau                 | $\tau^-$   | $\tau^+$         | $\frac{1}{2}$ | 1777                         | 0 | 0  | 0  | +1 | 0 | $2.91 \times 10^{-13}$ | $\mu^-\bar{\nu}_\mu\nu_\tau, e$ |
|                                  |                    | Neutrino ( $\tau$ ) | $\nu_\tau$ | $\bar{\nu}_\tau$ | $\frac{1}{2}$ | $0 (<0.14 \text{ eV}/c^2)^1$ | 0 | 0  | 0  | +1 | 0 | Stable                 |                                 |

Quarks w, em, str (see Table 32-3)

## Hadrons (composite), selected

|                  |             |            |          |                   |               |           |               |       |    |   |    |                         |                                      |
|------------------|-------------|------------|----------|-------------------|---------------|-----------|---------------|-------|----|---|----|-------------------------|--------------------------------------|
| Mesons           | str, em, w  | Pion       | $\pi^+$  | $\pi^-$           | 0             | 139.6     | 0             | 0     | 0  | 0 | 0  | $2.60 \times 10^{-8}$   | $\mu^+ \nu_\mu$                      |
|                  |             |            | $\pi^0$  | Self              | 0             | 135.0     | 0             | 0     | 0  | 0 | 0  | $0.85 \times 10^{-16}$  | $2\gamma$                            |
|                  |             | Kaon       | $K^+$    | $K^-$             | 0             | 493.7     | 0             | 0     | 0  | 0 | +1 | $1.24 \times 10^{-8}$   | $\mu^+ \nu_\mu, \pi^+$               |
|                  |             |            | $K_S^0$  | $\bar{K}_S^0$     | 0             | 497.6     | 0             | 0     | 0  | 0 | +1 | $0.895 \times 10^{-10}$ | $\pi^+ \pi^-, 2\pi$                  |
|                  |             |            | $K_L^0$  | $\bar{K}_L^0$     | 0             | 497.6     | 0             | 0     | 0  | 0 | +1 | $5.12 \times 10^{-8}$   | $\pi^+ e^+ \bar{\nu}_e, \pi^0 \pi^0$ |
|                  |             | Eta        | $\eta^0$ | Self              | 0             | 547.9     | 0             | 0     | 0  | 0 | 0  | $5.1 \times 10^{-19}$   | $2\gamma, 3\pi^0, \pi^+ \pi^- \pi^0$ |
|                  |             | Rho        | $\rho^0$ | Self              | 1             | 775       | 0             | 0     | 0  | 0 | 0  | $4.4 \times 10^{-24}$   | $\pi^+ \pi^-, 2\pi$                  |
|                  |             |            | $\rho^+$ | $\rho^-$          | 1             | 775       | 0             | 0     | 0  | 0 | 0  | $4.4 \times 10^{-24}$   | $\pi^+ \pi^0$                        |
|                  |             | and others |          |                   |               |           |               |       |    |   |    |                         |                                      |
|                  |             | w          |          | Proton            | p             | $\bar{p}$ | $\frac{1}{2}$ | 938.3 | +1 | 0 | 0  | 0                       | 0                                    |
| Neutron          | n           |            |          | $\bar{n}$         | $\frac{1}{2}$ | 939.6     | +1            | 0     | 0  | 0 | 0  | 882                     | $pe^+ \bar{\nu}_e$                   |
| Lambda           | $\Lambda^0$ |            |          | $\bar{\Lambda}^0$ | $\frac{1}{2}$ | 1115.7    | +1            | 0     | 0  | 0 | -1 | $2.63 \times 10^{-10}$  | $p\pi^-, n\pi^+$                     |
| Sigma            | $\Sigma^+$  |            |          | $\bar{\Sigma}^-$  | $\frac{1}{2}$ | 1189.4    | +1            | 0     | 0  | 0 | -1 | $0.80 \times 10^{-10}$  | $p\pi^0, n\pi^+$                     |
|                  | $\Sigma^0$  |            |          | $\bar{\Sigma}^0$  | $\frac{1}{2}$ | 1192.6    | +1            | 0     | 0  | 0 | -1 | $7.4 \times 10^{-20}$   | $\Lambda^0 \gamma$                   |
|                  | $\Sigma^-$  |            |          | $\bar{\Sigma}^+$  | $\frac{1}{2}$ | 1197.4    | +1            | 0     | 0  | 0 | -1 | $1.48 \times 10^{-10}$  | $n\pi^-$                             |
| Xi               | $\Xi^0$     |            |          | $\bar{\Xi}^0$     | $\frac{1}{2}$ | 1314.9    | +1            | 0     | 0  | 0 | -2 | $2.90 \times 10^{-10}$  | $\Lambda^0 \pi^0$                    |
|                  | $\Xi^-$     |            |          | $\bar{\Xi}^+$     | $\frac{1}{2}$ | 1321.7    | +1            | 0     | 0  | 0 | -2 | $1.64 \times 10^{-10}$  | $\Lambda^0 \pi^-$                    |
| Omega and others | $\Omega^-$  |            |          | $\bar{\Omega}^+$  | $\frac{1}{2}$ | 1672.5    | +1            | 0     | 0  | 0 | -3 | $0.82 \times 10^{-10}$  | $\Xi^0 \pi^-, \Lambda^+ \pi^-$       |

$\gamma c\tau \rightarrow 10^{-10}s$   
~1 m

Usually inside  
detector  
volumes

ID particles  
by track  
vertex  
topology

# Mass → Other Properties Baryon Number Lepton Number

- Conservation of Baryon Number Lepton Number Angular momentum Charge, determines what decays are allowed
- Griffiths problem 2.7
  - Eliminate half/most with easy things like that

| Spin $\frac{1}{2}$ |                                                                                 |                                                                     | Integer Spin                                                         |                                                    |                                                 |
|--------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------|
| LEPTONS            | UP<br>mass 2,3 MeV/c <sup>2</sup><br>charge $\frac{2}{3}$<br>spin $\frac{1}{2}$ | CHARM<br>1,275 GeV/c <sup>2</sup><br>$\frac{2}{3}$<br>$\frac{1}{2}$ | TOP<br>173,07 GeV/c <sup>2</sup><br>$\frac{2}{3}$<br>$\frac{1}{2}$   | GLUON<br>0<br>0<br>1                               | HIGGS BOSON<br>126 GeV/c <sup>2</sup><br>0<br>0 |
|                    | DOWN<br>4,8 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$               | STRANGE<br>95 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$ | BOTTOM<br>4,18 GeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$ | PHOTON<br>0<br>0<br>1                              |                                                 |
|                    | ELECTRON<br>0,511 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$                     | MUON<br>105,7 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$             | TAU<br>1,777 GeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$               | Z BOSON<br>91,2 GeV/c <sup>2</sup><br>0<br>1       |                                                 |
|                    | ELECTRON NEUTRINO<br><2,2 eV/c <sup>2</sup><br>0<br>$\frac{1}{2}$               | MUON NEUTRINO<br><0,17 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$     | TAU NEUTRINO<br><15,5 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$       | W BOSON<br>80,4 GeV/c <sup>2</sup><br>$\pm 1$<br>1 |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     |                                                                      |                                                    |                                                 |
|                    |                                                                                 |                                                                     | GAUGE BOSONS                                                         |                                                    |                                                 |

# Griffiths Chapter 4

- Clebsch Gordon Coefficients – That's especially what I want to skip
- So far you've used them to combine quantized angular momentum states.
- These are really just an extension of the (complex-number) linear algebra of "tensor products" – could work for any finite dimensional vector spaces
- "Group Theory (in Physics?)" : Just "tricks" for making higher dimensional vector spaces out of tensor products of lower dimensional ones?
- As we stated before, the vector elements we want to manipulate represent various properties of the SM particles (e.g. "flavor", charges, ang momentum, helicity, chirality others...)
- Mathematical way to make these properties so that they can be combined in a way that won't lose some aspect of their separateness (instead of scalar charge if just plus/minus will cancel out—spins can combine but usually the each "spinning" particle remains distinct.
  - combining "orthogonal" properties in linear combinations for satisfying new quantum operators
  - "properties" → quantum states/ kets
- Isospin Redux
- Parity, CP, CPT



# Symmetries – Noethers Theorem

Every symmetry of nature is associated to a conservation law, and *vice versa*.



- ✓ Translations in time → conservation of energy
- ✓ Translations in space → conservation of linear momentum
- ✓ Rotations in space → conservation of angular momentum
- ✓ Gauge transformations → conservation of charge

e.g.  $\psi \rightarrow e^{i\alpha} \psi$

Here  $\psi$  is the electron wavefunction.

Gauge transformation is “global” if  $\alpha = \text{const}$ , “local” if  $\alpha = \alpha(x)$

Our fundamental theories (e.g. those forming the Standard Model) are all based on the principle of local gauge invariance.

# Symmetries?

- Experimentally we're not using these much / phenomenology
- Some cases where it's interesting to know many of these rules
  - E.g. how many pions or photons a meson can decay to
- ...But in practice we can just use the PDG to “discover” or predict these things
- Maybe relevant if you want to look for (or you find!) a new resonance peak
- But this is it's own subfield I should mention I've heard seminars on some Jlab experiments (CLAS - Hicks)
  - Complex method of statistical subtractions of cross sections over wide angles/momenta (phase space) using constraints from these kind of spectroscopic properties (e.g. disentangling angular distributions) – can allow you to find certain resonances that would be buried otherwise. -- NAME ?!?
  - $P, C, G-P, J, L, S$  all built into such a model.
- THEORIES!!!! need the symms built in



# Review Spin Operators: Matrices

- Experimentally it's observed that only two values of Energy/ $S_z$  are observed/measured:
    - spin up and spin down :  $S_z = -\frac{\hbar}{2}$  &  $+\frac{\hbar}{2}$
  - Use Unit Vectors  $\chi_{\pm}$  to Represent the states up and down
    - Then  $\hat{H} / \hat{S}_z$  is Matrix w/ Eigenvalues  $\pm\hbar/2$
- $$\begin{array}{cc} \hbar/2 & -\hbar/2 \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{array}$$

$$H \propto S_z = \begin{pmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
- $\sigma_z = \sigma_3$
- Test:  $\overset{\text{just proportionality const}}{C} S_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = C \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow H \chi_+ = E_+ \chi \quad \text{TISE!}$
  - General State can have any prob  $P(+)$  and  $P(-) = 1 - P(+)$  to be in  $+/-$  states
  - If  $P(+)=|\alpha|^2$  and  $P(-)=|\beta|^2$ :
  - General state:  $\vec{\chi} \longrightarrow \vec{\chi} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$   
 $|\alpha|^2 + |\beta|^2 = 1$

# Review: Eigenkets / Expectation Vals

- Eigenkets  $\hat{A}|a_n\rangle = a_n|a_n\rangle$

– Examples:

$$S_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$S_x \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\hat{N}^{SHO} |1\rangle = 1|1\rangle$$

$$H|1\rangle = \frac{3}{2}\hbar\omega|1\rangle$$



$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ orthonormal basis}$$

$$\lambda = 1$$

$$\lambda = -1$$

$$a^\dagger a \psi_1^{SHO} \propto \left(\xi + \frac{d}{d\xi}\right) \left(\xi - \frac{d}{d\xi}\right) \psi_1(\xi) = 1 \psi_1(\xi)$$

- Generalized Easy Expectation Value Formula:

$$\langle \hat{O} \rangle = \sum_n \langle 0_n | c_n |^2$$

– where these  $c_n$ 's are the coeff's in  $\hat{O}$ 's orthonorm'd eigenbasis

$$\psi(x,0) = A [3\psi_0(x) + 4\psi_1(x)] \quad |4\rangle = 3A|0\rangle + 4A|1\rangle$$

$$\langle H \rangle = \langle 4 | H | 4 \rangle = (3A^* \langle 0 | + 4A^* \langle 1 |) H (3A | 0 \rangle + 4A | 1 \rangle)$$

$$= \left( \begin{matrix} 3A^* \langle 0 | & 4A^* \langle 1 | \end{matrix} \right) \left( \frac{\hbar\omega}{2} 3A | 0 \rangle + \frac{3\hbar\omega}{2} 4A | 1 \rangle \right)$$

$$= E_0 9A^2 \langle 0 | 0 \rangle + E_1 16A^2 \langle 1 | 1 \rangle = \sum E_n |c_n|^2$$

# Review Generalized Expectation Formula

$$\langle \hat{O} \rangle = \sum_n O_n |c_n|^2$$

• Spin examples:  $\langle \hat{H}_{\text{spin}} \rangle = \langle C \hat{S}_z \rangle$

$$\langle \hat{S}_z \rangle = \langle \chi | S_z | \chi \rangle$$

$$(\alpha^* \beta^*) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$= (\alpha^* \beta^*) S_z \left( \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$= (\alpha^* \beta^*) \left[ \alpha \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \left( -\frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \right]$$

$$= (\alpha^* \beta^*) \begin{pmatrix} \alpha \hbar/2 \\ -\beta \hbar/2 \end{pmatrix}$$

$$= |\alpha|^2 \frac{\hbar}{2} + |\beta|^2 \left( -\frac{\hbar}{2} \right)$$

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}^T = (\alpha^*, \beta^*) = a^* \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\langle S_x \uparrow |} + b^* \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\langle \chi | = a^* \langle S_x \uparrow | + b^* \langle S_x \downarrow |$$

$$\begin{cases} a = \frac{\alpha}{2} + \frac{\beta}{2} \\ b = \frac{\alpha}{2} - \frac{\beta}{2} \end{cases} \frac{1}{\sqrt{2}}$$

Example NOT  $\propto H$ :

$$\begin{aligned} \langle \chi | S_x | \chi \rangle &= (\alpha^* \beta^*) S_x \left[ a(\alpha, \beta) \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b(\alpha, \beta) \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right] \\ &= \left[ |a|^2 1 + |b|^2 (-1) \right] \frac{\hbar}{2} \end{aligned}$$

$\langle S_x \downarrow \rangle$

# New: Review Spin $\frac{1}{2}$ system $s = 1/2$

- Spin  $S_i = \frac{\hbar}{2} \sigma_i$   $\sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$   $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$$\sigma_i^2 = \mathbb{I} = 1 \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$S^2 = \sum_i S_i^2 = S_x^2 + S_y^2 + S_z^2 = \left(\frac{\hbar}{2}\right)^2 (1+1+1) = \frac{3}{4} \hbar^2$$

- States:

$$|s, m\rangle \Rightarrow \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow |\uparrow\rangle, |\downarrow\rangle$$

$$\langle \uparrow | S^2 | \uparrow \rangle = \frac{3}{4} \hbar^2 \langle \uparrow | \uparrow \rangle \quad \langle \chi | S^2 | \chi \rangle \quad \chi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\sum |\kappa_n|^2 = 1$$

$$\chi^\dagger S^2 \chi = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix} \frac{3}{4} \hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\frac{3}{4} \hbar^2 (|\alpha|^2 - |\beta|^2) = \frac{3}{4} \hbar^2$$

# Spin (/Isospin?)

- For example for Baryons

and 3 about the center of mass of all three (B).

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2$$

the latter can be achieved in two distinct ways. Specifically,

$$\left. \begin{aligned} | \frac{3}{2} \frac{3}{2} \rangle &= (\uparrow\uparrow\uparrow) \\ | \frac{3}{2} \frac{1}{2} \rangle &= (\uparrow\uparrow\downarrow + \uparrow\downarrow\uparrow + \downarrow\uparrow\uparrow)/\sqrt{3} \\ | \frac{3}{2} - \frac{1}{2} \rangle &= (\downarrow\downarrow\uparrow + \downarrow\uparrow\downarrow + \uparrow\downarrow\downarrow)/\sqrt{3} \\ | \frac{3}{2} - \frac{3}{2} \rangle &= (\downarrow\downarrow\downarrow) \end{aligned} \right\} \quad \text{spin } \frac{3}{2} (\psi_4) \quad (5.50)$$

$$\left. \begin{aligned} | \frac{1}{2} \frac{1}{2} \rangle_{12} &= (\uparrow\downarrow - \downarrow\uparrow)\uparrow/\sqrt{2} \\ | \frac{1}{2} - \frac{1}{2} \rangle_{12} &= (\uparrow\downarrow - \downarrow\uparrow)\downarrow/\sqrt{2} \end{aligned} \right\} \quad \text{spin } \frac{1}{2} (\psi_{12}) \quad (5.51)$$

$$\left. \begin{aligned} | \frac{1}{2} \frac{1}{2} \rangle_{23} &= \uparrow(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2} \\ | \frac{1}{2} - \frac{1}{2} \rangle_{23} &= \downarrow(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2} \end{aligned} \right\} \quad \text{spin } \frac{1}{2} (\psi_{23}) \quad (5.52)$$

The spin- $\frac{3}{2}$  combinations are completely *symmetric*, in the sense that interchanging any two particles leaves the state untouched. The spin- $\frac{1}{2}$  combinations are partially *antisymmetric* – interchange of two particles switches the sign. The first set is antisymmetric in particles 1 and 2 – hence the subscript; the second is antisymmetric in 2 and 3. We could also, of course, construct a pair of states antisymmetric in 1 and 3:

$$| \frac{1}{2} - \frac{1}{2} \rangle_{13} = (\uparrow\uparrow\downarrow - \downarrow\uparrow\uparrow)/\sqrt{2}$$

HOWEVER< IT IS WELL KNOWN that whenever probing the proton to reveal the quark, gluon and other substructure of the proton, we find that the three quark spin assignments are not what causes most of the spin of th proton. !!!! – the spin crises—some J caused by L

and 3 about the center of mass of all three (B).

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2$$

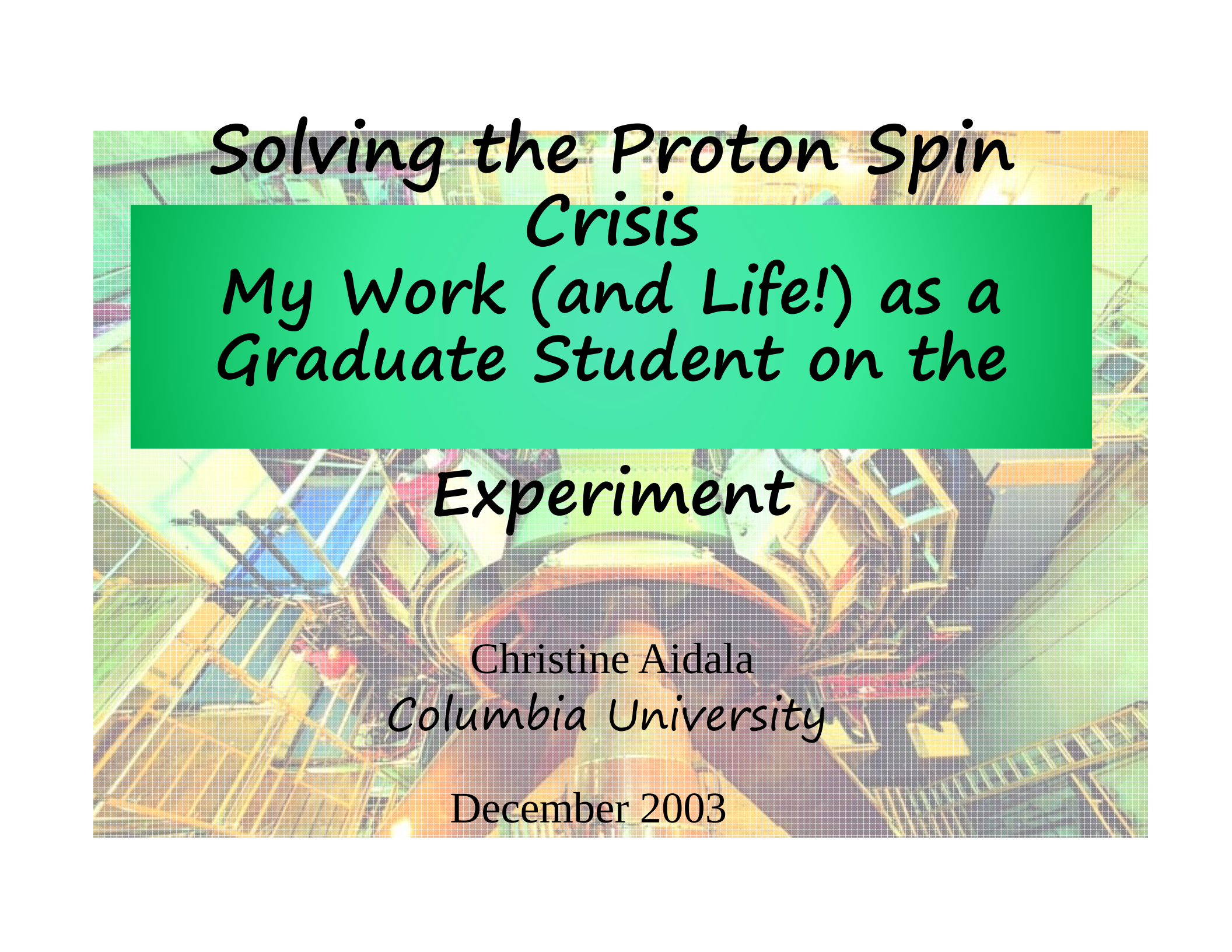
the latter can be achieved in two distinct ways. Specifically,

$$\left. \begin{aligned} | \frac{3}{2} \frac{3}{2} \rangle &= (\uparrow\uparrow\uparrow) \\ | \frac{3}{2} \frac{1}{2} \rangle &= (\uparrow\uparrow\downarrow + \uparrow\downarrow\uparrow + \downarrow\uparrow\uparrow)/\sqrt{3} \\ | \frac{3}{2} - \frac{1}{2} \rangle &= (\downarrow\downarrow\uparrow + \downarrow\uparrow\downarrow + \uparrow\downarrow\downarrow)/\sqrt{3} \\ | \frac{3}{2} - \frac{3}{2} \rangle &= (\downarrow\downarrow\downarrow) \end{aligned} \right\} \quad \text{spin } \frac{3}{2} (\psi_4) \quad (5.50)$$

$$\left. \begin{aligned} | \frac{1}{2} \frac{1}{2} \rangle_{12} &= (\uparrow\downarrow - \downarrow\uparrow)\uparrow/\sqrt{2} \\ | \frac{1}{2} - \frac{1}{2} \rangle_{12} &= (\uparrow\downarrow - \downarrow\uparrow)\downarrow/\sqrt{2} \end{aligned} \right\} \quad \text{spin } \frac{1}{2} (\psi_{12}) \quad (5.51)$$

$$\left. \begin{aligned} | \frac{1}{2} \frac{1}{2} \rangle_{23} &= \uparrow(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2} \\ | \frac{1}{2} - \frac{1}{2} \rangle_{23} &= \downarrow(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2} \end{aligned} \right\} \quad \text{spin } \frac{1}{2} (\psi_{23}) \quad (5.52)$$





# Solving the Proton Spin Crisis

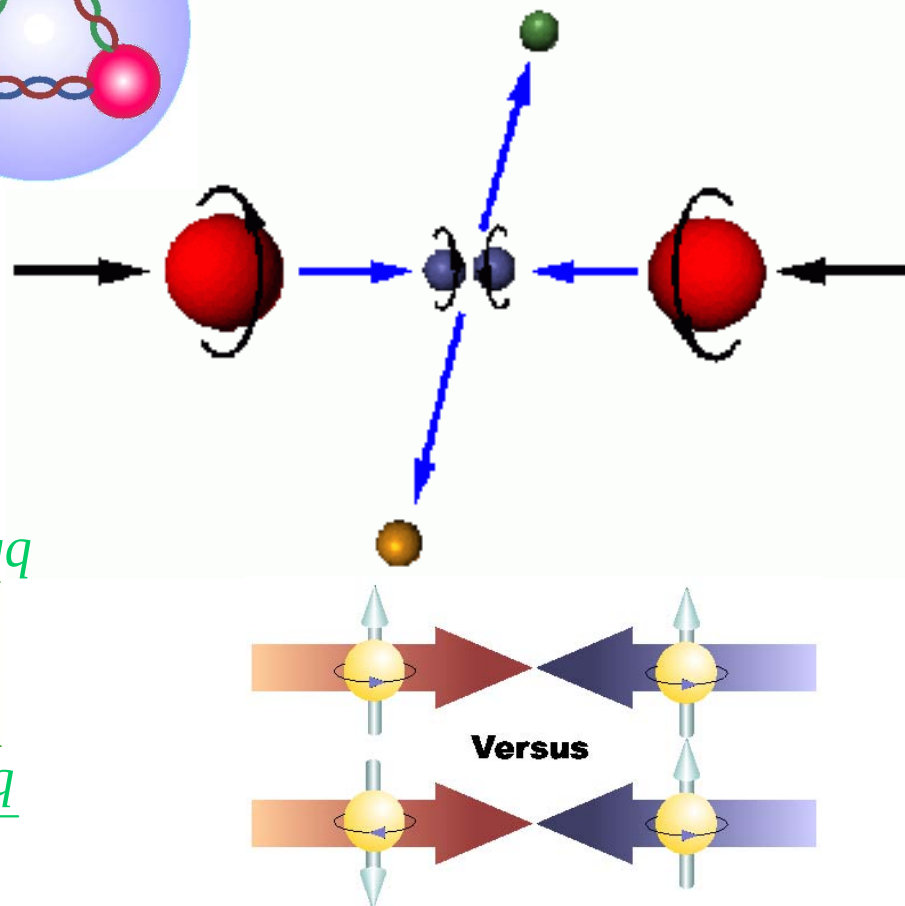
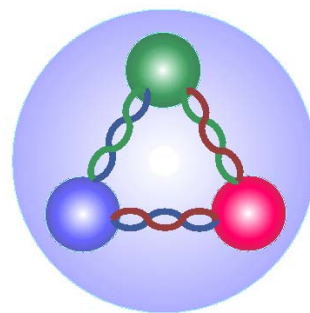
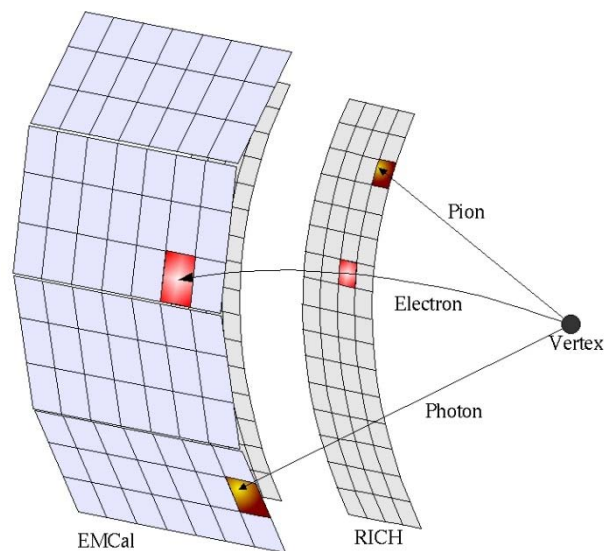
My Work (and Life!) as a  
Graduate Student on the

Experiment

Christine Aidala  
Columbia University

December 2003

# Part I: Some Physics



$$gg \rightarrow gg$$

$$\propto \frac{\Delta G}{G} \frac{\Delta G}{G}$$

$$gq \rightarrow gq$$

$$\propto \frac{\Delta q}{q} \frac{\Delta G}{G}$$

$$qq \rightarrow qq$$

$$\propto \frac{\Delta q}{q} \frac{\Delta q}{q}$$

# What is spin?

Spin is a quantum mechanical property of fundamental particles or combinations of particles.

It's called “spin” because it's a type of angular momentum and is described by equations treating angular momentum.



The units of angular momentum are the same as Planck's constant,  $\hbar$ , and can only have values that are integer : 0, 1, 2, 3, ...  
or half-integer: 1/2, 3/2, 5/2, ...

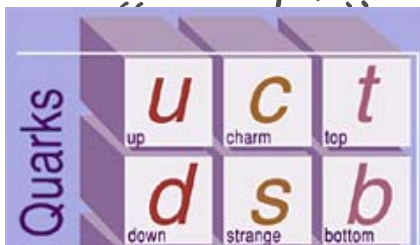
In a magnetic field, different spin states have different energies but have the same magnitude of the angular momentum.



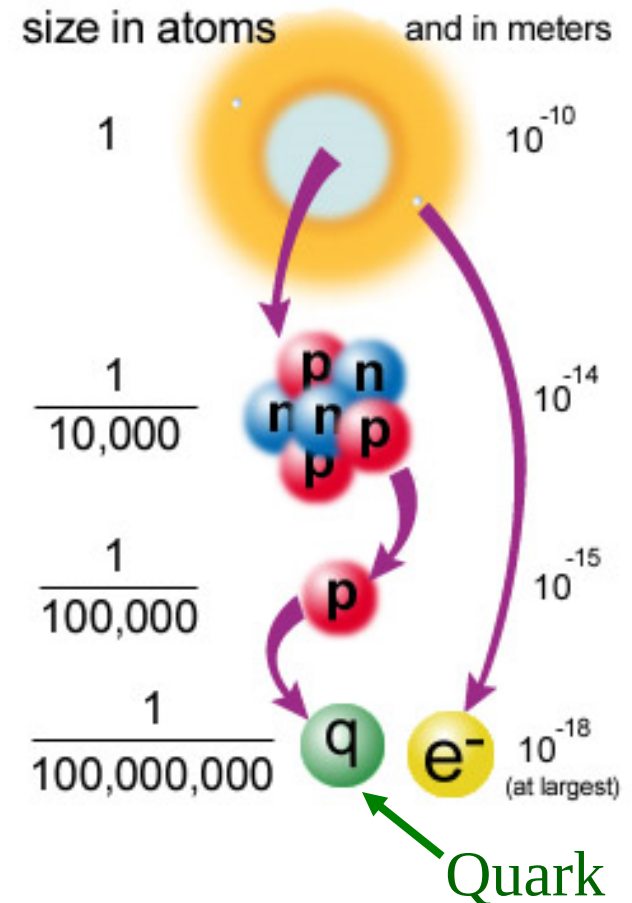
# The Quark-Parton Model

- Similarly to Rutherford's 1911 experiment in which the scattering of alpha particles at large angles off of gold revealed a hard atomic core (the nucleus), in the late 1960's at SLAC, scattering of electrons at large angles off of protons revealed "hard" subcomponents in the proton

- Protons weren't solid lumps of positive charge as previously believed!
- The pointlike constituents that make up the proton are called "quarks" or slightly more



**Quarks are fermions with spin 1/2.**



# Quark-Parton Model (cont.)



The simplest model says a proton's made of three "valence" quarks: 2 up quarks and 1 down quark.



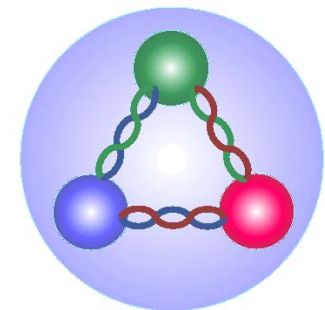
• *But these quarks are not completely free in the nucleon!*

- Bound by force-carrier particles called "gluons."

- "Sea quarks" are also present: short-lived quark-antiquark pairs from quantum mechanical fluctuations.



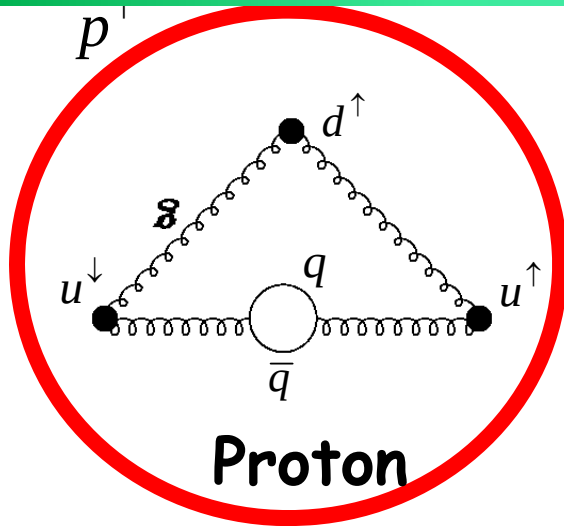
- As you hit the proton harder, you resolve shorter-lived fluctuations: gluons and sea quarks.



# Symmetries?

- Experimentally we're not using these much / phenomenology
- Some cases where it's interesting to know many of these rules
  - E.g. how many pions or photons a meson can decay to
- ...But in practice we can just use the PDG to “discover” or predict these things
- Maybe relevant if you want to look for (or you find!) a new resonance peak
- But this is it's own subfield I should mention I've heard seminars on some Jlab experiments (CLAS - Hicks)
  - Complex method of statistical subtractions of cross sections over wide angles/momenta (phase space) using constraints from these kind of spectroscopic properties (e.g. disentangling angular distributions) – can allow you to find certain resonances that would be buried otherwise. -- NAME ?!?
  - $P, C, G-P, J, L, S$  all built into such a model.
- THEORIES!!!! need the symms built in

# The Proton Spin Crisis



Say you have a proton with total spin  $+1/2$  along some axis. You'd expect it to contain two quarks with spin  $+1/2$  and one with spin  $-1/2$ .

$$1/2 + 1/2 - 1/2 = +1/2$$

Surprising data from polarized muon-nucleon scattering in late 1980s!

1987: Only 12%  $\pm$  16% of proton's spin carried by quarks!

**The proton spin crisis begins!!**

The rest now expected to be from gluon spin and orbital angular momentum of quarks and gluons, but this hasn't been easy to measure!

**Spin**

$$\frac{1}{2} = \frac{1}{2} \Delta\Sigma + \Delta G + L_z$$

Quark Spin      Gluon Spin

Orbital Angular Momentum

# Spin of the proton: Current state of affairs

- $S = 1/2 = (1/2) \Delta q + \Delta G + L_q + L_G$
- $\Delta q = 0.2 \pm 0.1$
- $\Delta G = 1.8 \pm 1.0$
- Experiments hunting for  $\Delta G$  include
  - PHENIX and STAR at RHIC:  $p-p$
  - COMPASS at CERN:  $\mu-N$
  - HERMES at DESY:  $e-N$

# Isospin

- These slides are from Krieger – U Toronto

## Pion-Nuclon Scattering Amplitudes

$$\pi^+ + p: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad \frac{3}{2} \right\rangle$$

$$\pi^0 + p: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^- + p: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^+ + n: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^0 + n: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^- + n: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad -\frac{3}{2} \right\rangle$$

See Perkins Introduction to High Energy Physics section on "Isospin in the pion nucleon system" for more detail. (Section number varies from edition to edition; it is 3.13 in the 4th edition)

$$(a) \pi^+ + p \rightarrow \pi^+ + p$$

$$(f) \pi^- + n \rightarrow \pi^- + n$$



Both are pure  $I = 3/2$  on either side of the reaction. Both proceed purely via the  $I = 3/2$  amplitude  $\mathcal{M}_3$

# ISOSPIN not a charge but similar effects

- $I, I_3$  not  $I_3$ ,  $m_{I_3} \rightarrow S, m_S$
- ( $I_3 \equiv "m_{I_3}"$ )
- At start I made an analogy to isospin being a new type of charge but I also claimed it was just flavor which isn't really a "charge"?
- Related to flavor which is like a charge in Weak force
- But it is a charge for residual nuclear strong force interactions:
- Chiral EFT explicit isospin terms in Lagrangian  $\rightarrow$  effective potentials, isospin dependent

# Isospin - quarks

$$u = \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle \quad d = \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle \quad s = |0 \ 0\rangle$$

Other quark flavours (c,b,t) are also isospin singlets, like the strange quark

Mesons containing (only) u,d quarks can therefore have isospin 0 or 1

$$\pi^- = \bar{u}d \quad \pi^0 = u\bar{u}, d\bar{d} \quad \pi^+ = u\bar{d} \quad \text{isospin triplet } I = 1$$

$$\eta = u\bar{u}, d\bar{d} \quad \text{isospin singlet } I = 0$$

Mesons containing only one u,d quark can only have isospin =  $\frac{1}{2}$

$$K^0 = u\bar{s} \quad K^+ = d\bar{s} \quad \text{form an isospin doublet}$$

$$\bar{K}^0 = \bar{u}s \quad K^- = \bar{d}s \quad \text{form a separate isospin doublet}$$

More on this later



# Isospin

- These slides are from Krieger – U Toronto

## Pion-Nuclon Scattering Amplitudes

$$\pi^+ + p: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad \frac{3}{2} \right\rangle$$

$$\pi^0 + p: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^- + p: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^+ + n: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^0 + n: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^- + n: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad -\frac{3}{2} \right\rangle$$

See Perkins Introduction to High Energy Physics section on "Isospin in the pion nucleon system" for more detail. (Section number varies from edition to edition; it is 3.13 in the 4th edition)

$$(a) \pi^+ + p \rightarrow \pi^+ + p$$

$$(f) \pi^- + n \rightarrow \pi^- + n$$



Both are pure  $I = 3/2$  on either side of the reaction. Both proceed purely via the  $I = 3/2$  amplitude  $\mathcal{M}_3$

This is the part we will come back to after Ch3 Griffiths

# Isospin – Like the Deuteron case

## Application example: pion-nucleon scattering

Six elastic  $\pi N \rightarrow \pi N$  processes (e.g. same particles in and out of scattering)

$$(a) \pi^+ + p \rightarrow \pi^+ + p \quad (b) \pi^0 + p \rightarrow \pi^0 + p$$

$$(c) \pi^- + p \rightarrow \pi^- + p \quad (d) \pi^+ + n \rightarrow \pi^+ + n$$

$$(e) \pi^0 + n \rightarrow \pi^0 + n \quad (f) \pi^- + n \rightarrow \pi^- + n$$

and 4 charge-exchange processes

$$(g) \pi^+ + n \rightarrow \pi^0 + p \quad (h) \pi^0 + p \rightarrow \pi^+ + n$$

$$(i) \pi^0 + n \rightarrow \pi^- + p \quad (j) \pi^- + p \rightarrow \pi^0 + n$$

Pions carry  $I = 1$  and nucleons,  $I = 1/2$  so the total isospin can be either  $1/2$  or  $3/2$

Since isospin is conserved (in these strong interactions) there are two distinct (quantum-mechanical) amplitudes:

$\mathcal{M}_1$  for  $I = 1/2$

Takes an isospin  $1/2$  initial state  
to an isospin  $1/2$  final state.

and

$\mathcal{M}_3$  for  $I = 3/2$

Takes an isospin  $1/2$  initial state  
to an isospin  $1/2$  final state.

# More info: What does this mean?

- Two Matrix elements :

$$M = \langle \text{final} | \hat{O} | \text{initial} \rangle$$

Conserving  
isospin  
const.

- Strong force doesn't affect isospin
  - for flavor: means (QCD interaction) doesn't depend on flavor (only on color)
- But this isn't entirely true for nucleons / hadronic interactions  
AFAICT ie for the "residual" effective nuclear force mediated by pions
- Consider this exact case :  $\hat{O}$  doesn't change isospin but interaction depends on  $I$  (total  $I$ , not  $I_3$ )
  - eigenvalue of  $\hat{O}$  apparently,  $M_3$  corresponds to different eigenvalues apparently compared to  $M_1$

# Isospin

- Just slightly more complicated than the deuteron case

$$\sigma_a : \sigma_c : \sigma_j = 9|\mathcal{M}_3|^2 : |\mathcal{M}_3 + 2\mathcal{M}_1|^2 : 2|\mathcal{M}_3 - \mathcal{M}_1|^2$$

At  $E_{\text{CM}} = 1232 \text{ MeV}$  there is a dramatic bump (increase) in the pion-nucleon scattering cross-section, due to a  $\pi N$  resonance (the  $\Delta$ ).

This is listed amongst the particles on the inside cover of your text

[we will discuss resonance in scattering in a future lecture]

$\Delta$  carries isospin  $3/2$ , therefore:

Expect the scattering at this energy to be dominated by the isospin  $3/2$  channel

$$\sigma_a : \sigma_c : \sigma_j = 9 : 1 : 2$$

My continued “bashing”:

-notice again, this is LOW ENERGY scattering

-Ratios ~work mostly elsewhere, not as good though, but there is a big background under the peaks – example of inv. mass  
-FRANTZ: but...the peaks are broad, and the “bkg” is still QCD – (Lattice QCD vs isospin)

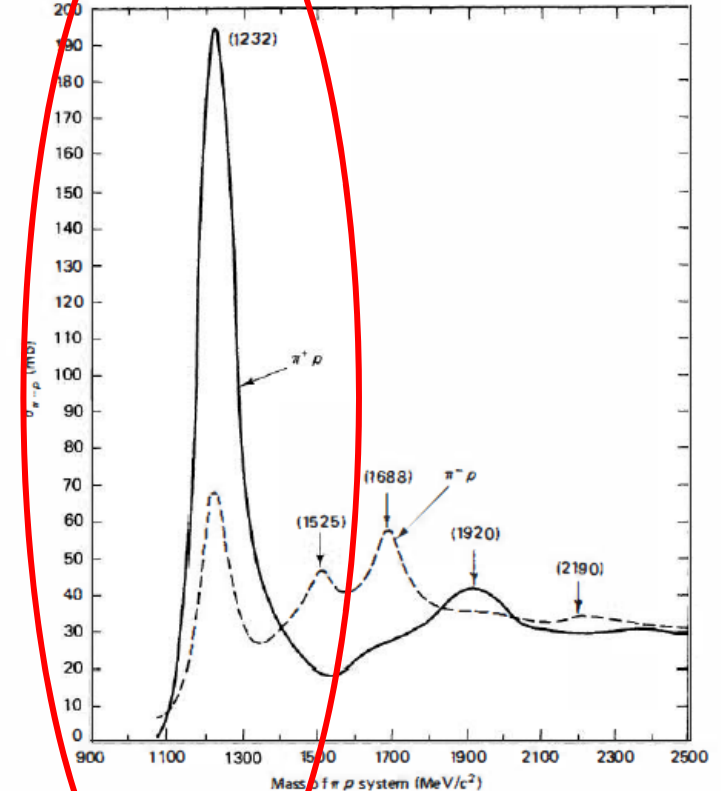


Fig. 4.6 Total cross sections for  $\pi^+ p$  (solid line) and  $\pi^- p$  (dashed line) scattering. (Source: Gasiorowicz, S. (1966) Elementary Particle Physics, John Wiley & Sons, New York, p. 294. Reprinted by permission of John Wiley and Sons, Inc.)

# Isospin conservation in a decay (allowed)

- $\rho \rightarrow \pi^+ \pi^-$
- $\rho \rightarrow \pi^0 \pi^0$ ?

# Parity

- Parity operation inverts the sign of all spatial coordinates:
- Position vector  $(x, y, z)$  goes to  $(-x, -y, -z)$  (eg  $P(\mathbf{r}) = -\mathbf{r}$ )
- Regular (polar) vectors transform in this way under parity transformation
- Regular scalars ( $a = \mathbf{b} \cdot \mathbf{c}$ ) transform like  $P(a) = a$  (e.g. they are unaffected)
- However there are scalars which do change sign!!!
- **Axial-vector** (also called **Pseudo-vector**) does not change sign under parity transformation.
- e.g. cross-product of two polar vectors  $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ ,  $\mathbf{B}$  ( $= \text{curl } \mathbf{A}$ )
- Clearly  $P^2 = I$  (so eigenvalues are  $\pm 1$ )
- **Pseudo-scalar**:  $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$  does change sign under a parity transformation
- (here,  $\mathbf{a}$ ,  $\mathbf{b}$  and  $\mathbf{c}$  are all polar vectors)

# Parity

- Pseudoscalar Mesons (Pions)
  - Spin 0
  - But Parity Operation change sign of Wave Functions
- Hmm WAVEFUNCTIONS!?!?!?

# PDG

$$\pi^\pm$$

$$J^G(J^P) = 1^-(0^-)$$

$$\text{Mass } m = 139.57018 \pm 0.00035 \text{ MeV} \quad (S = 1.2)$$

$$\text{Mean life } \tau = (2.6033 \pm 0.0005) \times 10^{-8} \text{ s} \quad (S = 1.2)$$

$$c\tau = 7.8045 \text{ m}$$

$\pi^\pm \rightarrow \ell^\pm \nu \gamma$  form factors [a]

$$F_V = 0.0254 \pm 0.0017$$

$$F_A = 0.0119 \pm 0.0001$$

$$F_V \text{ slope parameter } a = 0.10 \pm 0.06$$

$$R = 0.059^{+0.009}_{-0.008}$$

$\pi^-$  modes are charge conjugates of the modes below.

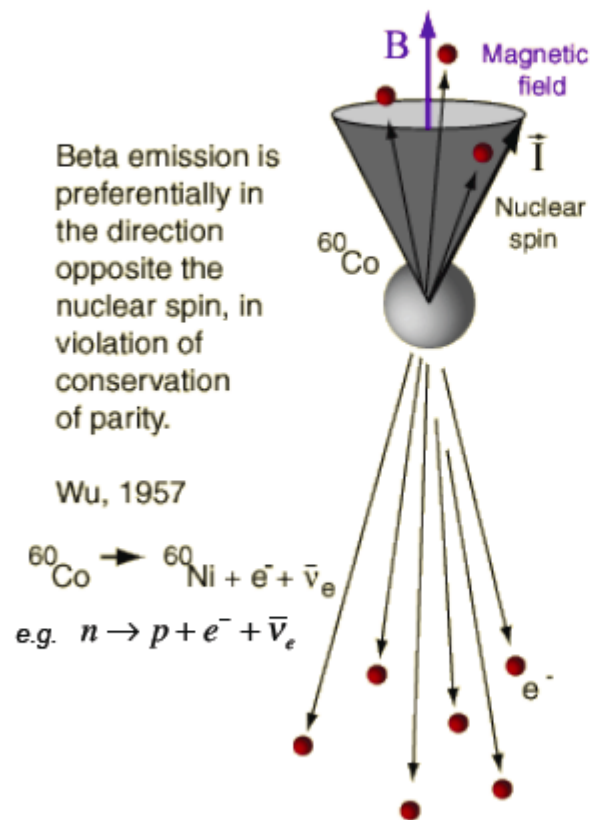
For decay limits to particles which are not established, see the section on Searches for Axions and Other Very Light Bosons.

| $\pi^+$ DECAY MODES                                                   | Fraction ( $\Gamma_i/\Gamma$ )           | Confidence level       | $P$<br>(MeV/c) |
|-----------------------------------------------------------------------|------------------------------------------|------------------------|----------------|
| $\mu^+ \nu_\mu$                                                       | [b] (99.98770 ± 0.00004) %               |                        | 30             |
| $\mu^+ \nu_\mu \gamma$                                                | [c] ( 2.00 ± 0.25 ) × 10 <sup>-4</sup>   |                        | 30             |
| $e^+ \nu_e$                                                           | [b] ( 1.230 ± 0.004 ) × 10 <sup>-4</sup> |                        | 70             |
| $e^+ \nu_e \gamma$                                                    | [c] ( 7.39 ± 0.05 ) × 10 <sup>-7</sup>   |                        | 70             |
| $e^+ \nu_e \pi^0$                                                     | ( 1.036 ± 0.006 ) × 10 <sup>-8</sup>     |                        | 4              |
| $e^+ \nu_e e^+ e^-$                                                   | ( 3.2 ± 0.5 ) × 10 <sup>-9</sup>         |                        | 70             |
| $e^+ \nu_e \nu \bar{\nu}$                                             | < 5                                      | × 10 <sup>-6</sup> 90% | 70             |
| <b>Lepton Family number (LF) or Lepton number (L) violating modes</b> |                                          |                        |                |
| $\mu^+ \bar{\nu}_e$                                                   | L [d] < 1.5                              | × 10 <sup>-3</sup> 90% | 30             |
| $\mu^+ \nu_e$                                                         | LF [d] < 8.0                             | × 10 <sup>-3</sup> 90% | 30             |
| $\mu^- e^+ e^+ \nu$                                                   | LF < 1.6                                 | × 10 <sup>-6</sup> 90% | 30             |



# parity violation exp

## Parity Violation: Experiment (Madame Wu, 1956)



# More on parity experiment

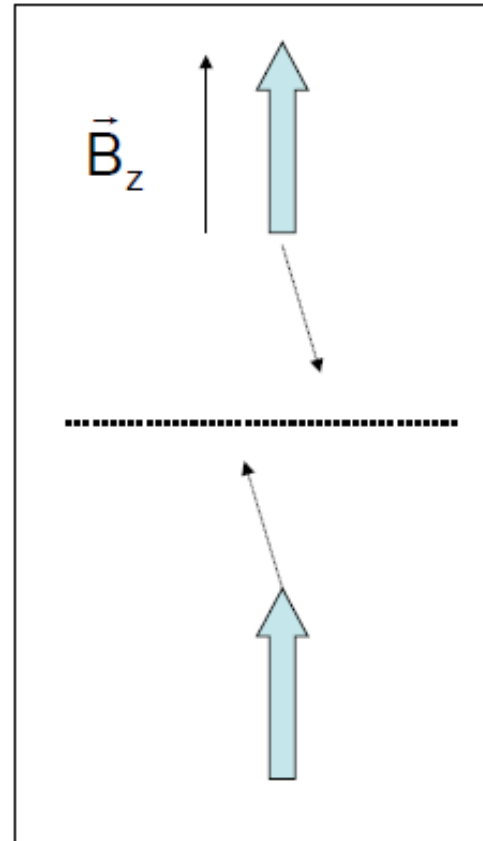
N.B. Spin is an axial vector which does NOT change sign under parity inversion.

Reflect through xy plane (here  $\vec{B}$  and nuclear spin are along z).

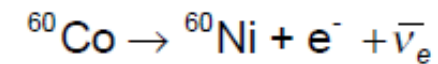
$$\vec{p}_e \rightarrow -\vec{p}_e$$

Spin does NOT reflect  
(it's a pseudovector)

Parity is NOT conserved  
in weak decays



$^{60}\text{Co}$  nuclear spin



electrons preferentially  
emitted in direction  
opposite to nuclear spin

electrons now preferentially  
emitted in direction of the  
nuclear spin. This is NOT  
experimentally observed.

# Helicity

- Discussed before
- Use  $v$  direction as  $z$ -axis

Helicity defined as  $m_s/s$ .

Particle of spin  $\frac{1}{2}$  can therefore have helicity of  $\pm 1$

Helicity +1 is referred to as right-handed



Helicity -1 is referred to as left handed.

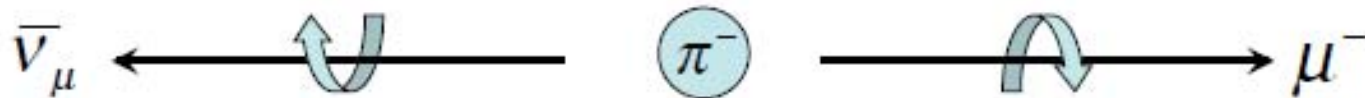


Note that helicity is NOT Lorentz invariant unless the particle is massless

If the particle has mass, one can always make Lorentz transformation into an inertial frame with velocity  $> v$ , and thus “flip” the helicity.

# Neutrino/ Parity

Imagine the decay (at rest) of a charged pion  $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$



Pion has spin 0, so spins of final state particles must be anti-aligned

Final state particles therefore have the same helicity.

A measurement of the muon helicity therefore gives a measurement of the neutrino helicity (which we have no easy way of measuring directly).

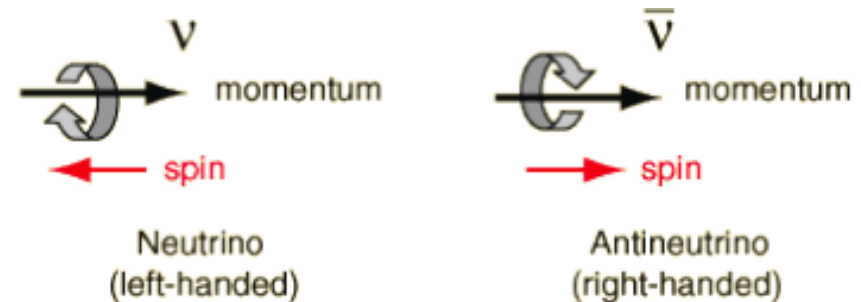
If parity were conserved, in this decay we would expect to see left-handed anti-neutrinos 50% of the time and right-handed anti-neutrinos 50% of the time.

Experimentally ONLY right-handed anti-neutrinos (e.g. as determined from the muon helicity) are observed.

# actual result

All neutrinos are left-handed

All anti-neutrinos are right-handed



This absolute statement is of course not true in the case where neutrinos have mass, which we now know they do.

However, in the rest frame of the pion (as an example) it is still true that the outgoing anti-neutrino is ALWAYS right-handed.

We say that parity is maximally violated in weak decays.

(i.e. there are not simply more left-handed neutrinos than right-handed neutrinos. There are **NO** right-handed neutrinos at all.)

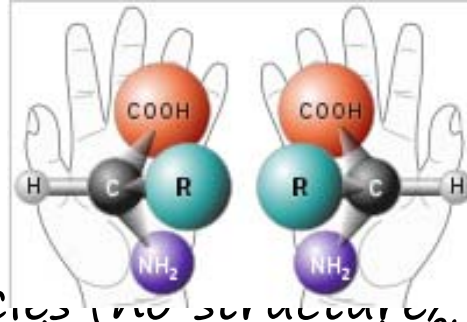
We will see that this parity violation is built into the dynamics of the weak interaction.

Parity operation applied to  $\nu_L$  gives  $\nu_R$  which does not exist.

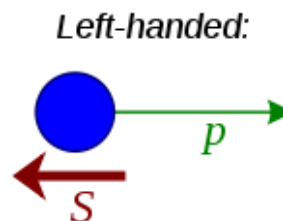
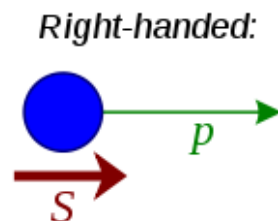
[Momentum vector reverses but spin does not]

# Chirality vs Helicity

- **Chirality** has to do with parity operation “mirror symmetry”
  - Does vector flip direction in the mirror



- For elementary particles (no structure, 'points') the only thing we can imagine is “spin” that cause such flips
- **Helicity**: relative orientation spin and momentum  $p$
- For non-massless particles boost faster than  $p$  – depends on reference frame
- For massless particles → “Locked in” by speed  $c$  limit! (photon spin 1)
- Chirality “=” Helicity (creates 1 to 1 matching)
- **NEUTRINOS → ALMOST!** – still practically use “equality” for calculations



# Charges (Chirality)

- Weak Charge (?) “Hyper charge” “Weak charge” ?
  - It's complicated Weak interactions always involve “mixing”
  - coupling parameter  $g$  divided  $g_{\text{Axial}}$   $g_{\text{Vecto}}$
  - charge  $\rightarrow$  depends on chirality

**Electroweak charges of Standard Model particles**

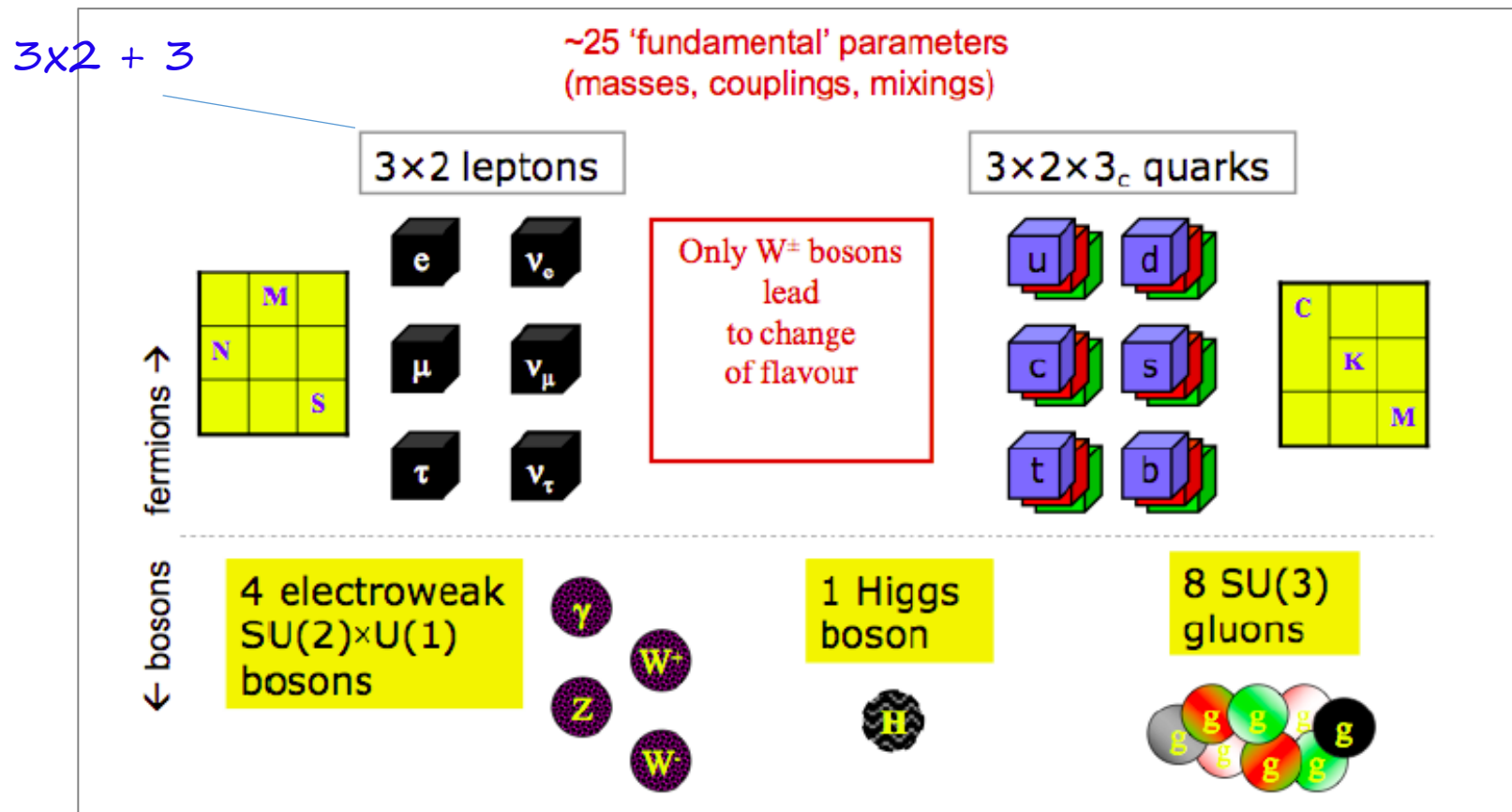
| Particle(s)                                         | Electric charge | Weak charge                                                  | Weak isospin   |       | Weak hypercharge |                | Z boson coupling                                                        |                                                           |
|-----------------------------------------------------|-----------------|--------------------------------------------------------------|----------------|-------|------------------|----------------|-------------------------------------------------------------------------|-----------------------------------------------------------|
|                                                     |                 |                                                              | LEFT           | RIGHT | LEFT             | RIGHT          | LEFT                                                                    | RIGHT                                                     |
| u, c, t<br>up, charm, top                           | $+\frac{2}{3}$  | $+1 - \frac{8}{3} \sin^2 \theta_w$<br>$\approx +\frac{1}{3}$ | $+\frac{1}{2}$ | 0     | $+\frac{1}{3}$   | $+\frac{4}{3}$ | $+\frac{1}{2} - \frac{2}{3} \sin^2 \theta_w$<br>$\approx +\frac{1}{3}$  | $-\frac{2}{3} \sin^2 \theta_w$<br>$\approx -\frac{1}{6}$  |
| d, s, b<br>down, strange, bottom                    | $-\frac{1}{3}$  | $-1 + \frac{4}{3} \sin^2 \theta_w$<br>$\approx -\frac{2}{3}$ | $-\frac{1}{2}$ | 0     | $+\frac{1}{3}$   | $-\frac{2}{3}$ | $-\frac{1}{2} + \frac{1}{3} \sin^2 \theta_w$<br>$\approx -\frac{5}{12}$ | $+\frac{1}{3} \sin^2 \theta_w$<br>$\approx +\frac{1}{12}$ |
| $\nu_e, \nu_\mu, \nu_\tau$<br>neutrinos             | 0               | +1                                                           | $+\frac{1}{2}$ | 0 [i] | -1               | 0 [i]          | $+\frac{1}{2}$                                                          | 0 [i]                                                     |
| $e^-, \mu^-, \tau^-$<br>electron, muon, tauon       | -1              | $-1 + 4 \sin^2 \theta_w$<br>$\approx 0$                      | $-\frac{1}{2}$ | 0     | -1               | -2             | $-\frac{1}{2} + \sin^2 \theta_w$<br>$\approx -\frac{1}{4}$              | $\sin^2 \theta_w$<br>$\approx +\frac{1}{4}$               |
| $Z^0, \gamma, g$<br>photon, Z boson, and gluon [ii] | 0 [iii]         |                                                              |                |       |                  |                |                                                                         |                                                           |
| $W^+$<br>W boson [iv]                               | +1              | $+2 - 4 \sin^2 \theta_w$<br>$\approx +1$                     | +1             |       | 0                |                | $+1 - \sin^2 \theta_w$<br>$\approx +\frac{3}{4}$                        |                                                           |
| $H^0$<br>Higgs boson                                | 0               | -1                                                           | $-\frac{1}{2}$ |       | +1               |                | $-\frac{1}{2}$                                                          |                                                           |

-0.99 per [neutron](#), and +0.07 per [proton](#)



# Charge/Chirality – Different Particles!

- Three different color charges really imply a different picture in terms of numbers of fundamental particles
- Chirality actually would double all numbers...except for neutrinos (all left handed)





# CKM (P-NMS) Matrix – Oscillations

- Weak Interaction States  $\neq$  Mass “Eigenstates”
  - (“What is a particle”?)
- Many effects: E.g. any weak process that produces “one” weak flavor (e.g. “ $\nu_\mu$ ”) produces mixed mass eigenstates ( $\nu_\mu + \nu_e \dots$ )
  - Neutrinos: “free particles” -- Time oscillation phase of “wave function components” go with mass eigenstates ... hence “neutrino oscillations”
  - For quarks, always in bound states, but “oscillations” still seen e.g. in alternate decay modes

$$K_1 \rightarrow 2\pi, \quad K_2 \rightarrow 3\pi \quad (4.70)$$

Now, the  $2\pi$  decay is much faster, because the energy released is greater. So if we start with a beam of  $K^0$ 's

$$|K^0\rangle = \left(\frac{1}{\sqrt{2}}\right) (|K_1\rangle + |K_2\rangle) \quad (4.71)$$

*This section in Chapter 4: coolest in the book?*

AND THESE THINGS about the Weak force we will come back to

# CKM (P-NMS) Matrix - Oscillations

And this then too

- Matrix -- Unitary I think this is main constraint against fourth generation, though some of the measured numbers may assume unitarity

- CKM: Cabibbo-Kobayashi-Maskawa - quarks

$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix}$$

CKM Strongly diagonal

$$\begin{bmatrix} 0.97370 & 0.2245 & 0.00382 \\ 0.221 & 0.987 & 0.0410 \\ 0.0080 & 0.0388 & 1.013 \end{bmatrix}$$

remember  
effects will  
with the  
squares of  
these  
numbers

- P-NMS: not as diagonal

- Pontecorvo-Maki-Nakagawa-Sakata - neutrinos

$$|U| = \begin{bmatrix} |U_{e1}| & |U_{e2}| & |U_{e3}| \\ |U_{\mu1}| & |U_{\mu2}| & |U_{\mu3}| \\ |U_{\tau1}| & |U_{\tau2}| & |U_{\tau3}| \end{bmatrix} = \begin{bmatrix} 0.801 \dots 0.845 & 0.513 \dots 0.579 & 0.143 \dots 0.156 \\ 0.233 \dots 0.507 & 0.461 \dots 0.694 & 0.631 \dots 0.778 \\ 0.261 \dots 0.526 & 0.471 \dots 0.701 & 0.611 \dots 0.761 \end{bmatrix}$$

## PREVIEW:

# Relativistic Kine: The Basics $p$ and $E$

- Starting Relations

$$E = mc^2 \quad [\text{at rest}]$$

$$E^2 = \vec{p}^2 + m^2$$

$$E^2 - \vec{p}^2 = m^2$$

$$E = \pm \sqrt{p^2 + m^2} \quad \checkmark$$

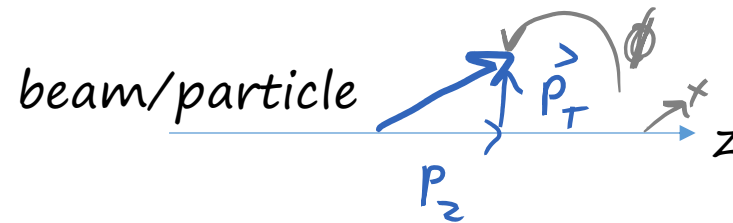
# Relativistic Kinematics - Basics

- Energy-mom  $(E, \vec{p}) = p^\mu$
- Time Space  ~~$(t, \vec{x}) = x^\mu$~~
- Velocity  ~~$(\gamma, \gamma\vec{B}) = v^\mu$~~

$$\vec{p} = (p_x, p_y, p_z) = (\vec{p}_T, p_z)$$

$$\vec{\beta} = \frac{\vec{p}}{E}$$

(not used)



Scalar Product

$$p_1 \cdot p_2 = p_1^\mu \cdot p_{2\mu} = p_1^\mu p_2^\nu g_{\mu\nu}$$

$$p^2 = p \cdot p = E^2 - |\vec{p}|^2$$

$$\mathbf{A} \cdot \mathbf{B} = (A^0 \ A^1 \ A^2 \ A^3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} B^0 \\ B^1 \\ B^2 \\ B^3 \end{pmatrix}$$